Algebra

Applications and Strategies

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Preliminary Edition

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Preface to the student

I want your experience in math to be as smooth and as enjoyable as possible. I want this for you, even if you can recall being frustrated with mathematics. To achieve that goal, I am deliberately writing this book for you the student, and not for your teacher. You'll see that the language I'm using in this book is deliberately informal. There is as much attempt to use really clear explanations. You may never have carefully read a math book before (or you may have tried and found it hard to comprehend), but I hope you'll really give it your full effort in this book.

I hope to take each part of algebra that has the potential to be challenging and *really* break it down step-by-step. This does mean that I may ask you to try something different from the way you have done it in the past. I might also ask you to think about things that you haven't really thought about much before. I hope you'll give it a shot: what do you have to lose by trying this subject in a new way?

Math might not be your favorite thing. You might even hate it. You might describe your self as "not a math person." I hope that by the end of this course, some of these things change, even if just a little. You can end this course and still not have math as your favorite thing, but my hope is that you can walk away from this experience with an appreciation for mathematics: I hope that at the end of the semester, you'll look back and think "hey, that wasn't that bad!"

I want you to see math, and algebra in particular, as a useful tool with a variety of applications. Some of these applications of algebra are to everyday life situations. Some of these applications are to a variety of other subjects, including physics, economics, business, marketing, management, biology, chemistry, psychology, engineering, medicine, pharmacology, medical dosimetry, sociology, political science, and architecture. At the start of each new section of this book, I will introduce a variety of applications of the algebra that we're about to dig into. I can't promise you that each section has an application that's specific to your future career goals. (Besides, students change their majors, and even past graduation, lots of people change career sectors!) My hope is that when you see an application, instead of dismissing it as "not for me", I hope you'll see it and think "okay, so I might not use the algebra that's about to get discussed in that way, but it's nonetheless neat to see a way that somebody would use this mathematics!"

After briefly introducing the applications, we will dig into the mathematics itself, but then the end of the section will revisit the application questions and answer each of them! If this is not your cup of tea, and you just want to see the math, you can skip the application part at the beginning and end of each section. In choosing the applications to introduce, there is a fine balance in picking situations that are authentic and realistic, while at the same time ensuring that the mathematics problem is not overly complicated.

In addition to the applications, I really want this text to help demystify every algebra skill. To do that, there are several intentional decisions I made in authoring this book:

- Examples help us learn, and so there are many examples in this book. These examples have thorough explanations. My desire in writing is to address everything, especially those things that students have shown me are confusing.

- Strategies for answering certain types of problems are explicitly discussed. Boxes that are clearly marked Strategy will appear throughout the book. Each strategy box starts with a clear and direct statement of which situtaion(s) the strategy will apply in, and then provides step-by-step numbered instructions for how to carry out the strategy, much like a recipe in a cookbook.

- When reasonable, we give two or three different-looking answers to the same question. We can't always afford the time and space for this (nor would you want to see this for every problem), but we do this when it makes sense. There are a number of questions that naturally lend themselves to multiple approaches. We want to honor those approaches. If we only showed one answer, and you had thought about answering the question a different way, you might think "oh, I was wrong to think about it that way." Thus, we think there's something really valuable in taking a problem and being honest that there can often be multiple approaches to a problem.

- Already in the point above, I mentioned the word "answer", which I wish to distinguish from the concept of "the final answer." In math, it's the process that matters. It's important to show your work to your instructor, and so it is important for me to show my work to you. Thus, all of my work and explanation is what I call an answer, and just the final result is what I call the final answer. In fact, there will be quite a number of examples where I provide two or more answers which have the same final answer. Sometimes this is done simply because students will be split $50/50$ on how they would approach the problem. Other times, this is done to illustrate an important concept.

- Whether through advice, warnings, or extended explanations, I will write in a way that equips you to succeed in thinking about algebra successfully. For example, in Example 1.1.87, we discuss distributing a negative in the given expression $(16x + 8) - (6x - 6)$. A correct first step is $16x + 8 - 6x + 6$ while a common mistake for a first step is $16x + 8 - 6x - 6$. In my own experience, the explanations regarding this error (and really every math error) are much too short, with perhaps the shortest of explanations (which is when we're simply told "just don't do that") being the least helpful. Thus, in these situations where students have been often told too little information, we really dig into thorough explanations.

- We dig into language when it's important, and we don't when it won't matter. For example, the words term and factor are very important, and we'll carefully introduce both vocabulary words with many examples. This is useful vocabulary to master, because the words term and factor will be used in their technical sense very often. On the other hand, this is the one and only time in the book you'll see that the little 3 in $\sqrt[3]{7}$ is called the index. We will just refer to this 3 as the little number outside of the radical.

In addition to having this text as a resource for mathematical skill building through thorough explanations and many examples, and in addition to the applications we introduce, I want you to be successful. I hope that you learn math in a way that is enjoyable and empowering, and that you'll use in a future course, in your career, and in your life. And more immediately, I hope that you gain confidence in your ability to do math, and that you earn the grade that you want to. I think reading a fresh perspective on algebra can help in that process, which is why I invite you to read through this text as thoroughly as you can, with the goal of learning and understanding.

No book and no instructor can substitute for your own effort. Math is like learning how to play basketball or speak a new language. You're going to have to try things, learn what works and what doesn't, and make mistakes. This is all part of the growing process. Even if you like all of your instructor's explanations, it's not enough to just feel like you understood the explanations you've heard. You've got to try it, and I'm just letting you know now then when you venture out on your own, you're going to get lost. Think about it: when traveling on a safari in Kenya, tour groups don't get lost because the guide and driver know where to go. Now, in the setting of a safari, it's actually dangerous to get out of the vehicle and wander on your own, but in math, at some point in the class, you will be expencted to (perhaps there are quizzes and exams). The point is that during some of your unaccompanied explorations, you're going to get lost. That's a normal part of the learning process. When you do get lost, use resources that will help you learn what went wrong and will guide you to how to think about the mathematics in a way that makes sense to you and is mathematically accurate. Visit your professor in office hours, ask questions in class, and work together with peers when that's allowed. In the end, there is no replacement for your effort.

Finally, don't give up. When you fall down, find it in yourself to get back up again. Ask questions. Each and every topic in math is learnable. If you ever encounter mathematics that feels unlearnable, you may need a different explanation, a more thorough explanation, or there may be some prior knowledge that needs to be revisited. But that's it. There isn't anything in math that is a secret: everything has a reason, and this text is doing its best to set the standard of providing explanations. Hang in there! You can do it! Not only do I hope you gain confidence in your mathematics, I hope that this class becomes an opportunity for you to believe in yourself! You've got this!

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Preface to the instructor

As was mentioned in the preface to the student, the text herein is deliberately informal and geared toward the student. As the instructor, it may not meet your expectations in terms of personal taste to have casual language, but I believe that the kind of writing style found in this book has the best possible chance of increasing student readership. Even in this format, I fully believe that an instructor can gather all that is needed in preparing lessons.

One key ethos of this text is to treat these students like adults, and not shy away from explanations, even when the matter at hand is really subtle or tedious. Explanations on minute details can be challenging to write, but I think it's important that students have a written explanation that they can reference, and hopefully also in-class discussions that follow up as well. What do I mean by "treating students like adults?" Often, faced with a student's mistake, it's common to just say "oh, don't do it that way because that is an error, so do this instead", but we find that the same error gets made again, sometimes even by the same student. We don't discuss every possible error that could occur (and I do have a huge inventory of errors that I've catalogued), but when we do discuss an error, I feel it's important to provide students reasonable evidence of why what we claim is an error actually is an error. Often, this means taking the incorrect work and face value and guiding the student from the information into an incongruency. This is very analogous to what we do in the more formal setting of a proof by contradiction in logic.

If every student in your class is super-enthusiastic about the prospect of taking this algebra class, great! However, I suspect that there are people who are dreading the class, and only taking it for a gen ed requiremnt, or because the course is a prerequisit to some course they need for their major. I suspect that a significant number of students may have a rocky relationship with mathematics in general, and perhaps a distaste for algebra specifically. This text takes a number of unorthodox decisions to serve these students. For a new and better experience in mathematics, something has to change, so why not start with the textbook that's selected?

Applications are introduced, and this won't appeal to every student, but I believe that inculding this in the text is a useful way of meeting the students where they are. While I encourage selecting one or two applications from each section to show in class, even if these are not explicitly done in class, they are available in the text to help keep students motivated.

Examples are included, and there is a deliberate attempt to really scale up the difficulty of exercises progressively. As instructors, we might say that examples are not necessary and we should only present theorems and proofs, but I can tell you that I really needed examples to make sense of Hilbert spaces, sigma algebras, normal subgroups, and tensor products of vector spaces. I think, more than we care to admit, we all learn from examples.

Explicit strategies are described for many types of problems, because, well, why shouldn't we make this clear to students? Multiple answers may be given to a question, to honor very different approaches to the same problem. In fact, as we progress the difficulty of examples within a section, it becomes easier to introduce exercises with very different looking answers, and this we set a norm of full explanations early on so that by the time students see multiple answers, there is a mechanism for comprehension and a familiarity in the style of explanations given thusfar.

There are many things related to the culture of reading, writing, speaking, and hearing mathematics that are covered proactively in the text. For example, we discuss language, giving students sample verbiage for what words they should speak (aloud or silently) when reading 5^x or $f(x)$ or $\log_2(x)$. We also extensively discuss notation, including exactly when we should write equal signs. We discuss ambiguities and oddities of writing mathematics, and how to write mathematics clearly. For example, in Principle 1.2.92, we discuss the potential ambiguities that can occur with writing fractions using a slanted fraction bar.

The kinds of exposition we get into regarding reading, writing, speaking, and hearing mathematics makes clear that mathematics is a culture. The likelihood of success in mathematics is increased when we invite our students into that culture. Learning a new culture is not easy, since it confronts thet habits that we already have. These are things that you know all too well if you've noticed that countries around the world have different expectations in hand gestures, eye contact, practices in transferring objects from one person to another, and so on. Each culture has its expectations.

The culture of mathematics has its own expectations too, and these feel peculiar to someone who hasn't been introduced to the culture. Because culture can be hard to pick up, we make these cultural aspects of mathematics explicit, with the belief that being proactive and clear is the best thing we can do for our students. For example, in Principle 1.2.79, we thoroughly explain the cultural norm of preferring improper fractions over mixed fractions, and time the presentation of this at the point in the text when we are able to fully justify to the student why we have this expectation.

This text works to establish clear foundations on things that often never get discussed or get glossed over. For example, when a student takes the equation $\frac{x}{4}+3=\frac{x}{2}$, there is a subtle issue when the student's next step is to write $4\cdot\frac{x}{4}+3=4\cdot\frac{x}{2}$. From the viewpoint of experience, we know that $4\cdot(\frac{x}{4}+3)$ is not the same as $4\cdot\frac{x}{4}+3$. I would explain this as an analysis of expressions based on the Order of Operations, but experience has shown me that many students view the Order of Operations having the very narrow scope of being applicable only when simplifying an expression that has no variables. Thus, in Principle 1.1.19, we introduce how to analyze expressions containing variables via the Order of Operations. Historically, students have found this task very challenging, but it's extremely important, so I have included many examples. Imagine how challenging it is to understand algebra with the misconception that the Order of Operations has nothing to say about comparing $4\cdot(\frac{x}{4}+3)$ and $4\cdot\frac{x}{4}+3$, and is only applicable in expressions such as $4\cdot(\frac{80}{4}+3)$ is not the same as $4\cdot\frac{08}{4}+3$.

One of the first areas where students may feel like they are going through the motions without complete confidence is in manipulating fractions. We launch into a thorough discussion of fractions, including what can cancel in a fraction. To discuss this fully, we use examples like $\frac{x+6}{x+12}$ to point out cancellation cannot occur alongside examples like $\frac{3(x+6)}{5(x+6)}$ where cancellation can occur. This discussion brings out the importance of discussing what is a factor, noting that it is not quite enough to say "when we see plus or minus signs in the numerator and denominator, we can't cancel." This leads to Principle 1.2.83, an important discussion that the entire numerator and entire denominator are each in a set of parentheses that we often do not explicitly write in, called "hidden parentheses."

To ensure that all students are invited into the conversation, we discuss additional hidden items, such as the hidden denominator of one in non-fractions in Note 1.2.4. We also discuss that a missing exponent is a hidden 1 in Note 1.5.2. While every algebra text introduces $a^1 = a$, it is often presented without much discussion. It is often the the concepts where there are few "moving parts" which are the hardest to grasp, so we discuss this property and how it applies. (By contrast, $a(b+c) = ab+ac$ has a sufficient number of moving parts, and so it is easier to apply this, at least in its initial examples.)

As another example of a mathematical concept with few moving parts, consider $a^{\log_a(x)} = x$. This property gets brought up very early on in introducing logarithms since it is an immediate consequence of the composition of inverse functions. However, by bringing this property up so soon after logarithms are introduced, students are only introduced to examples where the rather-large expression $5^{\log_5(c)}$ simplifies to just c. To the student, it just feels like there are not enough moving parts to make sense of this. Thus, we introduce $a^{\log_a(x)} = x$ only after we have introduced and practiced the Laws of Logarithms so that we can follow up with examples like $25^{\log_5(c)}$.

The traditional pedagogy for exponents introduces all the typical identities for exponents, but then stops a bit too early. It does not really set students up for success in thinking about in larger expressions like $\frac{a^3b^5c^7d^{10}}{a^8c^7d^6}$ and $\left(\frac{y^{3/2}}{y^{-1/3}}\right)^{-3}$. The traditional approach feels like we give students plenty of hand tools but expect them to build a car without power tools. When a fraction has many exponents, and the base appears in the numerator and denominator, and in the presence of negative exponents, we tend to think about this problems in a much faster manner, which is of course based on the primitive exponent rules. So, in Principle 1.5.107, we bring up the fact that a fraction with a factor of x^m in the numerator and a factor of x^n in the denominator can be replaced with a single factor of x^{m-n} in the numerator, and in Principle 1.5.106, we also bring up the facts that a factor in the denominator can be turned into a factor of the numerator by negating its exponent, and a factor in the numerator can be turned into a factor of the denominator by negating its exponent. We complete exercises where one solution uses these principles and one does not, so that students see by example why these principles are valid. It serves our students well to articulate the ways we tend to practically think about exponents, giving them the power tools (no pun intended!) and giving them clear connections between these powerful tools and the primitive exponent rules.

One of the decisions made while writing was to think about what details should get included. Included in deciding what gets included also brings up what doesn't get included. For example, when introducing $a^0 = 1$ in Principle 1.5.36, the requirement that $a \neq 0$ was not mentioned. In the midst of learning material, this detail can be distracting. In fact, it is not typical for textbooks to list any requirement regarding the base when introducing $a^m a^n = a^{m+n}$. We are working from the philosophy of "why stop there?"

You may notice quickly that content is not organized in the usual manner, with some traditional content moved to appendices because their inclusion broke the narrative arc of the text. Great care was taken to order the content in a way that builds progression of skills, and builds procedural fluency from conceptual understanding. Warnings and cautions are included without being overdone, and where practical, we launch into discussions of exactly why something doesn't work instead of just saying "don't do that". When viewing the text through the lens of an instructor, there are probably going to be many places where the language I use diverges from your personal preference. I believe that this is an indictment on us in the mathematical community for our lack of having universal terminology on the subtle aspects of algebra. (For example, when we label the axes of the xy-plane, draw the line for the equation $y = 3x + 4$, and include the point $(2, 10)$ on the line, is $x = 2$ because we zoom in to the point $(2, 10$, is x the "generic" value of x that could be any number in $y = 3x + 4$, or is x any real number by looking at the axis label?) Your language and my language on some of these details may differ, but it my hope that the explanations that happen in diversions, explicit descriptions of strategies, warnings, and discussions of language and notation give you many talking points that lead to student success!

Contents

CHAPTER 1

EXPRESSIONS AND EQUATIONS

This chapter reviews expressions, with a careful emphasis on how the Order of Operations applies beyond arithmetic in an algebraic context. Expressions involving fractions, exponents, and radicals are covered. In addition, we solve linear equations.

1.1 OPERATIONS

Objectives

In this section, we learn how to:

1. Simplify an expression using the Order of Operations.
2. Analyze a non-constant expression using the Order of Operations.
3. Describe expressions and their parts using the words sum, term, product, and factor.
4. Construct geometric representations of addition and multiplication.
5. Justify major algebraic formulas geometrically.
6. Connect collecting like terms to the Distributive Law.

Try it 1.1.1

AAAAAAAA Edit

1. As a cashier, a customer gives you \$20 for their purchase of \$15.75. You input into the cash register, and just as you're about to make the \$4.25 change, the customer hands you a one-dollar bill they found. What do you do?
2. A taxi charges \$3 to start plus \$2 for every mile. If you ride 7 miles, what is the total cost? $(3 + 2 \times 7)$
3. A movie ticket costs \$12, and popcorn costs \$5. If you buy 3 tickets and 1 popcorn, how much do you spend? $(3 \times 12 + 5)$
4. A plumber charges \$50 per hour plus a \$30 service fee. If the plumber works for 4 hours, what is the total bill? $(30 + 50 \times 4)$

5. A shirt costs \$20 and jeans cost \$30. If you buy 2 shirts and 3 pairs of jeans, how much do you spend? $(2 \times 20 + 3 \times 30)$
6. A train travels 60 miles per hour. How far does it go in 2.5 hours, and then return halfway back? $(60 \times 2.5 - 0.5 \times (60 \times 2.5))$
7. A gym charges \$40 per month plus \$10 per guest pass. If you keep the membership for 6 months and buy 4 guest passes, what is the cost? $(40 \times 6 + 10 \times 4)$
8. A baker sells cakes for \$15 each. If she sells 10 cakes and then spends \$25 on ingredients, what is her profit? $(10 \times 15 - 25)$
9. A ride at an amusement park charges \$4 admission plus \$2 per ride. If someone goes on 5 rides, how much do they pay? $(4 + 2 \times 5)$
10. A cell phone plan costs \$25 per month plus \$0.10 per text. If you keep the plan for 3 months and send 200 texts total, what is the cost? $(25 \times 3 + 0.10 \times 200)$
11. A pizza shop sells large pizzas for \$12 and small pizzas for \$8. If you buy 2 large and 3 small pizzas, how much do you spend? $(2 \times 12 + 3 \times 8)$
12. A car rental costs \$40 per day plus \$20 for insurance. If you rent it for 5 days, what is the total cost? $((40 \times 5) + 20)$
13. A taxi charges \$2 per mile, but there is also a \$5 starting fee. If you ride 10 miles, how much do you pay? $(5 + (2 \times 10))$
14. A runner trains by running 3 miles each morning and 5 miles each evening for 7 days. How many miles total? $(7 \times (3 + 5))$
15. A concert ticket is \$60, and a service fee of \$5 is added to each ticket. If you buy 4 tickets, how much do you pay? $(4 \times (60 + 5))$
16. A plumber charges \$40 per hour, plus a \$25 trip fee. If a job takes 3 hours, what is the total bill? $((40 \times 3) + 25)$
17. A baseball team sells hot dogs for \$3 and sodas for \$2. If 50 hot dogs and 70 sodas are sold, how much money is made? $((50 \times 3) + (70 \times 2))$
18. A ride service charges \$2 per mile, with a \$10 discount applied at the end. If the trip is 15 miles, what is the total? $((2 \times 15) - 10)$
19. A theme park charges \$50 admission and \$5 per ride. If someone goes on 6 rides, how much do they pay? $(50 + (5 \times 6))$
20. A farmer sells apples for \$2 per pound and pears for \$3 per pound. If she sells 15 pounds of apples and 10 pounds of pears, how much does she earn? $((15 \times 2) + (10 \times 3))$

Algebra provides a powerful way of solving many kinds of questions Algebra achieves this by enhancing arithmetic. Arithmetic focuses on the result of operations like addition and multiplication when using numbers that are constant. The contribution of algebra is to enhance arithmetic with variables and with geometry, with a focus on how the numbers appearing a problem *relate* to each other. With these extra ways of thinking, algebra allows us to answer many questions that are really hard to think about using arithmetic alone.

In the preface to you, the student, I mentioned that you may be encouraged in the book to try something different from the way you have done it in the past. The first of these encouragements is to think about the language used in mathematics. This might or might not have been on your radar before, but we will intentionally take time to talk about how to speak and think about mathematics using standardized terminology, since this will help frame our thinking:

Note 1.1.2 Pay attention to the language of mathematics.

As an example of this, when writing 2^x be sure to say one of the following:

- "2 to the x"
- "2 to the xth power"
- "2 raised to the x"
- "2 raised to the xth power"

If we only say "2" then slightly pause to say "x" this focuses on the specific individual symbols. The bigger problem is that saying "2" followed by "x" is taken to mean $2x$. Why is this a problem? Because 2^x and $2x$ are not equal. In fact, when $x = 3$ then 2^x simplifies to 8, but $2x$ simplifies to 6.

1.1.1 Order of Operations

We will take a little time to make sure that everyone is on the same page regarding the Order of Operations. Before we go any further, we should describe what we mean by an expression.

Definition 1.1.3

An **expression** is mathematical notation representing a number.

By the way... An expression may consist of just a single constant such as 3 or $\frac{4}{7}$, or can be a variable representing a number such as x or y, or can be a combination of constants and variables connected by operations such as addition, subtraction, multiplication, division, and exponentiation.

Example 1.1.4

Both -40 and $\sqrt{23}$ are examples of expressions. Both of these expressions are *constants*.

Example 1.1.5

Writing $-40 + \sqrt{64}$ is another expressions. This is an expression even if we haven't simplified this to the value -32. Like the two expressions in the previous example, the expression shown here (both the unsimplified and the simplified versions) is a *constant*, since the expression lacks any variable(s).

Example 1.1.6

Both $9x - 8$ and $x^2 + 3x + 31$ are examples of expressions which mention the variable x. In the first expression, the variable x is written once. In the second expression, the variable x is written twice.

Example 1.1.7

The expression $x^2 + 2x + y^2 - 6y$ mentions two variables. Since one or more variables appear, this expression is *not* a constant.

Please note that an expression does not contain an equal sign. For example, $x^2 + 2x + y^2 - 6y = 22$ is not an expression. The notation that we just wrote instead states that one exrpession is equal to another expression.

We need the Order of Operations because this it is easy to misinterpret an expression if we do not all agree on how to read expressions. The Order of Operations is a set of rules that tells us the order in which to evaluate (and more generally read) an expression.

Principle 1.1.8 Order of Operations.

*An expression must always be simplified and read by following the **Order of Operations**:*

1. *Parentheses*
2. *Exponents*
3. *Multiplication and Division (from left to right)*
4. *Addition and Subtraction (from left to right)*

In addition, every time you write an expression, ensure your writing is based on the Order of Operations.

Remark 1.1.9 By the time we get to the third part of the Order of Operations, all exponents would have been evaluated. At this point, we look for any multiplication or division. If there is both multiplication and division, we evaluate them from left to right. It is not true that multiplication must be done before division. All divisions and multiplications that we see have the same level of precedence, and we evaluate them scanning from left to right in that order.

Remark 1.1.10 Similarly, by the time we get to the last part of Order of Operations, all multiplications and divisions would have been handled. That means that what remains of our expressions should only have addition and subtraction operations remaining. These should be handled from left to right.

Remark 1.1.11 The Order of Operations is sometimes remembered by the acronym PEMDAS, which stands for Parentheses, Exponents, Multiplication, Division, Addition, and Subtraction.

Because of the way that PEMDAS is often taught, many people mistakenly believe that multiplication must be done before division, and addition must be done before subtraction. This is not true. For this reason, some people prefer the acronym GEMA, which stands for Grouping symbols, Exponents, Multiplication and Division, Addition and Subtraction. In the acronym GEMA, the "G" is used to indicate that there are many kinds of grouping symbols, not just parentheses. Also, the multiplication and division are addressed together in the "M", and addition and subtraction are addressed together in the "A".

Example 1.1.12

Simplify the expression $5^2 - 30 \div 3 + 2 \cdot 6$.

Solution. We will simplify the expression by following the Order of Operations. First, notice that there are no parentheses, so we move on to the next part of the Order of Operations. There is a place where the expression has exponens, so we zoom in on 5^2 and simplify this portion of the expression to 25. So, the expression given to us becomes $25 - 30 \div 3 + 2 \cdot 6$.

Now there is no more exponents. Next, we look for any multiplication or division. We see both multiplication and division, so we evaluate them from left to right. The left-most multplication or division we see is the division $30 \div 3$, which simplifies to 10. This gives us the expression $25 - 10 + 2 \cdot 6$. Continuing to scan from left to right for any multiplications or divisions, we see the multiplication $2 \cdot 6$, which simplifies to 12. This gives us the expression $25 - 10 + 12$.

Now there are no more multiplications or divisions, we look for any additions or subtractions, starting from the left. The left-most addition or subtraction we see is the subtraction $25 - 10$, which simplifies to 15. This gives us the expression $15 + 12$. Continuing to scan from left to right for any additions or subtractions, we see the addition $15 + 12$, which simplifies to 27.

Because this was our first example of applying the Order of Operations, we wanted to be very thorough to explain each step. To present our work, we start from the original expression and after writing an equal sign (to indicate that what we will write next is equal) write a simplified version of the expression. We continue this process until we reach the final simplified expression. For this example, we have $5^2 - 30 \div 3 + 2 \cdot 6 = 25 - 30 \div 3 + 2 \cdot 6 = 25 - 10 + 2 \cdot 6 = 25 - 10 + 12 = 15 + 12 = 27$.

It is also acceptable to write each expressions on their own lines, as follows:

$$\begin{aligned} 5^2 - 30 \div 3 + 2 \cdot 6 &= 25 - 30 \div 3 + 2 \cdot 6 \\ &= 25 - 10 + 2 \cdot 6 \\ &= 25 - 10 + 12 \\ &= 15 + 12 \\ &= 27 \end{aligned}$$

Note that when presenting our work *vertically* we still include the equal signs to indicate that each expression is equal to the previous expression.

Principle 1.1.13 Expectation.

When simplifying any expression, it is important to include equal signs to indicate that each expression is equal to the previous expression.

Principle 1.1.14 Expectation.

While simplifying expressions, ensure that the next expression you write is truly equal to the previous expression, instead of just writing the portion of the expression that is changing.

Example 1.1.15

Simplify the expression $3 \cdot (6-4)^2 \div 2$.

Solution 1. We can present our work horizontally, continuing to always write to the right of an equal sign like this $3 \cdot (6-4)^2 \div 2 = 3 \cdot 2^2 \div 2 = 3 \cdot 4 \div 2 = 12 \div 2 = 6$.

Solution 2. We can instead present our work vertically

$$\begin{aligned} 3 \cdot (6-4)^2 \div 2 &= 3 \cdot 2^2 \div 2 \\ &= 3 \cdot 4 \div 2 \\ &= 12 \div 2 \\ &= 6 \end{aligned}$$

Example 1.1.16

Simplify the expression $200 - 4^2 \div 8 \times 5 + 6$.

Solution 1. We can present our work horizontally, always writing to the right of the equal sign like this: $200 - 4^2 \div 8 \times 5 + 6 = 200 - 16 \div 8 \times 5 + 6 = 200 - 2 \times 5 + 6 = 200 - 10 + 6 = 190 + 6 = 196$.

Solution 2. We can instead present our work vertically:

$$\begin{aligned} 200 - 4^2 \div 8 \times 5 + 6 &= 200 - 16 \div 8 \times 5 + 6 \\ &= 200 - 2 \times 5 + 6 \\ &= 200 - 10 + 6 \\ &= 190 + 6 \\ &= 196 \end{aligned}$$

Principle 1.1.17 Habit.

Always read expressions based on the Order of Operations.

Principle 1.1.18 Habit.

Always write expressions based on the Order of Operations.

1.1.2 Order of Operations with variables

It is important for us to apply the Order of Operations not *only* to simplify expressions that contain only constants, but to also apply the Order of Operations when to interpret expressions that contain variables. This is important because one of the main contributions of algebra over arithmetic is to use variables to represent numbers that we do not know yet. So even when expressions contain variables, we still must read and write based on the Order of Operations. Doing this might feel new and totally weird, but let's explain how it's done and walk through examples together.

Principle 1.1.19 How to apply Order of Operations to expressions with variables.

Given an expression:

1. *Identify each operation that is written in the given expression.*
2. *Apply the Order of Operations to identify in which order the operations* would *be performed.*

Instead of simplifying an expression, we would hypothetically perform the operations.

Example 1.1.20

What order are the operations performed in the expression $z^3 - xy$?

Solution. The expression $z^3 - xy$ has three operations. Scanning from left to right, we see the following operations present: exponentiation, subtraction, and multiplication. When there is no symbol written between two variables, there is a hidden multiplication sign. In fact, to make it clearer, we can rewrite the expression as $z^3 - x \cdot y$.

Now that we have identified which operations are present in this expression (in this example, three of them) we will apply the Order of Operations to determine in which order these operations would be performed.

- Since there are no parentheses in our expression, the first operation that we would perform is the exponentiation. That is, if we knew the value of z, then we would first simplify z^3.
- Next, we would perform the multiplication $x \cdot y$. In other words, if we knew the value of x and we knew the value of y, then our work in this second step would be to simplify, and we'd have xy, "whatever the value of that is"
- Finally, we would perform the subtraction: we would take whatever the value of z^3 and subtract from this whatever the vaule of $x \cdot y$ is to get $z^3 - (x \cdot y)$.

Notice that we never simplified the expression $z^3 - xy$ or any part of this expression. We couldn't, because we did not know the numerical value of any of the variables. It seems like what we're doing is lazy, but I want to spin this into something positive. I encourage you to think of it this way: since we don't have the numbers behind the variables, we *get* to be lazy!

To perform this kind of analysis, it is helpful to say phrases like "whatever the value of *that* is". We are discussing in which order we would *hypothetically* perform the operations, if we knew the values of the variables. It seems like what we're doing is a bit silly, but it is important to practice this way of thinking. It is rare that I'll encourage the following, but I encourage you to say out loud the full text of the next several examples, pausing right before any phrase that looks similar to "whatever the value of *that* is" when this kind of phrase appears right after an expression followed by a comma.

Example 1.1.21

What order are the operations performed in the expression $(a+b)\cdot c^6$?

Solution. First, we identify the operations present, which are an addition, a multiplication written as a dot, and exponentiation. In addition, a part of the expression is contained inside parentheses.

- Since $a+b$ is in parentheses, we would perform the addition first. So, if we hypothetically knew the values of a and b, we'd replace this with the value of $a+b$, whatever value that is.
- After this, there would be no parentheses. Since all that would be left is a multiplication and an exponentiation, we would perform the exponentiation. We would take whatever the value of c is and raise this value to the 6th power. This would give us c^6, whatever the value of *that* is.
- Finally, we'd take whatever the value of $(a+b)$ is and whatever the value of c^6 is and multiply these values together.

Notice that we never simplified the expression $(a+b)\cdot c^6$ or any part of this expression. This process may feel weird to you because you may never have been asked about analyzing expressions in this way. Digging into the careful details of how to analyze an expression like this often gets overlooked but this next-level type of problem in applynig the Order of Operations sets up an important foundation for reading and writing expressions in algebra correctly. It's weird because it's new, but it's important because algebra enhances arithmetic by having variables, so I want you to be comfortable with reading and writing expressions with variables.

Example 1.1.22

What order are the operations performed in the expression $a(b-c)^4+7d$?

Solution. To make it clearer where the operations are, let's rewrite the expression as $a\cdot(b-c)^4+7\cdot d$.

- Since $b-c$ is in parentheses, we would perform this subtraction first.

- Next, there would be no more parentheses, so we would perform the exponentiation: we would take $(b-c)$, whatever the value of $(b-c)$ is, and raise this to the 4th power. Thus, we'd have whatever the value of $(b-c)^4$ is.

- Next, we would perform the multiplication $a \cdot (b-c)^4$. That is, if we knew the value of a we would multiply this by the value of value of $(b-c)^4$ from the previous step, which would give us $a \cdot (b-c)^4$, "whatever the value of that is".

- Then, we would perform the multiplication $7 \cdot d$. That is, if we knew the value of d, we would simplify $7 \cdot d$.

- Finally, we would perform the addition: we would take whatever the value of $a \cdot (b-c)^4$ is and add to this whatever the value of $7 \cdot d$ is.

The activity we just went through leads up to the following new type of activity. We will be given two expressions which will only differ from each other in the inclusion or removal of a set of parentheses. Then, we will determine if the two expressions are equal, or if we don't know based only on the Order of Operations.

Example 1.1.23

Based only on the Order of Operations, are $x-(y \cdot z)$ and $x-y \cdot z$ equal or do we not have enough information?

Solution.

- In the first expression $x-(y \cdot z)$, we perform the multiplication $y \cdot z$ first because the multiplication is in parentheses. Then we perform the subtraction of x and $y \cdot z$ to get $x-(y \cdot z)$.

- In the second expression $x-y \cdot z$, there are no parentheses. We perform the multiplication $y \cdot z$ first. Then we perform the subtraction of x and $y \cdot z$ to get $x-y \cdot z$.

In both expressions, we perform the multiplication $y \cdot z$ first, and then we perform the subtraction of x and $y \cdot z$ next. Based only on the Order of Operations, the two expressions are equal.

Example 1.1.24

Based only on the Order of Operations, are $(a+b) \cdot c^6$ and $a+b \cdot c^6$ equal or do we not have enough information?

Solution.

- In the first expression $(a+b)\cdot c^6$, we perform the addition $a+b$ first because the the content is in parentheses. We do exponentiation next to get c^6. Our final operation is the multiplication of the value of $a+b$ and the value of c^6.

- In the second expression $a+b\cdot c^6$, we perform the exponentiation first to get the value of c^6. Next, we perform the multiplication of b and c^6. Finally, we perform the addition of a and $b\cdot c^6$.

In the first expression, we perform the addition first, then the exponentiation, and finally the multiplication. In the second expression, we perform the exponentiation first, then the multiplication, and finally the addition. Based only on the Order of Operations, we do not have enough information to determine if the two expressions are equal. (These expressions might be equal or they might not, but we cannot determine this based only on the Order of Operations. If these expressions happened to be equal, this would have to be explained by something *other* than the Order of Operations.)

Example 1.1.25

Based only on the Order of Operations, are $a+b\cdot c-d$ and $a+(b\cdot c)-d$ equal or do we not have enough information?

Solution.

- In the first expression $a+b\cdot c-d$, we perform the multiplication $b\cdot c$ first. Then, we perform the addition of a and the value of $b\cdot c$. Finally, we perform the subtraction of the value of $a+b\cdot c$ and d.

- In the second expression $a+(b\cdot c)-d$, we perform the multiplication $b\cdot c$ first because this is in parentheses. Then, we perform the addition of a and the value of $b\cdot c$. Finally, we perform the subtraction of the value of $a+b\cdot c$ and d.

In both expressions, we perform the multiplication first, then the addition, and finally the subtraction. Based only on the Order of Operations, the two expressions are equal.

Example 1.1.26

Based only on the Order of Operations, are $a\div(b+c)\times 7$ and $a\div b+c\times 7$ equal or do we not have enough information?

Solution.

- In the first expression $a\div(b+c)\times 7$, we perform the addition $b+c$ first because the addition is in parentheses. Next, we perform the division of a by the value of $b + c$. Finally, we perform the multiplication of the value of $a \div (b + c)$ and 7.

- In the second expression $a \div b + c \times 7$, we perform the division $a \div b$ first. Next, we perform the multiplication $c \times 7$. Finally, we perform the addition of the value of $a \div b$ and the value of $c \times 7$.

In the first expression, we perform the addition first, then the division, and finally the multiplication. In the second expression, we perform the division first, then the multiplication, and finally the addition. Based only on the Order of Operations, we do not have enough information to determine if the two expressions are equal.

Example 1.1.27

Based only on the Order of Operations, are $(w + x) \div (y + z)$ and $w + x \div y + z$ equal or do we not have enough information?

Solution.

- In the first expression $(w + x) \div (y + z)$, we perform the addition $w + x$ first because this is in parentheses. Next, we perform the addition $y + z$ because this is in parentheses. Finally, we perform the division of the value of $w + x$ by the value of $y + z$.
- In the second expression $w + x \div y + z$, we perform the division $x \div y$ first. Next, we perform the addition of w and the value of $x \div y$. Finally, we perform the addition of the value of $w + x \div y$ and z.

In the first expression, we perform the addition $w + x$ first, then the addition $y + z$, and finally do the division of whatever the value of $w + x$ is by whatever the value of $y + z$ is. In the second expression, we perform the division first, then an addition, and finally another addition. To be clear, in the first expression the value of $w + x$ is divided by whatever the value of $y + z$ is, and in the second expression the value of x is divided by whatever the value of y is. Based only on the Order of Operations, we do not have enough information to determine if the two expressions are equal.

Example 1.1.28

Based only on the Order of Operations, are $a \cdot (b - c)^4 + 7 \cdot d$ and $a \cdot b - c^4 + 7 \cdot d$ equal or do we not have enough information?

Solution.

- In the first expression $a \cdot (b-c)^4 + 7 \cdot d$, we perform the subtraction $b - c$ first because the subtraction is in parentheses. Next, we perform the exponentiation to get $(b-c)^4$. Then, we perform the multiplication of a and $(b-c)^4$. Next, we perform the multiplication of 7 and d to get $7 \cdot d$. Finally, we perform the addition of the value of $a(b-c)^4$ and the value of $7 \cdot d$.

- In the second expression $a \cdot b - c^4 + 7 \cdot d$, we perform the exponentiation first to get the value of c^4. Next, we perform the multiplication of a and b. Then, we perform the multiplication of 7 and d. Finally, we perform the subtraction of the value of $a \cdot b$ and the value of c^4, and then add to this whatever the value of $7 \cdot d$ is.

In the first expression, we perform the subtraction first, then the exponentiation, then two multiplications, and finally an addition. In the second expression, we perform an exponentiation first, then two multiplications, then a subtraction, and finally an addition. Based only on the Order of Operations, we do not have enough information to determine if the two expressions are equal.

1.1.3 Language

Before we move on, let's be sure that we're on the same page regarding some language used to describe expressions.

Definition 1.1.29

An expression that is the result of adding and/or subtracting two or more expressions together is called a **sum**. Each individual piece of the sum is called a **term**.

Informally, a *sum* is an expression that has additions and/or subtractions as its last operations. Consider using "mental scissors" to cut at each plus sign and at each minus sign. Then each of the pieces that we have is called a *term*.

Example 1.1.30

The expression $3x + 5y - 7$ is the result of adding and subtracting expressions together, so is called a sum. The three terms in this sum are $3x$, $5y$, and 7.

Definition 1.1.31

An expression that is the result of multiplying two or more expressions together is called a **product**. Each individual piece of the product is called a **factor**.

Informally, a *product* is an expression that has multiplications as its last operations. Consider using "mental scissors" to cut at each multiplication sign. (Before doing this, you may wish to draw in any hidden multiplication signs.) Then each of the pieces that we have is called a *factor*.

Example 1.1.32

The expression $4(x+y)z$ is the result of multiplying expressions together, so is called a product. The three factors in this product are 4, $x+y$, and z.

Example 1.1.33

Is $2a(b+c)(d+e)$ a sum, a product, or neither? If it is a sum, what are the terms? If it is a product, what are the factors?

Solution. The expression $2a(b+c)(d+e)$ is a product. The three factors in this product are $2a$, $(b+c)$, and $(d+e)$.

Example 1.1.34

Is $4x^2+3y-7\div z$ a sum, a product, or neither? If it is a sum, what are the terms? If it is a product, what are the factors?

Solution. The expression $4x^2+3y-7\div z$ is a sum. The three terms in this sum are $4x^2$, $3y$, and $7\div z$.

Example 1.1.35

Is $(a-b)^2+(c+d)^2$ a sum, a product, or neither? If it is a sum, what are the terms? If it is a product, what are the factors?

Solution. The expression $(a-b)^2+(c+d)^2$ is a sum. The two terms in this sum are $(a-b)^2$ and $(c+d)^2$.

1.1.4 Pictures

In algebra, we will often use pictures to represent expressions. This is a powerful way to think about expressions, and it is important to be able to translate between pictures and expressions. This may be new to you, but I encourage you to give it a try! Practicing this now will make future concepts go smoother!

Principle 1.1.36 Representing Addition Geometrically.

Suppose a and b are positive real numbers. Then $a+b$ is geometrically represented by the length of the stick made by gluing a stick of length a to a stick of length b .

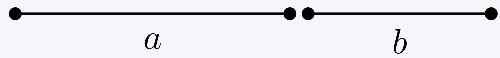

***Figure 1.1.37** A picture of $a + b$*

We have drawn the sticks slightly separated so that we can see them individually, but in reality we should imagine them pushed together. In the drawing, I made the choice to represent a as a slightly larger number than b. In addition, I chose to draw the stick representing a on the left and the stick representing b on the right because in the expression $a + b$, the a appears to the left of the plus sign, while the b appears to the right of the plus sign.

Example 1.1.38

Represent $3 + 4$ geometrically.

Solution.

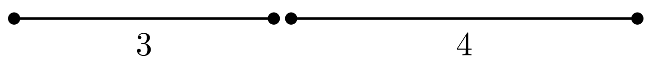

Figure 1.1.39 A picture of $3 + 4$

Example 1.1.40

Represent $2 + 0.5$ geometrically.

Solution.

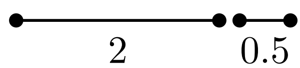

Figure 1.1.41 A picture of $2 + 0.5$

Example 1.1.42

Represent $x + 2$ geometrically.

Solution.

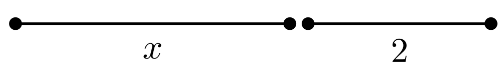

Figure 1.1.43 A picture of $x + 2$

We can apply the geometric representation of addition to learn several important algebra facts about addition.

Principle 1.1.44 Commutative Property of Addition.

$a + b = b + a$

Example 1.1.45

Give a geometric explanation of why $a + b = b + a$ is true.

Solution. To see why this is true, we can represent both sides of the equation geometrically.

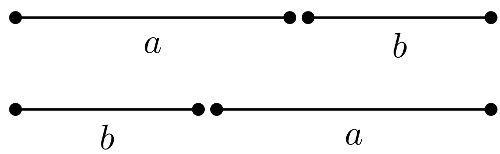

Figure 1.1.46 A picture of $a + b$ and $b + a$

The left side, $a + b$, is represented by a stick of length a glued to a stick of length b. The right side, $b + a$, is represented by a stick of length b glued to a stick of length a. In both cases, the resulting stick has the same length, so the two expressions are equal. (Note that in providing our geometric explanation, we *never* plug in numbers for a or b.)

In the Commutative Property of Addition $a + b = b + a$, we can substitute any expression we want for the a and any expression we want for the b. If we can for a moment explain what the Commutative Property of Addition is saying informally, the result of "this" plus "that" is the same as "that" plus "this". In other words, it is saying that when we add two things together, the order in which we add them doesn't matter. For example the Commutative Property of Addition tells us that $x^7 + 31y$ is equal to $31y + x^7$.

Principle 1.1.47 Associative Property of Addition.

$(a + b) + c = a + (b + c)$

Example 1.1.48

Give a geometric explanation of why $(a + b) + c = a + (b + c)$ is true.

Solution. To see why this is true, we can represent both sides of the equation geometrically.

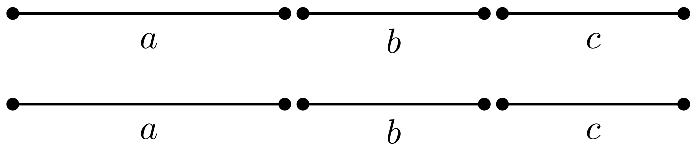

Figure 1.1.49 A picture of $(a+b)+c$ and $a+(b+c)$

The left side, $(a+b)+c$, is represented by a stick of length a glued to a stick of length b, and then this resulting stick is glued to a stick of length c. The right side, $a+(b+c)$, is represented by a stick of length b glued to a stick of length c, and then this resulting stick is glued to a stick of length a. (What the drawing doesn't indicate is the order of the gluing, so we need to clarify this with words: in the first diagram the stick of length a is glued to the stick of length b first, but in the second diagram the stick of length b is glued to the stick of length c first. If doing a hand drawing arrows can be drawn with labels like "glue here first" and "glue here next".) In both cases, the resulting stick has the same length, so the two expressions are equal. (Note that in providing our geometric explanation, we *never* plug in numbers for a or b or c.)

Having the Commutative Property of Addition and the Associative Property of Addition together basically tells us that when the expression we have is a sum, we can rearrange the terms in any order we want and simplify the addition of any terms in any order we want. For example $5a+3b-7c+12d^8$ is equal to $12d^8-7c+5a+3b$ and is also equal to $-7c+3b+12d^8+5a$. When rearranging, be sure that any minus sign that was in front of a term stays in front of that term.

Principle 1.1.50 Representing Multiplication Geometrically.

Suppose a and b are positive real numbers. Then $a \cdot b$ is geometrically represented by the area of a rectangle with height a and width b .

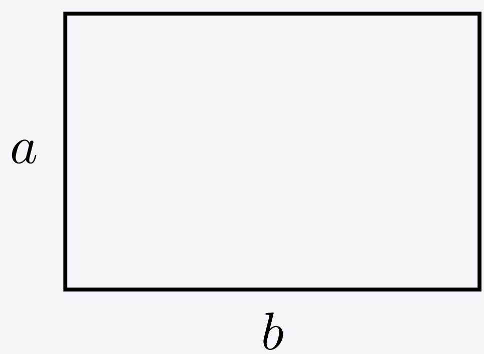

Figure 1.1.51 *The area of the rectangle is a picture of $a \cdot b$*

We will draw with the convention that the factor before the muliplication symbol is the height of the rectangle and the factor after the multiplication symbol is the width. In the drawing, I made the choice to represent a as a slightly smaller number than b.

Note 1.1.52 What we're introducing regarding the geometric representation of multiplication is an idea that we've seen before in geometry: we often write $A = \ell w$ to represent the area A of a rectangle with length ℓ and width w. Here, we're just using a and b instead of ℓ and w as the two factors.

Example 1.1.53

Represent $3 \cdot 4$ geometrically.

Solution.

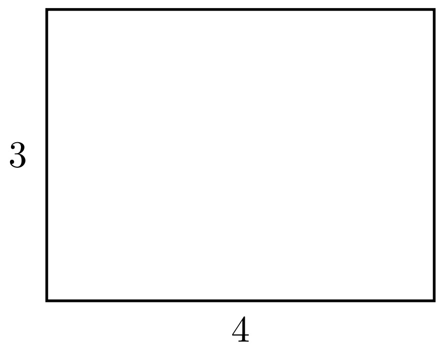

Figure 1.1.54 A picture of $3 \cdot 4$

Example 1.1.55

Represent $2 \cdot 0.5$ geometrically.

Solution.

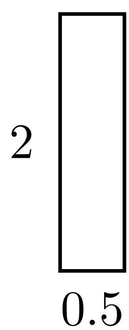

Figure 1.1.56 A picture of $2 \cdot 0.5$

We can apply the geometric representation of addition to learn several important algebra facts about addition.

Principle 1.1.57 Commutative Property of Multiplication.

$ab = ba$

Example 1.1.58

Give a geometric explanation of why $ab = ba$ is true.

Solution. To see why this is true, we can represent both sides of the equation geometrically.

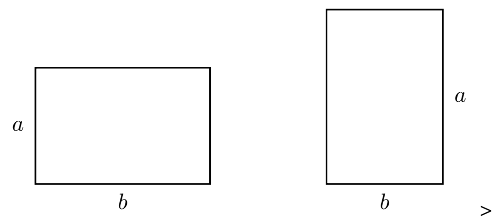

Figure 1.1.59 A picture of ab and ba

The left side, ab, is represented by a rectangle with height a and width b. The right side, ba, is represented by a rectangle with height b and width a. Both rectangles have the same area, since we can get from one rectangle to the other by rotation. Since the two areas are equal, the two expressions they represent, namely ab and ba, are equal. (Note that in providing our geometric explanation, we *never* plug in numbers for a or b.)

Principle 1.1.60 Associative Property of Multiplication.

$(ab)c = a(bc)$

We will skip the geometric explanation of the Associative Property of Multiplication, since we'd have to enhance our geometric interpretation of multiplication to include three factors. If you're curious, try to think through how you might represent $(ab)c$ and $a(bc)$ geometrically. As a hint, you'll need three-dimensional objects to do this.

Note 1.1.61 I just want to take a moment to keep encouraging you to think about addition and multiplication geometrically. It may be new and strange to you. It may seem like a waste of time to you. So far, it may just seem like a silly to give a geometric explanation of why certain facts (that might even feel obvious to you) are explained using geometry. However, setting up this foundation will make several challenging concepts in the future become a lot easier to digest.

Principle 1.1.62 How do I prevent confusing the two geometric representations?

The geometric representation of addition is the gluing together of sticks. The geometric representation of multiplication is the area of a rectangle.

One technique to help recall which is which is to recall that a usual presentation of the area formula is $A = \ell w$. Because this formula multiplies together two quantities (named ℓ and w), we can remember that multiplication is represented by area. Since addition is not represented by area, it must be represented by the other geometric idea we've seen, which is gluing sticks together.

Here's another techinque! Pick two numbers where the sum of the two numbers and the product of the two numbers is different. What I mean is that we wouldn't want to pick 2 and 2, since both the sum and the product are 4.

1. *If we pick 2 and 3, we can ask ourselves what geometric object has some aspect of having size 5 and what geometric object has some aspect of having size 6. The object with size 5 is a stick of length 5, which is the result of gluing together a stick of length 2 and a stick of length 3. The object with size 6 is a rectangle with area 6, which is the result of multiplying together 2 and 3.*

2. *If we pick 4 and 7, we can ask ourselves what geometric object has some aspect of having size 11 and what geometric object has some aspect of having size 28. The object with size 11 is a stick of length 11, which is the result of gluing together a stick of length 4 and a stick of length 7. The object with size 28 is a rectangle with area 28, which is the result of multiplying together 4 and 7.*

You can pick any two numbers you want, as long as the sum and product are different

We are now about to talk about an important algebra property that mentions both addition and multiplication. When we look at the geometric representation, we will need both geometric representations: gluing together of sticks *and* area of rectangles will *both* appear.

Principle 1.1.63 Distributive Law, version 1.

$a(b + c) = ab + ac$

Example 1.1.64

Give a geometric explanation of why $a(b + c) = ab + ac$ is true.

Solution. To see why this is true, we can represent both sides of the equation geometrically.

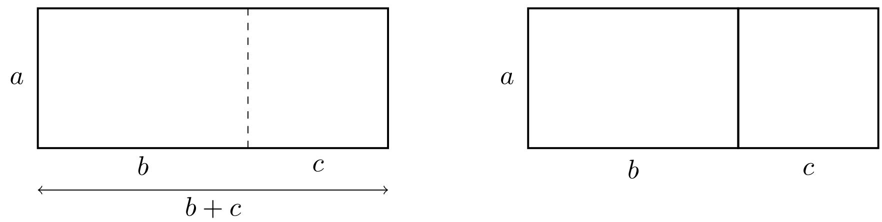

Figure 1.1.65 A picture of $a(b + c)$ and $ab + ac$

Before digging in, let's note that in the diagram the left, we see a stick of length b glued to a stick of length c, making a stick of length $b + c$. (This makes use of the geometric representation of addition.) The left side, $a(b+c)$, is represented by a rectangle with height a and width $b + c$. The right side, $ab + ac$, is represented by two rectangles: one with height a and width b, and the other with height a and width c. The area of the big rectangle on the left is equal to the sum of the areas of the two rectangles on the right, since we can get from the rectangle on the left to the two rectangles on the right by cutting the rectangle on the left vertically into two pieces. Since the area of the big rectangle on the left is equal to the sum of the areas of the two rectangles on the right, the two expressions they represent, namely $a(b + c)$ and $ab + ac$, are equal. (Note that in providing our geometric explanation, we *never* plug in numbers for a or b or c.)

In the Distributive Law $a(b + c) = ab + ac$, we can substitute any expression we want for the a, any expression we want for the b, and any expression we want for the c. If we can for a moment explain what the Distributive Law is saying informally, the result of "this" times the sum of "that" and "the other thing" is the same as "this" times "that" plus "this" times "the other thing". For example the Distributive Law tells us that $x(3y + 5)$ is equal to $3xy + 5x$.

Note 1.1.66 When you see $x(3y + 5)$, I encourage you to think about the Distributive Law as we wrote it $a(b + c) = ab + ac$, and relate the a and b and c in the formula to the x and $3y$ and 5 in the expression $x(3y + 5)$.

It is easy to dismiss this advice and just think "What's the point of all this? I can just see that I should distribute x." However, it is important to see exactly what this Distributive Law $a(b + c) = ab + ac$ is saying, and what it is *not* saying. The left side $a(b + c)$ addresses *only* an expression that has addition on the inside of the parentheses, and multiplication outside. For example, the formula $a(b+c) = ab+ac$ has *nothing* to say about the expression $j + (k \cdot \sqrt{m})$. It would be tempting to look at $j + (k \cdot \sqrt{m})$ and try to "distribute" the j somehow. But when we read the left side of $a(b + c) = ab + ac$ and we see that the left side says $a(b + c)$, we have to carefully note that addition is inside the parentheses with multiplication outside. The problem with $j + (k \cdot \sqrt{m})$ is that multiplication is inside the parentheses with the addition outside. So the Distributive Law $a(b+c) = ab + ac$ does not apply to $j + (k \cdot \sqrt{m})$.

Example 1.1.67

Use the Distributive Law to rewrite $5(x+2)$.

Solution. Using the Distributive Law $a(b+c) = ab+ac$, we can let $a = 5$, $b = x$, and $c = 2$. Then we have

$$5(x+2) = 5 \cdot x + 5 \cdot 2 = 5x + 10.$$

Example 1.1.68

Use the Distributive Law to rewrite $10(h^2 + t^3)$

Solution. Using the Distributive Law $a(b+c) = ab+ac$, we can let $a = 10$, $b = h^2$, and $c = t^3$. Then we have

$$10(h^2 + t^3) = 10 \cdot h^2 + 10 \cdot t^3 = 10h^2 + 10t^3.$$

In the example above, we informally say that we "**distributed** 10". We can also turn an expression of the form $ab + ac$ into an expression of the form $a(b+c)$. This process is called **factoring**. (Factoring is the reverse of the process of distributing.)

Example 1.1.69

Use the Distributive Law to factor $4c + 4d$.

Solution. Using the Distributive Law $a(b+c) = ab+ac$, we can let $a = 4$, $b = c$, and $c = d$. Then we have

$$4c + 4d = 4(c+d).$$

Example 1.1.70

Use the Distributive Law to factor $12x^2 + 18x$.

Solution. Using the Distributive Law $a(b+c) = ab+ac$, we can let $a = 6x$, $b = 2x$, and $c = 3$. Then we have

$$12x^2 + 18x = 6x \cdot 2x + 6x \cdot 3 = 6x(2x+3).$$

The selection of $6x$ occured by looking for the greatest common factor of $12x^2$ and $18x$.

Example 1.1.71

Use the Distributive Law to factor $15xy + 6y$.

Solution. Using the Distributive Law $a(b + c) = ab + ac$, we can let $a = 3y$, $b = 5x$, and $c = 2$. Then we have

$$15xy + 6y = 3y \cdot 5x + 3y \cdot 2 = 3y(5x + 2).$$

The selection of $3y$ occured by looking for the greatest common factor of $15xy$ and $6y$.

Example 1.1.72

Use the Distributive Law to factor $9x + 9$.

Solution. Using the Distributive Law $a(b + c) = ab + ac$, we can let $a = 9$, $b = x$, and $c = 1$. Then we have

$$9x + 9 = 9(x + 1).$$

Warning 1.1.73 When factoring $9x + 9$ it is common to make the mistake of saying that this expression is equal to $9(x)$ instead of $9(x + 1)$. When we see $9x + 9$, to successfully factor out a 9 we will write an expression in the format $9(?+?)$, and the first question mark is filled in by asking "9 times *what* is the first term $9x$", while the second question mark is filled in by asking "9 times *what* is the second term 9".

To provide a little more convincing, you can always check your factoring by taking your answer and distributing. Note that distributing in the expression $9(x + 1)$ gives us $9x + 9$.

Example 1.1.74

Factor the expression $71x + 71$.

Solution. Using the Distributive Law $a(b + c) = ab + ac$, we can let $a = 71$, $b = x$, and $c = 1$. Then we have

$$71x + 71 = 71(x + 1).$$

As we factor out 71, the reason that there is a 1 as the second term in parentheses is because this answers "71 times *what* equals the second term 71 in the original expression", which applies the same reasoning for why an x appears as the first term in parentheses because this answers "71 times *what* equals the first term $71x$ in the original expression".

Principle 1.1.75 Distributive Law, version 2.

$a(b - c) = ab - ac$

This version has subtraction inside the parentheses instead of addition inside the parentheses. Since we didn't provide a geometric representation of subtraction, we won't provide a geometric explanation of why this version of the Distributive Law is true. (However, for a mental challenge, we can use the first version of the Distributive Law to explain why this version is true. We can think of $b - c$ as $b + (-c)$, and then apply the first version of the Distributive Law.)

Example 1.1.76

Use the Distributive Law to rewrite $4(x - 3)$.

Solution. Using the Distributive Law $a(b - c) = ab - ac$, we can let $a = 4$, $b = x$, and $c = 3$. Then we have

$$4(x - 3) = 4 \cdot x - 4 \cdot 3 = 4x - 12.$$

Example 1.1.77

Use the Distributive Law to rewrite $30x - 42x^2$.

Solution. Using the Distributive Law $a(b - c) = ab - ac$, we can let $a = 6x$, $b = 5$, and $c = 7x$. Then we have

$$30x - 42x^2 = 6x \cdot 5 - 6x \cdot 7x = 6x(5 - 7x).$$

The selection of $6x$ occured by looking for the greatest common factor of $30x$ and $42x^2$.

Principle 1.1.78 Distributive Law, version 3.

$(b + c)a = ba + ca$

This version has the multiplication on the right instead of the left.

Try it 1.1.79

Spend a few minutes providing the geometric explanation of why $(b + c)a = ba + ca$ is true. When doing this, remember we shouldn't select specific numbers for a, b, or c.

Principle 1.1.80 Distributive Law, version 4.

$(b - c)a = ba - ca$

Example 1.1.81

Use the Distributive Law to rewrite $7 - 14y$.

Solution. Using the Distributive Law $(b - c)a = ba - ca$, we can let $a = 7$, $b = 1$, and $c = 2y$. Then we have

$$7 - 14y = 1 \cdot 7 - 2y \cdot 7 = (1 - 2y)7.$$

Example 1.1.82

Use the Distributive Law to rewrite $13x + 7x$.

Solution. Using the Distributive Law $(b + c)a = ba + ca$, we can let $a = x$, $b = 13$, and $c = 7$. Then we have

$$13x + 7x = (13 + 7)x = 20x.$$

Example 1.1.83

Use the Distributive Law to rewrite $13x - 7x$.

Solution. Using the Distributive Law $(b - c)a = ba - ca$, we can let $a = x$, $b = 13$, and $c = 7$. Then we have

$$13x - 7x = (13 - 7)x = 6x.$$

The last two examples show that the process known as **collecting like terms** is actually justified by the Distributive Law. In fact, the only reason why collecting like terms works at all is because of the Distributive Law! Collecting like terms is actually factoring out the variable from several terms (though we often skip writing that "middle step" such as $(13 - 7)x$ and just simplify the part that is in parentheses in our head.)

Note 1.1.84 Collecting like terms is justified by the Distributive Law. It may be tempting to ignore this comment about the connection between collecting like terms and the Distributive Law. If we ignore the connection, it would be easy to see an expression like $3x + \sqrt{7}\,x$ and feel stuck thinking that we "can't" collect like terms. However, if we remember that collecting like terms is actually factoring, then we can see that

$$3x + \sqrt{7}\,x = (3 + \sqrt{7})x.$$

In another example like $3x + 5x = (3 + 5)x = 8x$ we very often skip writing the middle step $(3+5)x$ and just directly go from $3x + 5x$ to $8x$. However, it's good to remember that there is a middle step, and that middle step is justified by the Distributive Law. Recalling this allows us to not get intimidated by expressions like $3x + \sqrt{7}\,x$.

We have shown four versions of the Distributive Law. Here is an extended version of the Distributive Law:

Principle 1.1.85 Distributive Law, one extended version.

$a(b + c + d) = ab + ac + ad$

Try it 1.1.86

Come up with other extended versions of the Distributive Law. Be careful to pay attention to where the addition and subtraction signs are, and where the multiplication signs are (including hidden multiplication signs).

We will need to watch out for some subtle problems involving distribution.

Example 1.1.87

Simplify $(16x + 8) - (6x - 6)$.

Solution 1. The first set of parentheses can be dropped. So, the given expression is equal to $16x + 8 - (6x - 6)$. Now, the minus sign in front of the parentheses is often casually called "distributing the minus sign". The previous expression is equal to $16x + 8 - 6x + 6$. We have provided explanations, but to now provide a good presentation of work which includes equal signs where they should belong, we should write $(16x+8)-(6x-6) = 16x + 8 - (6x - 6) = 16x + 8 - 6x + 6 = 10x + 14$.

Solution 2. For some, the explanation given above was sufficient. However, I think that if I had seen this for the first time (or if I've always been confused by this in the past), then I'd love to have a more detailed explanation. So here is a more detailed explanation. When subtracting any value, we can think of replacing what's being subtracting by adding the value multiplied by -1. That sounds confusing, so let's describe this on several examples, starting small. We can reinterpret $9-5$ as $9+(-1)(5)$. We can reinterpret $a-b$ as $a+(-1)(b)$. We can reinterpret $a-(b+c)$ as $a+(-1)(b+c)$. In the same way, we can treat $16x + 8 - (6x - 6)$ as $16x + 8 + (-1)(6x - 6)$, again reintrepreting by saying that instead of subtracting a quantity, we'll add the negative of the quantity (and the negative of a quantity is obtained when multiplying the quantity by -1.) Then in $16x + 8 + (-1)(6x - 6)$, we can distribute -1 but even here, let's choose to do this slowly, so that we can see what's going on. What I mean is that instead of disturbiting and simplifynig all in one step, let's just distribute without simplifying any multiplications. So, the previous expression is equal to $16x + 8 + (-1)(6x) + (-1)(-6)$. We can change the adding of $(-1)(6x)$ into subtracting $6x$ instead, and for the last term, we happen to be adding, due to that plus sign, but adding what? We are adding a negative times a negative. So, we could copy down that final plus sign first, and replace $(-1)(-6)$ with 6. So the previous expression is equal to $16x + 8 - 6x + 6$. Putting this altogether: $(16x + 8) - (6x - 6) = 16x + 8 + (-1)(6x - 6) = 16x + 8 + (-1)(6x) + (-1)(-6) = 16x + 8 - 6x + 6 = 10x + 14$.

Example 1.1.88

Simplify $10x + 9 - (3x + 5)$

Solution 1. One view is to "distribute the minus sign" and have $10x + 9 - (3x + 5) = 10x + 9 - 3x - 5 = 7x + 4$

Solution 2. We can turn $a - b$ as $a + (-1)(b)$, or in reverse turn $a + (-1)(b)$ into $a - b$. Using both of these ideas (and really, the second idea is the first idea in reverse), we write $10x + 9 - (3x + 5) = 10x + 9 + (-1)(3x + 5) = 10x + 9 + (-1)(3x) + (-1)(5) = 10x + 9 - 3x - 5 = 7x + 4$.

While we advertised this concept of "distributing the minus sign" in its most subtle form (when we are distributing -1), a similar idea applies when we are distributing any negative number.

Example 1.1.89

Simplify $7x - 3(2x + 4)$.

Solution 1. One view is to distribute -3 $7x - 3(2x + 4) = 7x - 3(2x) - 3(4) = 7x - 6x - 12 = x - 12$.

Solution 2. The answer we gave may be unsatisfying to some folks, so here's a version where the subtracting is rewritten as adding a negative. Visually, we still will have the same amount of minus signs: where the minus sign used to be will be replaced with a plus sign followed by a minus sign (it's just that the minus sign will be right in front of the 3). We have $7x - 3(2x + 4) = 7x + (-3)(2x + 4) = 7x + (-3)(2x) + (-3)(4) = 7x - 6x - 12 = x - 12$.

Example 1.1.90

Simplify $7x - 3(2x - 4)$.

Solution 1. One view is to distribute -3 $7x - 3(2x + 4) = 7x - 3(2x) + 3(4) =$

$7x - 6x + 12 = x + 12$.

Solution 2. Here's a version where the subtracting is rewritten as adding a negative. We have $7x - 3(2x - 4) = 7x + (-3)(2x - 4) = 7x + (-3)(2x) + (-3)(-4) = 7x - 6x + 12 = x + 12$.

Perhaps the slightly longer second answer convinces you that the first answer worked. In any case, the question in this example is only slightly different than the example right before that, so it makes sense that the final expressions are very similar: the question we just answered had a constant of $+12$ while the question before had a constant of -12. In particular, it makes sense that the two different questions had two different answers. I think it would have been a little weird if we got the same answer for the two differnt questions.

1.1.5 Summary

Expressions with and without variables can be interpreted and analyzed using the Order of Operations, and expressions without variables can be simplified down to a single constant using the Order of Operations. The result of adding and subtracting expressions called terms results in a sum, while the result of multiplying expression called factors results in a product. The geometric representation of addition by gluing together sticks and the geometric representation of multiplication by the area of a rectangle can be used to justify major algebraic formulas, including the Distributive Law. It is the Distributive Law which actually justifies the shortcut process known as collecting like terms.

1.1.6 Exercises

1. Write in your own words how each expression could be spoken aloud with clarity.

   (a) $8 + a$

   (b) $8a$

   (c) 8^a

   (d) $5x + 6^x$

   (e) $3^x - 3x$

   (f) $5(x + y)^2$

   (g) $5x + y^2$

   (h) $\dfrac{3}{a + b}$

   (i) $\dfrac{3}{a} + b$

2. Simplify the following expressions. Because there are no variables, each expression can be simplified based only on the Order of Operations. Because

the intention of this exercise is to practice the Order of Operations, avoid applying the Distributive Law, even if you are certain that the Distributive Law could be applied.

(a) $4(2+6)$

(b) $4(2 \cdot 6)$

(c) $4-(3+5 \cdot 2)+1$

(d) $6-2^3$

(e) $1-2-3-4-5-6$

(f) $(2 \cdot 3)^3$

(g) $100-2 \cdot 3 \times 4$

(h) $5(4+3)^2$

(i) $10 \div 2+3 \cdot 4$

(j) $9-2 \times 3+5 \times 10$

(k) $20-12 \div 4+2 \times 5$

(l) $7^2-3(4+1)+10 \cdot 9$

3. In each expression below, identify which operation is performed first, which is performed next, and so on.

(a) $a(b+c)^2$

(b) $p \div q+r \cdot s$

(c) $a-b \div c+d \times e$

(d) $a^b-c(d+e)+x \cdot y$

(e) $a \div b-(c+d)^m$

(f) $p \div q-r \cdot s+2^m$

4. In each expression below, identify which operation is performed first.

(a) $5+2^x$

(b) $8(x-1)$

(c) $8x-1$

(d) $9 \cdot(x-2)+4$

(e) $a-b \div c+(x-y)^z$

(f) $a-b \div c+x-y^z$

(g) $x \div y-z+a^b$

(h) $x \div(y-z)+a^b$

(i) $a \cdot b+c \div d-e \cdot f+g \cdot h$

5. In each expression below, identify which operation is performed last.

(a) $5 + 2^x$

(b) $8(x - 1)$

(c) $8x - 1$

(d) $9 \cdot (x - 2) + 4$

(e) $a - b \div c + (x - y)^z$

(f) $a - b \div c + x - y^z$

(g) $x \div y - z + a^b$

(h) $x \div (y - z) + a^b$

(i) $a \cdot b + c \div d - e \cdot f + g \cdot h$

6. In each part below, you are given two expressions. Based only on the Order of Operations, are the two expressions equal or do we not have enough information? Notice how similar the expressions are: in most cases, one expression just has more parentheses. (Base your answer *only* on the Order of Operations, and not on whether anything "should" or "shouldn't" distribute.)

(a) $3x + 5$ and $3(x + 5)$

(b) $2 + (a \times 6)$ and $2 + a \times 6$

(c) $(a + b)(c + d)$ and $a + b(c + d)$

(d) $(p \times q) + (r \times s)$ and $p \times q + r \times s$

(e) $(x^2 + 3)(2) - (5x - 7)(4)$ and $(x^2 + 3)2 - (5x - 7)4$

(f) $(t^4 - 3) \cdot (8) + (10)(x - 5)$ and $t^4 - 3 \cdot (8) + 10(x - 5)$

(g) $9 \cdot (x - 2) + 4$ and $9 \cdot x - 2 + 4$

(h) $a - b \div c + (x - y)^z$ and $a - b \div c + x - y^z$

(i) $x \div y - z + a^b$ and $x \div (y - z) + a^b$

(j) $a \cdot b + c \div d - e \cdot f + g \cdot h$ and $a \cdot b + c \div d - (e \cdot f + g \cdot h)$

(k) $p \div q + r \cdot s$ and $(p \div q + r) \cdot s$

7. Each expression below is a sum. State the terms.

(a) State the terms of $3x + 5 + 71y$

(b) State the terms of $4 + 2m + 3x + r$

(c) State the terms of $7a^2 - 3b + 4c - 9$

(d) State the terms of $5x - 7y + 2z + 9 + w$

8. What term(s) do $3x + 5 + 71y$ and $2m + 3x + r - 8y$ have in common?

9. Geometrically represent the following expressions.

(a) $x + 3$

(b) $p \cdot q$

(c) $a + b + 4$

(d) $p(q + r)$

(e) $(a + b)(c + d)$

(f) $(x + 3)(x + 3)$

10. For each geometric representation below, write an expression it represents. (Note, there is often more than one correct answer.)

    (a) $x + y + z$

    (b) $r \times s$

    (c) $p(q + r)$

    (d) $(p + q)r$

    (e) $(p + q)(r + s)$

    (f) $a(b + c + d)$

    (g) $a \times a$

    (h) $(x + 5)(x + 5)$

11. Factor each expression below.

    (a) $6 + 3x$

    (b) $10a - 30$

    (c) $9xy + 4xy$

    (d) $10x + 3x$

    (e) $10x + \sqrt{3}x$

    (f) $7x + x$

    (g) $23x + 23$

12. Simplify each expression below two ways: once by applying the appropriate Distributive Law to factor (showing all steps), and once by quickly collecting like terms. Why ask you to simplify each expression the "slightly slower way" and the "slightly faster way"? The purpose of this exercise is to reinforce the important idea that collecting like terms is actually justified by the Distributive Law.

    (a) $4x + 5x$

    (b) $13y - 2y$

    (c) $3a + 4a + 5a$

    (d) $9c - 2c + 4c$

    (e) $7z - z$

1.2 Expressions with Fractions

Objectives

In this section, we learn how to:

1. Add, subtract, multiply, and divide fractions.
2. Determine when common denominators are required and when they are not.
3. Determine when cancelling is permissible in a fraction and when it isn't.
4. Convert between mixed fractions and improper fractions.

1.2.1 Applications

Operations on fractions are a little more involved than operations on whole numbers, but what we learn will enable us to answer these questions:

1. A baking recipe calls for $5\frac{1}{4}$ cups of flour. The recipe says that it serves 10. You are hosting a party where 35 people will attend, so you need to make $3\frac{1}{2}$ times the recipe. How many cups of flour do you need?
2. You have a two-by-four piece of lumber that measures $2\frac{5}{8}$ feet long and another that is $1\frac{3}{4}$ feet long. You will glue the two pieces together to create one leg of a nightstand. But you'll need to go to the store to make three other legs of the same height. How long is each leg?
3. A swimming pool is being filled at a rate of $8\frac{1}{4}$ gallons per hour. How many gallons of water will be in the pool after $2\frac{1}{4}$ hour?
4. A car is traveling at a speed of 60 miles per hour. How far will the car travel in $4\frac{1}{2}$ hours?
5. A group of 5 friends go out to dinner. The bill comes to $\$48.75$. If they split the bill evenly, how much does each person pay?
6. A recipe calls for $\frac{3}{4}$ cup of sugar. You only have a $\frac{1}{3}$ cup measuring cup. How many $\frac{1}{3}$ cups of sugar do you need to use to get the correct amount?
7. A recipe calls for $\frac{3}{4}$ cup of sugar. You put in $\frac{1}{3}$ cup of sugar by accident. How much more sugar do you need to put in to get back on track in following the recipe?

1.2.2 Representation and Equal/Unequal Fractions

Principle 1.2.1 Representing Fractions Geometrically.

Suppose a and b are positive real numbers. We can represent the fraction $\frac{a}{b}$ as follows: Start with a pizza that has b equal-sized slices. Then $\frac{a}{b}$ is represented by eating a of those slices.

In this way, 1 represents taking/eating the whole pizza, and $\frac{1}{2}$ represents taking a pizza that only has 2 slices (those are *huge* slices, since there are only two of them) and eating one of those slices.

Example 1.2.2

The fraction $\frac{2}{7}$ represents taking a pizza that has been cut into 7 equal slices and eating 2 of those slices.

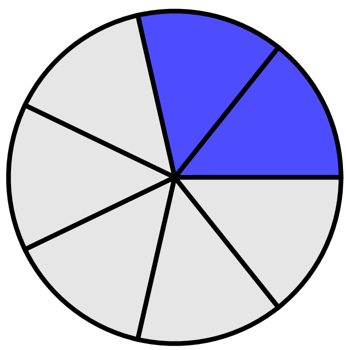

Figure 1.2.3 A picture of $\frac{2}{7}$

Note 1.2.4 Before looking at more examples, let's make the following observation. Say we are given the fraction $\frac{4}{1}$. To represent this, we can think of having a pizza that's been cut into 1 slice, but what does that mean? The slice is the whole pizza. So, to eat 4 of these huge slices, we need 4 whole pizzas. So $\frac{4}{1} = 4$.

In the same way, $\frac{5}{1} = 5$ and so on. Stated more generally, we write

$$\frac{a}{1} = a.$$

There will be times that it is convenient to simplify $\frac{a}{1}$ to a, but there will also be other times that we want to replace a with $\frac{a}{1}$ instead.

That is to say, we can turn the non-fraction a into the fraction $\frac{a}{1}$ if it is helpful for our work. We can say that there is a "hidden denominator of 1" when we are staring at writing that isn't a fraction. Moreover, even when we have a written without the "over 1" explicitly written in, we have $a = \frac{a}{1}$ but $a \neq \frac{1}{a}$. For this reason, we might informally refer to a as being content "in the numerator" (but definitely not "in the denominator") even when a fraction hasn't explicitly been written, and when we say this, we are prentending to see $\frac{a}{1}$ in place of a plain a.

We included this note above about the "hidden denominator of 1" just to make sure we've actually said it, instead of just assuming this later. This is in the name of making sure we have really tried to say everything.

Example 1.2.5

What fraction is equal to $\frac{1}{3}$ but has 6 in the denominator? Draw pictures of $\frac{1}{3}$ and the resulting fraction to convince yourself.

Solution. The fraction $\frac{1}{3}$ represents taking a pizza that has been cut into 3 equal slices and eating 1 of those slices.

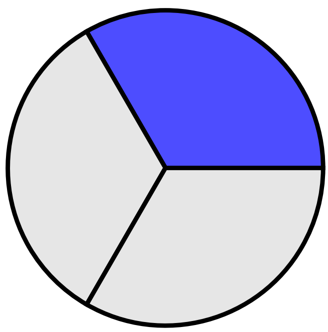

Figure 1.2.6 A picture of $\frac{1}{3}$

To get a denominator of 6, we need to cut each of the 3 slices in half. This gives us 6 equal slices, and we have eaten 2 of those slices. So the answer is $\frac{2}{6}$.

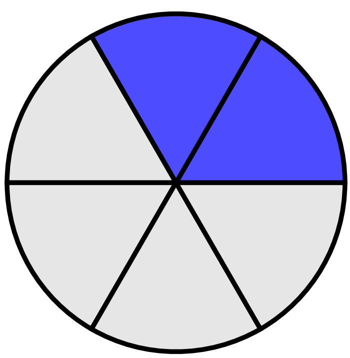

Figure 1.2.7 A picture of $\frac{2}{6}$

If we step back and ignore where the cuts were made by a pizza cutter, the blue shaded part (representing the eaten pizza) looks to be the same amount in both pictures. So $\frac{1}{3} = \frac{2}{6}$.

Example 1.2.8

What fraction is equal to $\frac{3}{4}$ but has 8 in the denominator? Draw pictures of $\frac{3}{4}$ and the resulting fraction to convince yourself.

Solution. The fraction $\frac{3}{4}$ represents taking a pizza that has been cut into 4 equal slices and eating 3 of those slices.

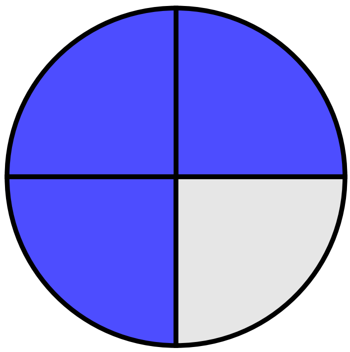

Figure 1.2.9 A picture of $\frac{3}{4}$

To get a denominator of 8, we need to cut each of the 4 slices in half. This gives us 8 equal slices, and we have eaten 6 of those slices. So the answer is $\frac{6}{8}$.

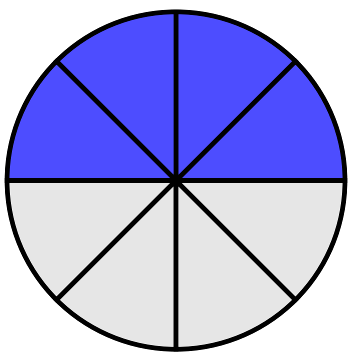

Figure 1.2.10 A picture of $\frac{6}{8}$

Try it 1.2.11

What fraction is equal to $\frac{2}{5}$ but has 15 in the denominator? Draw pictures of $\frac{2}{5}$ and the resulting fraction to convince yourself.

Note 1.2.12 In earlier examples, we had to cut each original slice of pizza into two equal-sized smaller pieces. In the *Try it* exercise, we need to cut each original slice of pizza into three equal-sized smaller pieces.

Note 1.2.13 In the *Try it* exercise, going from a total of 5 slices to a total of 15 slices meant that in the resulting pizza, every slice was very small. This makes intuitive sense to us: when the denominator is large (in other words, there are many slices), then the size of each slice is small. In addition, when the denominator is small (in other words, there are few slices), then the size of each slice is large.

To create an equal fraction to the fraction we're given but without drawing pictures, we can notice that we take our starting fraction and *multiply* both the numerator and denominator by the same number. To formally state this fact:

Principle 1.2.14 Equal Fractions, different denominators.

$$\frac{a}{b} = \frac{a \cdot c}{b \cdot c},$$

where b and c are nonzero.

Example 1.2.15

Using the Equal Fractions formula, What fraction is equal to $\frac{3}{5}$ but has 20 in the denominator?

Solution. To get a denominator of 20, we need to multiply the denominator of $\frac{3}{5}$ by 4. So we also need to multiply the numerator by 4.

$$\frac{3}{5} = \frac{3 \cdot 4}{5 \cdot 4} = \frac{12}{20}.$$

In what we just wrote, we went from the initial fraction to the middle expression to emphasize the Equal Fractions formula, though if we feel comfortable skipping this, we might write $\frac{3}{5} = \frac{12}{20}$ directly. That said, we may find a future example more complicated and writing the middle step (which *indicates* multiplication on top and on bottom, but does not *simplify* the multiplication) is good to practice, even in cases where we might not feel we need it.

Example 1.2.16

Using the Equal Fractions formula, What fraction is equal to $\frac{2}{3}$ but has 12 in the denominator?

Solution. To get a denominator of 12, we need to multiply the denominator of $\frac{2}{3}$ by 4. So we also need to multiply the numerator by 4.

$$\frac{2}{3} = \frac{2 \cdot 4}{3 \cdot 4} = \frac{8}{12}.$$

Warning 1.2.17 When writing our work, we must be careful how we write our work. It is possible to write something that can be interpreted incorrectly. In detail:

- We can write
$$\frac{2}{3} = \frac{2 \cdot 4}{3 \cdot 4} = \frac{8}{12}$$
exactly as the previous example showed.

- We can write the 4 that we were multiplying on top and on bottom before the original content like this:
$$\frac{2}{3} = \frac{4 \cdot 2}{4 \cdot 3} = \frac{8}{12}$$
and this just swaps the order of the factors in both multiplications.

- We can skip actually writing in the "times 4" on top and on bottom and just write in the resulting numerator and denominator like this:
$$\frac{2}{3} = \frac{8}{12}$$

- We *cannot* write
$$4\frac{2}{3}$$
to mean that we are multiplying by 4 on top and on bottom. There are two ways of interpreting $4\frac{2}{3}$. We will discuss one of the ways to interpret $4\frac{2}{3}$ by the end of this section, but a common way to interpret $4\frac{2}{3}$ is as a mixed fraction, whose value is clearly larger than 4 itself, while the original fraction $\frac{2}{3}$ is smaller than 1. So, we either jump to writing $\frac{8}{12}$ directly, or we need to write that we are multiplying by 4 on top and on bottom (and that really requires physically writing two 4s: one on top and one on bottom), but we cannot just write a single 4. Reading $4\frac{2}{3}$ as a mixed fraction has a diffrent value than the original fraction, and in the other interpretation will also result in a value that is not equal to $\frac{2}{3}$.

For another example of this warning, if we start with $\frac{2}{7}$ and we want to properly write about multiplying by 10 on top and on bottom, we can write $\frac{2\cdot 10}{7\cdot 10}$ or $\frac{10\cdot 2}{10\cdot 7}$ or just immediately write $\frac{20}{70}$, but we cannot write $10\frac{2}{7}$.

Example 1.2.18

What fraction is equal to $\frac{2}{3}$ but has $3x$ in the denominator?

Solution. We need to multiply the denominator by x, so we also multiply the numerator by x.

$$\frac{2}{3} = \frac{2x}{3x}$$

Example 1.2.19

What fraction is equal to $\frac{2}{3}$ but has $3(10x+2)$ in the denominator?

Solution. We need to multiply the denominator by $(10x+2)$ so we also multiply the numerator by $(10x+2)$.

$$\frac{2}{3} = \frac{2(10x+2)}{3(10x+2)}$$

Here, if we wanted, we can distribute in both the numerator and denominator. However, we stopped where we did because we wanted to highlight the role of the Equal Fractions formula.

Example 1.2.20

What fraction is equal to $\frac{5x}{2y}$ but has $10y$ in the denominator?

Solution. We need to multiply the denominator by 5, so we also multiply the numerator by 5.

$$\frac{5x}{2y} = \frac{5\cdot 5x}{5\cdot 2y} = \frac{25x}{10y}$$

In certain situations, we can also use the Equal Fractions formula to reduce the size of the denominator.

Principle 1.2.21 Reducing Fractions.

If we see the same factor in the numerator and denominator of a fraction, the process of removing this common factor is called ***cancelling*** *in a fraction or* ***reducing*** *a fraction. This is applying our earlier formula $\frac{a}{b} = \frac{a\cdot c}{b\cdot c}$ "backwards" in other words, this is turning $\frac{a\cdot c}{b\cdot c}$ into $\frac{a}{b}$*

Example 1.2.22

What fraction is equal to $\frac{8}{20}$ but has 5 in the denominator?

Solution. The numerator and denominator both have a factor of 4. For the first several examples, we will intentionally slow down and rewrite the numerator and denominator in factored form to highlight the common factor, which helps highlight exactly the role taken by the *Reducing Fractions* process we just described.

$$\frac{8}{20} = \frac{2 \cdot 4}{5 \cdot 4} = \frac{2}{5}.$$

We cancelled the common factor of 4 in the numerator and denominator.

Example 1.2.23

What fraction is equal to $\frac{12}{27}$ but has 9 in the denominator?

Solution.

$$\frac{12}{27} = \frac{4 \cdot 3}{9 \cdot 3} = \frac{4}{9}.$$

We cancelled the common factor of 3 in the numerator and denominator.

Example 1.2.24

What fraction is equal to $\frac{18x}{24y}$ but has $4y$ in the denominator?

Solution.

$$\frac{18x}{24y} = \frac{3x \cdot 6}{4y \cdot 6} = \frac{3x}{4y}.$$

Now we can cancel the common factor of 6 in the numerator and denominator.

Example 1.2.25

Reduce the fraction $\frac{3x^2}{6xy}$ as much as possible.

Solution.

$$\frac{3x^2}{6xy} = \frac{x \cdot 3x}{2y \cdot 3x} = \frac{x}{2y}.$$

Now we can cancel the common factor of $3x$ in the numerator and denominator.

Warning 1.2.26 While we can cancel common *factors* in a fraction, we cannot cancel common *terms*. In other words,

$$\frac{a+c}{b+c} \neq \frac{a}{b}.$$

Let's describe this frequent error through two examples:

- If someone took $\frac{2+1}{4+1}$ and tried to "cancel" the $+1$ in the numerator and denominator, they would get $\frac{2}{4}$, which is 0.5 in a calculator. However, the original fraction can naturally be rewritten as $\frac{3}{5}$, which is 0.6 in a calculator.
- The issue is more likely to occur when there are variables. (In fact, the previous example with its decimal representations of fractions was included to create a concrete and convincing example that we cannot cancel terms on top and bottom.) We cannot turn $\frac{x+2}{x+3}$ into $\frac{2}{3}$ by "cancelling" the common x in the numerator and denominator. This is because x is a *term* on top and bottom, and not a *factor* on top and bottom.

Example 1.2.27

Reduce the fraction $\frac{20x+60}{14x+30}$

Solution 1. There is a temptation to want to cancel part of the 60 with the 30, but these are terms. We can only cancel factors. First, we rewrite the numerator and denominator in factored form:

$$\frac{20x+60}{14x+30} = \frac{2(10x+30)}{2(7x+15)} = \frac{10x+30}{7x+15}.$$

We cancelled the common factor of 2 in the numerator and denominator.

Solution 2. You may prefer to factor a larger factor out of the numerator. (Note, we still cannot cancel terms.)

$$\frac{20x+60}{14x+30} = \frac{20(x+3)}{2(7x+15)} = \frac{10(x+3)\cdot 2}{(7x+15)\cdot 2} = \frac{10(x+3)}{7x+15}.$$

After factoring, we replaced the 20 that we factored out with $10 \cdot 2$ to really highight that it is a factor of 2 on top and bottom that we cancelled. To make the final expression in this solution look like the final expression in the previous solution, we could distribute the 10 in the numerator. (In this example, you may feel comfortable skipping the writing the third expression, jumping directly from the second expression to the fourth expression. We included the third expression because going from the third expression to the fourth expression highlights the role of the *Reducing Fractions* process we described earlier.)

1.2.3 Adding and subtracting fractions

Try it 1.2.28

The pictures of $\frac{5}{6}$ and $\frac{4}{9}$ are shown below. How can we use these pictures to represent a picture that represents the value of $\frac{5}{6} + \frac{4}{9}$ as a single fraction?

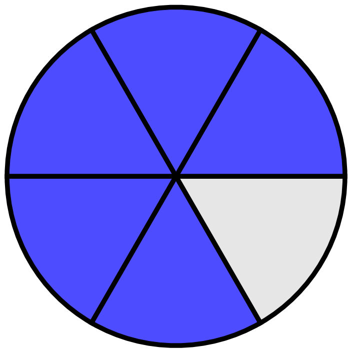

Figure 1.2.29 A picture of $\frac{5}{6}$

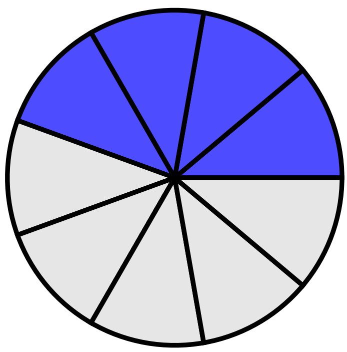

Figure 1.2.30 A picture of $\frac{4}{9}$

Did you try this exercise? What did you notice about the pictures? Because of the different denominators, the slices in the two pictures are different sizes, so it seems a little funny if we just tried to add the numerators across and add the denominators across when we write with notation. This hints at the following idea: To add fractions, we need a common denominator. In this case, we can use 18 as a common denominator. We can convert $\frac{5}{6}$ to a fraction with denominator 18 by multiplying the numerator and denominator by 3. We can convert $\frac{4}{9}$ to a fraction with denominator 18 by multiplying the numerator and denominator by 2. So:

$$\frac{5}{6} + \frac{4}{9} = \frac{5 \cdot 3}{6 \cdot 3} + \frac{4 \cdot 2}{9 \cdot 2} = \frac{15}{18} + \frac{8}{18} = \frac{15+8}{18} = \frac{23}{18}.$$

Note that the final answer is an improper fraction. This is perfectly fine. In algebra, it is often more helpful to leave answers as improper fractions. (By the end of this section, we'll explain exactly why improper fractions are preferred.)

Principle 1.2.31 Adding Fractions.

Simplifying the addition of fractions requires having a common denominator.

$$\frac{a}{c}+\frac{b}{c}=\frac{a+b}{c}.$$

Notice in the two fractions on the left side of the formula above, c is the denominator for both fractions. The fact that c is written for both denominators on the left is the formula communicating to us that both fractions have to have the same denominator. When we have that common denominator c, the right side of this formula which says $\frac{a+b}{c}$ is tell us that fraction that we get as a result from simplifying the addition copies that same denominator c, while the numerator of the new fraction is made by adding the numerators of the two fractions that had the same denominator. It is incorrect to try to “just add straight across”:

Warning 1.2.32

$$\frac{p}{q}+\frac{r}{s}\neq\frac{p+r}{q+s}$$

To show why this warning is here, in Try it 1.2.28, we saw $\frac{5}{6}+\frac{4}{9}$ simplifies to $\frac{23}{18}$ but if we added straight across (using the “fake formula” meantion in the warning), we would have gotten $\frac{9}{15}$. You can check that $\frac{23}{18}$ and $\frac{9}{15}$ are different answers with a calculator, or without a calculator, we can notice that $\frac{23}{18}$ is greater than 1, while $\frac{9}{15}$ is less than 1.

Example 1.2.33

Simplify $\frac{3}{5}+\frac{1}{4}$

Solution.

$$\frac{3}{5}+\frac{1}{4}=\frac{12}{20}+\frac{5}{20}=\frac{12+5}{20}=\frac{17}{20}$$

The third expression may be skipped if you feel comfortable, but this is a nice technique to practice in smaller situations (beacuse in some larger situations, it may be harder for us to simplify the addition of the numerators in our head). The purpose of showing the second expression $\frac{12}{20}+\frac{5}{20}$ being equal to the third expression $\frac{12+5}{20}$ is also helpful in seeing the Adding Fractions formula apply as literally as possible.

Example 1.2.34

Simplify $\frac{3}{10}+\frac{4}{15}$

Solution.

$$\frac{3}{10}+\frac{4}{15}=\frac{9}{30}+\frac{8}{30}=\frac{9+8}{30}=\frac{17}{30}$$

Because we are doing algebra, our fractions may have variables in them. The Adding Fractions formula still applies, and we still need a common denominator to simplify the addition of fractions.

Example 1.2.35

Simplify $\frac{2x}{3}+\frac{5x}{4}$

Solution.

$$\frac{2x}{3}+\frac{5x}{4}=\frac{8x}{12}+\frac{15x}{12}=\frac{8x+15x}{12}=\frac{23x}{12}$$

Example 1.2.36

Simplify $\frac{3}{2x^2y}+\frac{5}{4x}$

Solution. Recall that x^2 means $x\cdot x$. To achieve a common denominator of $4x^2y$, let's multiply both the top and bottom of the first fraction by 2, and multiply both the top and bottom of the second fraction by xy.

$$\frac{3}{2x^2y}+\frac{5}{4x}=\frac{6}{4x^2y}+\frac{5xy}{4x^2y}=\frac{6+5xy}{4x^2y}$$

Note that our final answer is $\frac{6+5xy}{4x^2y}$, and there is nothing that will cancel: the denominator is a product (the result of multiplication) so its pieces (called factors) are *eligible* to be cancelled, but the numerator is not a product, and so we cannot cancel any common factors on top and bottom (since the top is not written as a product).

Example 1.2.37

Simplify $\frac{4a}{3bcd}+\frac{2b}{5acd^2}$

Solution. Recall that d^2 means $d\cdot d$. To achieve a common denominator of $15abcd^2$, let's multiply both the top and bottom of the first fraction by $5d$, and multiply both the top and bottom of the second fraction by $3a$. (Be sure to read $15abcd^2$ using the Order of Operations: the exponent does not apply to all variables, just to d.)

$$\frac{4a}{3bcd}+\frac{2b}{5acd^2}=\frac{20ad}{15abcd^2}+\frac{6b^2}{15abcd^2}=\frac{20ad+6b^2}{15abcd^2}$$

How should we add a fraction and a non-fraction together? Recall from Note 1.2.4

that anything that presents as a non-fraction can be written as a fraction with denominator 1.

Principle 1.2.38 How to add a fraction and a non-fraction.

Turn the non-fraction into a fraction with denominator 1*. Then use the Adding Fractions formula, noting that this formula requires a common denominator.*

Example 1.2.39

Simplify $2+\frac{3}{5}$

Solution. We rewrite 2 as $\frac{2}{1}$. Then we can use the Adding Fractions formula, noting that we need a common denominator.

$$2+\frac{3}{5}=\frac{2}{1}+\frac{3}{5}=\frac{10}{5}+\frac{3}{5}=\frac{10+3}{5}=\frac{13}{5}.$$

Example 1.2.40

Simplify $\frac{2}{7}+3$

Solution. We rewrite 3 as $\frac{3}{1}$.

$$\frac{2}{7}+3=\frac{2}{7}+\frac{3}{1}=\frac{2}{7}+\frac{21}{7}=\frac{2+21}{7}=\frac{23}{7}.$$

Like adding fractions, subtracting requires a common denominator:

Principle 1.2.41 Subtracting Fractions.

Simplifying the subtraction of fractions requires having a common denominator.

$$\frac{a}{c}-\frac{b}{c}=\frac{a-b}{c}.$$

Example 1.2.42

Simplify $\frac{5}{6}-\frac{2}{9}$

Solution.

$$\frac{5}{6}-\frac{2}{9}=\frac{15}{18}-\frac{4}{18}=\frac{15-4}{18}=\frac{11}{18}$$

The third expression $\frac{15-4}{18}$ may be skipped if you feel comfortable, but we specifically included this step because going from the second expression $\frac{15}{18}-\frac{4}{18}$ to the third expression $\frac{15-4}{18}$ highlights the role of the *Subtracting Fractions* formula.

Example 1.2.43

Simplify $\frac{5}{12a} - \frac{3}{8ab}$

Solution.

$$\frac{5}{12a} - \frac{3}{8ab} = \frac{10b}{24ab} - \frac{9a}{24ab} = \frac{10b - 9a}{24ab}$$

In this example, to achieve a common denominator of $24ab$, the top and bottom of the first fraction both got multiplied by $2b$, while the top and bottom of the second fraction both got multiplied by $3a$.

Note that nothing cancels in $\frac{10b-9a}{24ab}$, because the denominator is a product (the result of multiplication) so its pieces (called factors) are *eligible* to be cancelled, but the numerator is not a product, and so we cannot cancel any common factors on top and bottom (since the top is not written as a product).

Example 1.2.44

Simplify $\frac{4a}{3bc} - \frac{2b}{5ac}$

Solution.

$$\frac{4a}{3bc} - \frac{2b}{5ac} = \frac{20a^2}{15abc} - \frac{6b^2}{15abc} = \frac{20a^2 - 6b^2}{15abc}$$

In this example, to achieve a common denominator of $15abc$, the top and bottom of the first fraction both got multiplied by $5a$, while the top and bottom of the second fraction both got multiplied by $3b$.

Principle 1.2.45 Subtraction involving a fraction and a non-fraction.

Turn the non-fraction into a fraction with denominator 1*. Then use the Subtracting Fractions formula, noting that this formula requires a common denominator.*

Example 1.2.46

Simplify $8 - \frac{3}{10}$

Solution. Rewrite 8 as $\frac{8}{1}$.

$$8 - \frac{3}{10} = \frac{8}{1} - \frac{3}{10} = \frac{80}{10} - \frac{3}{10} = \frac{80-3}{10} = \frac{77}{10}.$$

1.2.4 Multiplying fractions

The process for multiplying fractions feels really different from adding or subtracting fractions. To be convinced that the process we will describe in a moment works, we'll go back to the geometric interpretation of multiplication. Recall that $a \cdot b$ can be represented as the area of a rectangle with height a and width b.

Example 1.2.47

Represent 2×3 geometrically.

Solution.

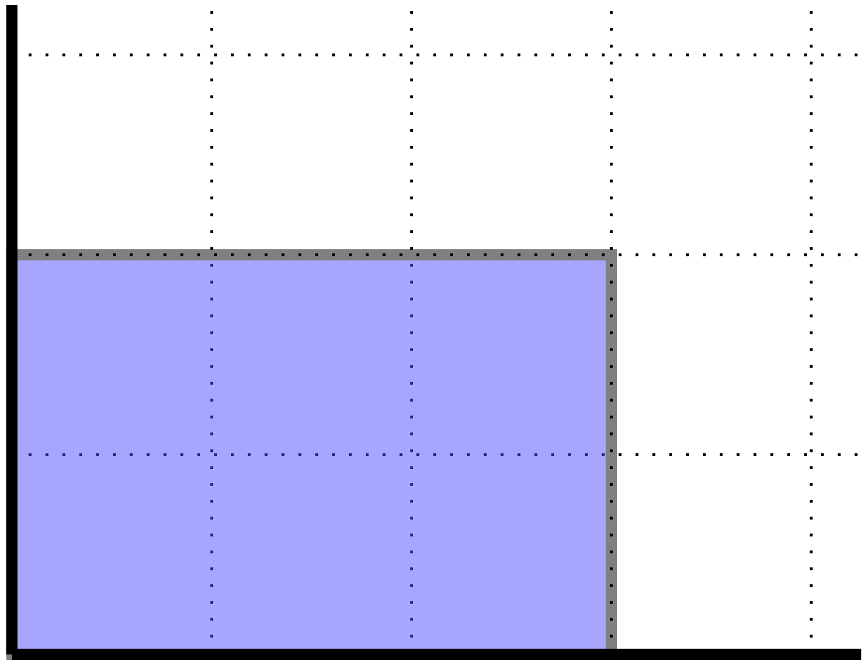

Figure 1.2.48 A picture of 2×3

The dotted lines in the picture above help us see that the rectangle has a height of 2 and a width of 3. We can be really convinced that the resulting area is 6 because we can notice the rectangle is made up of 6 unit squares. I understand that it probably still feels a little funny to represent multiplication as an area, but this is a really useful way to think about multiplication.

Example 1.2.49

Represent $\frac{1}{4} \cdot \frac{2}{3}$ geometrically. Use the picture to determine the value of $\frac{1}{4} \cdot \frac{2}{3}$.

Solution. It can be a little more challenging to represent the area. To standardize things, let's copy (to scale) one of the unit squares from the previous example. Since the first factor is $\frac{1}{4}$, we need a height of $\frac{1}{4}$. The entire square that we copied has a height of 1 so to get a height of $\frac{1}{4}$, we can divide the height of the square into 4 equal parts (achieved by creating 3> equally-spaced apart cuts in the interior of this line segment) and take 1 of those parts. Similarly, for a width of $\frac{2}{3}$, we can divide the width of the square into 3 equal parts (achieved by creating 2 equally-spaced apart cuts in the interior of this line segment) and take 2 of those parts.

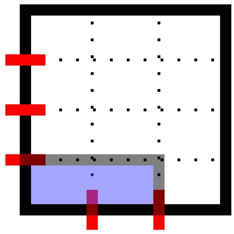

Figure 1.2.50 Picture of $\frac{1}{4} \cdot \frac{2}{3}$

By going with what we have said about the geometric interpretation of multiplication, $\frac{1}{4} \cdot \frac{2}{3}$ is the area of the blue rectangle. The portion shaded in blue is definitely less than 1 because shading the entire unit square would have had area 1. How can we determine the area of the blue rectangle exactly, though? Surprisingly, we can apply the geometric interpretation of a fraction! Recall that $\frac{a}{b}$ means to take a whole pizza and cut it into b equal slices and take a of those slices, but there was no requirement that the pizza had to be a circle. So, we can think of the entire unit square as a whole pizza. To get the area of the blue rectangle, we can see all of the dotted lines as cuts that divide the pizza into equal slices. From our starting square and uncut pizza, the cuts end up creating 12 equal slices (arranged in 4 rows and 3 columns). Then the blue rectange (representing the eaten part of the pizza) is made up of 2 of those 12 equal slices, so the area of the blue rectangle is $\frac{2}{12}$ which we can reduce to $\frac{1}{6}$.

This is a profound first example that uses a lot of ideas, so be patient with yourself! We used the geometric interpretation of multiplication to *create* the shaded region in the first place. Once we have the region we need the area of, we stopped thinking about the geometric interpretation of multiplication and instead thought about the geometric representation of a fraction. Imagine someone using a pizza cutter to cut horizontally along 3 lines, and to cut vertically in 2 lines, as shown as dotted lines in Figure 1.2.50. (The figure indicates in red the places where the pizza cutter began to cut the crust.) Out of 12 equal-sized pieces of pizza, the area of the blue rectangle is made up of 2 of those pieces, so the area of the blue rectangle is $\frac{2}{12}$ which reduces to $\frac{1}{6}$.

Example 1.2.51

Represent $\frac{1}{5} \cdot \frac{3}{4}$ geometrically. Use the picture to determine the value of $\frac{1}{5} \cdot \frac{3}{4}$.

Solution. Starting from the exact same size unit square used in the last examples, we draw inside a rectangle with height $\frac{1}{5}$ and width $\frac{3}{4}$. A height of $\frac{1}{5}$ is achieved by dividing square's height of 1 into 5 equal parts (achieved by creating 4 equally-spaced apart cuts in the interior of this line segment) and taking 1 of those parts. A width of $\frac{3}{4}$ is achieved by dividing the square's width of 1 into 4 equal parts (achieved by creating 3 equally-spaced apart cuts in the interior of this line segment) and taking 3 of those parts.

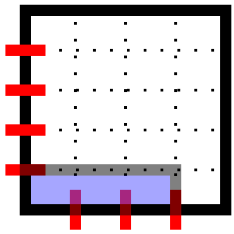

Figure 1.2.52 Picture of $\frac{1}{5} \cdot \frac{3}{4}$

The area of the blue shaded rectangle answers the question of what $\frac{1}{5} \cdot \frac{3}{4}$ is. The dotted lines represent cuts that divide the pizza into 20 equal slices (arranged in 5 rows and 4 columns). The shaded part represents taking 3 of those slices. Thus, the product is $\frac{3}{20}$.

Example 1.2.53

Represent $\frac{2}{5} \cdot \frac{3}{7}$ geometrically. Use the picture to determine the value of $\frac{2}{5} \cdot \frac{3}{7}$.

Solution. Starting from the exact same size unit square used in the last examples, we draw inside a rectangle with height $\frac{2}{5}$ and width $\frac{3}{7}$. A height of $\frac{2}{5}$ is achieved by dividing square's height of 1 into 5 equal parts (achieved by creating 4 equally-spaced apart cuts in the interior of this line segment) and taking 2 of those parts. A width of $\frac{3}{7}$ is achieved by dividing the square's width of 1 into 7 equal parts (achieved by creating 6 equally-spaced apart cuts in the interior of this line segment) and taking 3 of those parts.

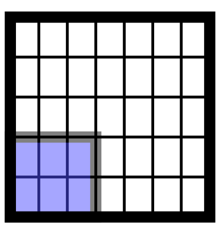

Figure 1.2.54 Picture of $\frac{2}{5} \cdot \frac{3}{7}$

The area of the blue shaded rectangle answers the question of what $\frac{2}{5} \cdot \frac{3}{7}$ is. The dotted lines represent cuts that divide the pizza into 35 equal slices (arranged in 5 rows and 7 columns). The shaded part represents taking 6 of those slices. Thus, the product is $\frac{6}{35}$.

The last example probably felt a little annoying, but I wanted us to explore the product of two fractions where neither fraction had 1 in the numerator, so that we can see if there's a pattern. (Can you guess what the pattern is?) In fact, I thought I'd show the example we just did, so that you can try a slightly less annoying example. If you have a guess for what the pattern is, can you confirm it with drawing a picture in the next example. If you don't have a guess yet, that's totally okay, and I encourage you to try making a drawing for the next example (so that you have another example to look at for the pattern).

Try it 1.2.55

Represent $\frac{3}{4} \cdot \frac{2}{5}$ geometrically. Use the picture to determine the value of $\frac{3}{4} \cdot \frac{2}{5}$.

Did you try this exercise? What did you find? Let's go back , where we saw that $\frac{2}{5} \cdot \frac{3}{7} = \frac{6}{35}$. Let's examine the resulting fraction $\frac{6}{35}$.

1. The denominator was the total number of pizza slices. Since the pizza was cut in such a way that there were 5 rows and 7 columns, there were a total of 35 slices. This number happens to be the product of the denominators of the two fractions we were originally given: $5 \cdot 7 = 35$,
2. The numerator was the number of pizza slices that were eaten, shown in the figure shaded in blue. The shaded rectangle is an arrangement of pizza slices in 2 rows and 3 columns, so there were a total of 6 slices eaten. This number happens to be the product of the numerators of the two fractions we were originally given: $2 \cdot 3 = 6$.

Did you find the same thing in Try it 1.2.55? Could you describe the numerator and denominator of the product by talking about the numerators and denominators of the two fractions we started with? If you didn't make a drawing, I encourage you to go back and try it. The point of drawing this a couple times is to be convinced that there is a pattern, and once you know that the pattern is always there (and even more awesomely, can see why it works by trying it hands on), then it won't be too surprising for you when we present the following procedure for simplifying the product of fractions.

Principle 1.2.56 Multiplying Fractions.

To simplify the multiplication of fractions, we use

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}.$$

Notice that this formula does not require a common denominator. This is a big difference from the Adding Fractions and Subtracting Fractions formulas. Even when we looked at , where we saw that $\frac{2}{5} \cdot \frac{3}{7} = \frac{6}{35}$, we didn't need a common denominator: the two fractions that we started with had denominators 5 and 7, which are different.

The formula $\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}$ communicates that common denominators are not required by having different letters in the denominators for the two fractions on the left side of the equation. The right side of the equation $\frac{a \cdot c}{b \cdot d}$ communicates that we just "multiply straight across". This is the opposite of what we saw in the Adding Fractions formula, where we were specifically warned not to "add straight across". If you take a quick minute to check that this formula applied to all the examples where we drew pictures, then it'll feel a lot more comfortable going through the next few examples where we don't draw pictures and just multiply straight across.

Example 1.2.57

Simplify $\frac{3}{7} \cdot \frac{10}{11}$

Solution. $\frac{3}{7} \cdot \frac{10}{11} = \frac{3 \cdot 10}{7 \cdot 11} = \frac{30}{77}$

In the example above, we showed the second expression $\frac{3 \cdot 10}{7 \cdot 11}$ becauuse going from the first expression to the second expression literally highlights the use of the Multiplying Fractions formula. If you feel comfortable, you can skip writing the second expression in future examples, going from the first expression to the third expression.

Example 1.2.58

Simplify $\frac{5}{9} \cdot \frac{4}{7}$

Solution 1. $\frac{5}{9} \cdot \frac{4}{7} = \frac{5 \cdot 4}{9 \cdot 7} = \frac{20}{63}$

Solution 2. If we felt comfortable skipping the second expression, our work would look like this: $\frac{5}{9} \cdot \frac{4}{7} = \frac{20}{63}$

How should we multiply a fraction and a non-fraction together? Will the technique we used for adding and subtracting a fraction and a non-fraction work here too? Yes! Anything that presents as a non-fraction can be written as a fraction with denominator 1.

Principle 1.2.59 Multiplying a Fraction and a Non-Fraction.

To multiply a fraction and a non-fraction, we can rewrite the non-fraction as a fraction with a denominator of 1. *Then we can use the Multiplying Fractions formula.*

Example 1.2.60

Simplify $2 \cdot \frac{3}{5}$

Solution. We rewrite 2 as $\frac{2}{1}$.

$$2 \cdot \frac{3}{5} = \frac{2}{1} \cdot \frac{3}{5} = \frac{2 \cdot 3}{1 \cdot 5} = \frac{6}{5}$$

Let's practice additional examples that illustrate the Multiplying Fractions formula.

Example 1.2.61

Simplify $\frac{4}{9} \cdot \frac{3}{8}$

Solution. $\frac{4}{9} \cdot \frac{3}{8} = \frac{12}{72} = \frac{1}{6}$

In this example, we reduced the fraction $\frac{12}{72}$ to $\frac{1}{6}$ in the final step. (Remember that we can only cancel common *factors*, not common *terms*. We canceled a factor of 12 from the numerator and denominator.)

Sometimes, we can simplify a multiplication of fractions by canceling common factors before we simplify the straight across multiplication. That is, we can *indicate* that multiplication is happening without *simplifying* the multiplication:

Example 1.2.62

Simplify $\frac{4}{9} \cdot \frac{3}{8}$

Solution. $\frac{4}{9} \cdot \frac{3}{8} = \frac{4 \cdot 3}{9 \cdot 8}$

Now we can see that there is a common factor of 4 in the numerator and denominator, so we can cancel that common factor:

$\frac{4 \cdot 3}{9 \cdot 8} = \frac{\cancel{4} \cdot 3}{9 \cdot \cancel{8}} = \frac{3}{9 \cdot 2} = \frac{3}{18} = \frac{1}{6}$

This is the same answer we got in the previous example, but this time we canceled a common factor before doing the multiplication. This technique can be really helpful especially when simplifying the multiplications in the numerator and/or denominator would lead to large numbers: we avoid needing to do large-number multiplication, and also, it can be easier to spot common factors before multiplying.

Example 1.2.63

Simplify $\frac{5}{12} \cdot \frac{8}{15}$

Solution. $\frac{5}{12} \cdot \frac{8}{15} = \frac{5 \cdot 8}{12 \cdot 15}$

Now we can see that there is a common factor of 4 in the numerator and denominator as well as a common factor of 5 in the numerator and denominator (or instead of seeing the 4s and 5s individuall

$\frac{5 \cdot 8}{12 \cdot 15} = \frac{\cancel{4} \cdot \cancel{5} \cdot 2}{\cancel{4} \cdot \cancel{5} \cdot 9} = \frac{2}{9}$

Please note that you don't have to do it this way, but it can be convenient. If we write that there is a multiplication happening but do not simplify multiplication, it might just be easier to spot common factors. Cancelling early on means that when we eventually do simplify the multiplication, we will be working with smaller numbers.

Example 1.2.64

Simplify $\frac{20}{27} \cdot \frac{9}{14}$

Solution 1. $\frac{20}{27} \cdot \frac{9}{14} = \frac{20 \cdot 9}{27 \cdot 14} = \frac{180}{378} = \frac{10}{21}$

It takes a bit of work to simplify the multiplications in the numerator and denominator to obtain $\frac{180}{378}$ in the first place. Then, due to the numbers being large, it takes considerable effort to reduce this fraction to its equal value $\frac{10}{21}$. This solution is completely valid, but the second solution below shows how we can avoid the large-number multiplications by canceling common factors before simplifying the multiplication.

Solution 2. $\frac{20}{27} \cdot \frac{9}{14} = \frac{20 \cdot 9}{27 \cdot 14}$

The numerator and denominator both have factors of 9 and 2, so we can cancel those common factors:

$$\frac{20 \cdot 9}{27 \cdot 14} = \frac{\cancel{9} \cdot \cancel{2} \cdot 10}{\cancel{9} \cdot 3 \cdot \cancel{2} \cdot 7} = \frac{10}{21}$$

Example 1.2.65

Simplify $\frac{3ab}{10c} \cdot \frac{55bc}{21a^2}$

Solution. First we indicate the multiplication without simplifying:

$$\frac{3ab}{10c} \cdot \frac{55bc}{21a^2} = \frac{3ab \cdot 55bc}{10c \cdot 21a^2} = \frac{3ab \cdot 55bc}{10c \cdot 21 \cdot a \cdot a}$$

Above, we rewrote a^2 as $a \cdot a$. The factors that appear in both the numerator and denominator are 5 and 3 and a and c.

$$\frac{3ab \cdot 55bc}{10c \cdot 21 \cdot a \cdot a} = \frac{\cancel{3} \cdot b \cdot \cancel{5} \cdot 11 \cdot b \cdot \cancel{c}}{\cancel{5} \cdot 2 \cdot \cancel{3} \cdot 7 \cdot \cancel{a} \cdot a} = \frac{11 \cdot b \cdot b}{14a} = \frac{11b^2}{14a}$$ where we rewrote $b \cdot b$ as b^2.

Warning 1.2.66 Please note that when multiplying fractions, we do not need a common denominator: sometimes people get overly cautious and try to treat multiplying fractions like adding fractions. Let's look at an example and see what the common error is:

- When asked to simplify $\frac{4}{9} \cdot \frac{5}{9}$ it is tempting to think about common denominators (because we notice that both fractions already have the same denominator of 9) but it is incorrect to state the result of simplying this multiplication as $\frac{20}{9}$.

- Before talking about what we *should* do and ignoring our work above, let's talk about the work above, to see if we can take the experience and foundation that we've built to question the reasonableness of our answer: Notice that the numerator is larger than the denominator in $\frac{20}{9}$, so this fraction is greater than 1, but both of the fractions we started with were less than 1, so it doesn't make sense that their product would be greater than 1. In fact, the pizza diagram for $\frac{4}{9} \cdot \frac{5}{9}$ would take just *some* of the slices from a square pizza that was cut into a 9-by-9 grid of 81 slices, so the resulting fraction should be less than 1.

- The correct simplification is $\frac{4}{9} \cdot \frac{5}{9} = \frac{4 \cdot 5}{9 \cdot 9} = \frac{20}{81}$. Notice that the denominator of the result is 81, not 9. The denominator of the result is the product of the denominators of the two fractions we started with: $9 \cdot 9 = 81$, even though the two denominators that we started with were the same. (This is different from adding fractions, where the denominator of the result was the common denominator.) The fact that we had common denominators is actually a distraction, and we should ignore it.

- Here is a silghtly different example: $\frac{4}{9} \cdot \frac{5}{10}$. While this is a different problem, because the only change was by slightly changing the denominator of the second fraction, it seems reasonable that the final answer to this question would be close to the final answer of the previous question. Here, the denominators are different, so there is no distraction. Following our Multiplying Fractions formula, we have $\frac{4}{9} \cdot \frac{5}{10} = \frac{20}{90}$ by multiplying straight across. Use a calculator to verify that $\frac{20}{90}$ is close to $\frac{20}{81}$, but far from $\frac{20}{9}$.

The takeaway is that when multiplying fractions just multiply straight across whether we have a common denominator or not!

1.2.5 Dividing fractions

Dividing by a fraction is the same as multiplying by its reciprocal.

Principle 1.2.67 Dividing Fractions formula.

To divide by a fraction, multiply by its reciprocal and follow the Multiplying Fractions formula to simplify the multiplication of fractions.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

Example 1.2.68

Simplify $\frac{2}{3} \div \frac{5}{7}$

Solution. $\frac{2}{3} \div \frac{5}{7} = \frac{2}{3} \cdot \frac{7}{5} = \frac{14}{15}$

Example 1.2.69

Simplify $\frac{\frac{4}{9}}{\frac{2}{3}}$

Solution. Although the question didn't write the division symbol $\div$, the fraction bar indicates division. (So we could rewrite the problem as $\frac{4}{9} \div \frac{2}{3}$, and we'll say this is asking the same question.) However, to practice this new format, let's leave the original question in this format and work from there.

$$\frac{\frac{4}{9}}{\frac{2}{3}} = \frac{4}{9} \cdot \frac{3}{2} = \frac{12}{18} = \frac{2}{3}.$$

From the second-to-last expression to the last expression, we reduced the fraction.

Warning 1.2.70 It is often tempting to see a question in this format $\frac{\frac{4}{9}}{\frac{2}{3}}$ and in the desire to multiply by the reciprocal, something unintentional remains in the writing. For example, looking to the previous computation, when starting with $\frac{\frac{4}{9}}{\frac{2}{3}}$ the next expression shouldn't read $\frac{\frac{4}{9} \cdot \frac{3}{2}}{\frac{2}{3}}$. The point is that multiplying by $\frac{3}{2}$ is what *replaces* dividing by $\frac{2}{3}$, so we should either have $\frac{2}{3}$ in the denominator or have the multiplication by $\frac{3}{2}$, but not both.

The analogy in the question before this one would be like if someone turned $\frac{2}{3} \div \frac{5}{7}$ into $\frac{2}{3} \div \frac{5}{7} \cdot \frac{7}{5}$ instead of $\frac{2}{3} \cdot \frac{7}{5}$. Multiplying by $\frac{7}{5}$ is what *replaces* dividing by $\frac{5}{7}$, so we should either divide by $\frac{5}{7}$ or multiply by $\frac{7}{5}$, but not both.

Example 1.2.71

Simplify $\frac{\frac{3}{4ac}}{\frac{5b}{6a}}$

Solution.

$$\frac{\frac{3}{4ac}}{\frac{5b}{6a}} = \frac{3}{4ac} \cdot \frac{6a}{5b} = \frac{18a}{20abc} = \frac{9}{10bc}$$

How would we divide between a fraction and a non-fraction? As usual, turn the non-fraction into a fraction with denominator 1.

Example 1.2.72

Simplify $2 \div \frac{3}{5}$.

Solution. We rewrite 2 as $\frac{2}{1}$. Then:

$$2 \div \frac{3}{5} = \frac{2}{1} \div \frac{3}{5} = \frac{2}{1} \cdot \frac{5}{3} = \frac{10}{3}$$

1.2.6 Additional practice

Example 1.2.73

Simplify $\frac{3}{4} + \frac{2}{5} \cdot \frac{7}{8}$

Solution.

$$\frac{3}{4} + \frac{2}{5} \cdot \frac{7}{8} = \frac{3}{4} + \frac{14}{40} = \frac{3}{4} + \frac{7}{20} = \frac{15}{20} + \frac{7}{20} = \frac{22}{20} = \frac{11}{10}.$$

Note that we had to simplify multiplication before simplifying addition. It is easy to fall into the trap of simplifying addition (that is, working on simplifying $\frac{3}{4} + \frac{2}{5}$) first, because we see the addition written first. However, we still need to read expressions based on the Order of Operations, even if the expression contains fractions.

Example 1.2.74

Simplify $\frac{2}{3} - \frac{6}{7} \div \frac{5}{2}$

Solution. The division must be simplified before the subtraction. We turn dividing by $\frac{5}{2}$ into multiplying by $\frac{2}{5}$ instead.

$$\frac{2}{3} - \frac{6}{7} \div \frac{5}{2} = \frac{2}{3} - \frac{6}{7} \cdot \frac{2}{5} = \frac{2}{3} - \frac{12}{35} = \frac{70}{105} - \frac{36}{105} = \frac{34}{105}.$$

Example 1.2.75

Simplify $\frac{2}{3x} - \frac{6y}{7} \cdot \frac{1}{2x}$.

Solution.

$$\frac{2}{3x} - \frac{6y}{7} \cdot \frac{1}{2x} = \frac{2}{3x} - \frac{6y \cdot 1}{7 \cdot 2x} = \frac{2}{3x} - \frac{6y}{14x} = \frac{2}{3x} - \frac{3y}{7x} = \frac{14 - 9y}{21x}.$$

Note that we cannot cancel part of the 14 with part of the 21 because 14 is not a factor of the numerator. Similarly, we cannot cancel part of the 9 with part of the 21 because 9 is not a factor of the numerator.

Example 1.2.76

Simplify $\frac{2x}{3y} + \frac{\frac{3x}{4y}}{\frac{5y}{6x}}$

Solution. We can simplify the division of fractions before simplifying addition:

$$\frac{2x}{3y} + \frac{\frac{3x}{4y}}{\frac{5y}{6x}} = \frac{2x}{3y} + \frac{3x \cdot 6x}{4y \cdot 5y} = \frac{2x}{3y} + \frac{18x^2}{20y^2} = \frac{2x}{3y} + \frac{9x^2}{10y^2}.$$

Now we need a common denominator to simplify addition of fractions:

$$\frac{2x}{3y} + \frac{9x^2}{10y^2} = \frac{2x \cdot 10y}{3y \cdot 10y} + \frac{9x^2 \cdot 3y}{10y^2 \cdot 3y} = \frac{20xy}{30y^2} + \frac{27x^2y}{30y^2} = \frac{20xy + 27x^2y}{30y^2}.$$

Finally, we can factor out y in the numerator, and having factors of y in both the numerator and denominator allows us to cancel y:

$$\frac{20xy + 27x^2y}{30y^2} = \frac{y(20x + 27x^2)}{30y^2} = \frac{20x + 27x^2}{30y}.$$

1.2.7 Clarifications and details

Let's clarify some details about fractions. It will be easier to discuss this using a concrete example, but the discussion here applies in general. What we discuss will lead to important clarifications and expectations. Along the way, we discuss convenient places to stop working on a problem (it's always good to know when to *stop* and not work further than you have to!), because some formats of writing will be easier to work with in algebra than other formats.

Note 1.2.77 First, we note that when writing a fraction whose value is negative, it is equivalent to present the fraction with the negative sign in the numerator, or in the denominator, or in front of the entire fraction. However, it is *not* equivalent to present the negative sign in both the numerator and denominator.

Example 1.2.78

Writing $\frac{4}{-7}$ and $\frac{-4}{7}$ and $-\frac{4}{7}$ are all legitimate ways to write the same negative fraction. You can check in a calculator that $4 \div -7$ and $-4 \div 7$ and $-(4 \div 7)$ all give the same decimal value.

However, $\frac{-4}{-7}$ is not the same as the other three ways of writing the negative fraction. Instead, $\frac{-4}{-7} = \frac{4}{7}$. You can check in a calculator that $-4 \div -7$ gives the same decimal value as $4 \div 7$. Alternatively, you can note that a negative divided by a negative results in a positive, or follow this cancellation of factors:

$$\frac{-4}{-7} = \frac{-1 \cdot 4}{-1 \cdot 7} = \frac{4}{7}.$$

Suppose somone wrote $2\frac{3}{5}$. Then:

- Someone might have written $2\frac{3}{5}$ with the intent of writing a **mixed fraction**. The *mixed* part of the name refers to the fact that there is a non-fraction part (2) and a fraction part ($\frac{3}{5}$) mixed together. Using the standard procedure for converting a mixed fraction into an improper fraction, the value of $2\frac{3}{5}$ is $\frac{2\cdot5+3}{5}$ which simplifies to $\frac{13}{5}$.

  This is a good time to mention that writing $2\frac{3}{5}$ to mean a mixed fraction is actually shorthand for $2 + \frac{3}{5}$. In fact, in Example 1.2.39 we had simplified $2 + \frac{3}{5}$ to get $\frac{13}{5}$.

- An entirely different view is possible. Someone might have written $2\frac{3}{5}$ to mean $2 \cdot \frac{3}{5}$, since a missing operation symbol is actually a hidden multiplication sign. In Example 1.2.60, we simplified $2 \cdot \frac{3}{5}$ to get $\frac{6}{5}$. Please note that $\frac{6}{5}$ is not the same as $\frac{13}{5}$. The two interpretations that we already have lead to different results: the mixed fraction intent leads to $\frac{13}{5}$ while treating a missing operation symbol as multiplication leads to $\frac{6}{5}$.

- We should bring up a third possibility, though what we are about to bring up is a common error of writing. Suppose we were in a situation where someone needs to start with the fraction $\frac{3}{5}$ and multiply the top and bottom by 2. (A situation where this might happen is when we are tasked with simplifying $\frac{3}{5} + \frac{7}{10}$, and we'd need a common denominator to simplify the addition of fractions.) In a situation where someone needs to start with the fraction $\frac{3}{5}$ and multiply the top and bottom by 2, they might just write a single 2 next to the fraction, which would look like this: $2\frac{3}{5}$. However, this is not a correct way to write what the writer intends. With this incorrect writing the writer would usually recover and correct for this in the next step by writing $\frac{6}{10}$ the result of actually multiplying by 2 on top and bottom. You can use a calculator to check that $\frac{6}{10}$ is not equal to either $\frac{13}{5}$ or $\frac{6}{5}$. In fact, $\frac{6}{10}$ is equal to $\frac{3}{5}$, which makes sense because of what the Equal Fractions formula tells us.

  When the desire is to multiply by two on top and on bottom, there are two ways to do this that are clear and unambiguous:

  1. Write the "times 2" both on top *and* on bottom: $\frac{3}{5} = \frac{2\cdot3}{2\cdot5} = \frac{6}{10}$.
  2. Write *neither* the 2 on top *nor* the 2 on bottom, but perform and simplify these actions nonetheless: $\frac{3}{5} = \frac{6}{10}$

  But what we *really* must avoid doing is writing a *single* 2 to mean that we are multiplying by 2 on top and bottom. So writing $2\frac{3}{5}$ cannot mean for us that we are multiplying by 2 on top and on bottom, even if there's a lot of spacing between the 2 and the fraction. If you wish to write what you're multiplying by, just be sure to write two copies: one on top and one on bottom.

Maybe it's shocking to see that the same excerpt of writing $2\frac{3}{5}$ can be interpreted two different ways leading to two different numbers. This discussion leads to some important ideas. First, let's present an expectation regarding our final answers for fractions:

Principle 1.2.79 Expectation for Fraction Format.

We expect our final answers for fractions to be in improper fraction form whenever possible. Unless specifically requested by the question, we will never present an answer as a mixed fraction. Whenever we write a mixed fraction, we should actually mention *that our writing should be interpreted as a mixed fraction.*

You may be more used to mixed fractions and feel offended that the expectation avoids mixed fractions and favors improper fractions instead. But improper fractions are easier to work with in algebra, because it is hard to directly add, subtract, multiply, or divide mixed fractions. (In fact, when given a mixed fraction and a second mixed fraction, to perform any operation, we first need to convert both mixed fractions into improper fractions.) Therefore, it's actually convenient to leave any fractions as improper fractions in their final answers.

Example 1.2.80

Simplify $2\frac{3}{5} \cdot 3\frac{7}{10}$ where we interpret $2\frac{3}{5}$ and $3\frac{7}{10}$ as mixed fractions.

Solution. We rewrite the first mixed fraction $2\frac{3}{5}$ as $\frac{13}{5}$. We rewrite the second mixed fraction $3\frac{7}{10}$ as $\frac{37}{10}$. Then, the product is

$$\frac{13}{5} \cdot \frac{37}{10} = \frac{481}{50}.$$

By going through this example, I hope you really saw that it's preferrable to work with improper fractions anyway. In this question, we only had the step of completing one opertaion, but in a larger problem, we might need to take the result and do something further. It would be more convenient to do any further task working the improper fraction $\frac{481}{50}$, so that's why we don't really bother converting this into a mixed fraction.

Note 1.2.81 Perhaps the history of calling $\frac{481}{50}$ an **improper fraction** is what makes us a little uneasy. We use this language here (as opposed to the fraction $\frac{47}{50}$) because the numerator is larger than the denominator in $\frac{481}{50}$. I guess I'd call $\frac{x}{y+3}$ an "improper fraction" only because I wouldn't call $\frac{x}{y+3}$ a mixed fraction, but because there are variables, I don't really know if x is larger than $y+3$ or not. However, $\frac{x}{y+3}$ behaves like it needs to, and we can apply our fraction operations as described. For example,

$$\frac{x}{y+3}+\frac{z-6}{y+3}=\frac{x+z-6}{y+3}.$$

Just like improper fractions are easier to work with than mixed fractions, improper fractions are are generally easier to work with than decimal represtations. Therefore, it is often more convenient for future steps to leave $\frac{9}{8}$ as is instead of presenting this as 1.125, and similarly, it is more convenient for us (and less work!) to leave $\frac{3}{8}$ as it is instead of presenting this as 0.375.

Principle 1.2.82 Well then, why have mixed fractions or decimals in the first place?

In everyday settings, people are used to seeing mixed fractions. Therefore, at the very end of an applied problem, when we are done and we are certain that the value will not be needed in further work, it would be reasonable to state a final answer (in this limited situation) as a mixed fraction.

The role of decimals is somewhat similar: when you have a final answer that is an improper fraction but you need to know a "ballpark figure" representation of that number, a decimal is helpful. For example, I don't really have a good sense of the value of $\frac{2711348695}{182638791}$*, but I can put this into a calculator to see the value is close to* 14.8454152601*, a decimal representation.*

Changing gears, there's one thing that we haven't mentioned yet that's worth clarifying: The top and bottom of a fraction are each in what I'd like to call hidden parentheses.

Principle 1.2.83 Fractions have hidden parentheses.

The top and bottom of a fraction are each in hidden parentheses.

Let's clarify what this means, and then talk about how this connects to the Order of Operations. The connection to the Order of Operations is really important in applied settings when a fraction-based expression on paper needs to be input into a calculator to get a decimal approximation.

Example 1.2.84

In the fraction $\frac{2+3}{5-1}$, the top is $2+3$ and the bottom is $5-1$. The top and bottom are each in hidden parentheses, so we can rewrite this as $\frac{(2+3)}{(5-1)}$. We can also write this as $(2+3)\div(5-1)$, but cannot write this as $2+3\div 5-1$ due to the Order of Operations. Therefore, to enter $\frac{2+3}{5-1}$ into a calculator, type

$(2+3) \div (5-1)$, including the parentheses.

Example 1.2.85

From a general chemistry class, a student got a value of $\frac{3.6 \cdot 10^{-5}}{(4.0 \cdot 10^{-2})(2.5 \cdot 10^{-2})}$ on paper. How should this be entered on a calculator?

Solution. There are two sets of parentheses in the denominator already, but we should insert a set of parentheses around the entire denominator. We should also insert a set of parentheses around the numerator. Therefore, we should enter $(3.6 \cdot 10^{-5}) \div ((4.0 \cdot 10^{-2})(2.5 \cdot 10^{-2}))$ into the calculator.

Example 1.2.86

If you deposited 100 dollars every month for 10 years into an account that ears 6% interest compounded monthly, we can compute what the balance will be at the end of 10 years by applying an equation in finance, and we would obtain

$$100 \cdot \frac{(1+\frac{0.06}{12})^{(12)(10)} - 1}{\frac{0.07}{12}}$$

as the balance. How should this be input into the calculator so that we can really understand what the balance will be?

Solution. Insert the entire numerator and denominator of the fraction in parentheses. In a calculator, we would type $100 * ((1+(0.06/12))^((12)(10)) - 1)/(0.06/12)$.

This fact that the entire numerator and the entire denominator are each in hidden parentheses, we can enhance and clarify what we've said about when we can cancel in fractions.

Example 1.2.87

Simplify $\frac{4(x+2)}{x+2}$

Solution. Let's draw in the hidden parentheses to surround the entire denominator.

$$\frac{4(x+2)}{x+2} = \frac{4(x+2)}{(x+2)} = \frac{4}{1} = 4$$

We canceled the factor of $(x+2)$ in the numerator and denominator

If it still feels hard to view $(x+2)$ in the denominator, that's okay! I think that means you're doing a great job of *really* digging into the idea that factors are the pieces of a multiplication (and where *is* multiplication in the denominator)? Let's rewrite a solution, noting that multiplying anything by 1 doesn't change a value, so $1 \cdot (x+2) = x+2$.

$$\frac{4(x+2)}{x+2} = \frac{4 \cdot (x+2)}{1 \cdot (x+2)} = \frac{4}{1} = 4$$

In fact, writing this level of detail explains why we have a 1 instead of a 0 in the denominator after the cancellation happened.

Example 1.2.88

Simplify $\frac{(x+8)^2}{x+8}$

Solution. Recall that squaring a thing means to multiply the thing by itself, even if that ``thing'' is longer to write, like $x+8$, so $(x+8)^2 = (x+8) \cdot (x+8)$. Let's make that replacement in the numerator, and then also surround the denominator in parentheses, making those hidden parentheses visible.

$$\frac{(x+8)^2}{x+8} = \frac{(x+8) \cdot (x+8)}{1 \cdot (x+8)} = \frac{x+8}{1} = x+8$$

In the second to last expression, which is the moment we just finished cancellation, we could have written $\frac{(x+8)}{1}$ instead of $\frac{x+8}{1}$. However, since the numerator was $(x+8)$, I wanted to practice the principle that the numerator of a fraction is always in a hidden set of parentheses, but "backwards" in a sense: just as much as we can draw in the hidden parentheses around the entire top or entire bottom, we can remove parentheses that surround the entire top or entire bottom.

Example 1.2.89

Simplify $\frac{x+8}{(x+8)^2}$

Solution.

$$\frac{x+8}{(x+8)^2} = \frac{(x+8)}{(x+8) \cdot (x+8)} = \frac{1}{(x+8)} = \frac{1}{x+8}$$

We canceled the factor of $(x+8)$ in the numerator and denominator. One new thing to note in this question: after cancellation, what's left is the factor of $(x+8)$ in the denominator, and we just removed the parentheses that surrounded the entire denominator. However, a common error that occurs right at the moment of cancellation is that it's temtping to put the factor of $(x+8)$ in the numerator. Note that after cancellation, the $(x+8)$ factor that didn't cancel was in the denominator, and shouldn't move up to the numerator.

Warning 1.2.90 After cancelling the same factor on top and bottom of a fraction, anything that remains uncanceled should stay in the same part of the fraction. Factors that are in the denominator should remain in the denominator, and shouldn't move up to the numerator. Factors that are in the numerator should remain in the numerator, and shouldn't move down to the denominator.

Example 1.2.91

Simplify $\frac{(x+3)(x+5)}{(x+3)(x+5)(x+7)}$

Solution. $\frac{(x+3)(x+5)}{(x+3)(x+5)(x+7)} = \frac{1}{(x+7)} = \frac{1}{x+7}$

Principle 1.2.92 Slanted fraction bars.

Generally speaking, we write fractions with a horizontal bar, and mostly avoid fractions written with a slanted bar. The reason for preferring horizontal bars is to avoid potential ambiguity.

Example 1.2.93

Let's examine a specific example of potential ambiguity. Suppose we have have to write the fraction $\frac{20}{2x}$. Written with a horizontal fraction bar (as shown in the typing we gave), there is no ambiguity. Written with a slanted fraction bar, our font doesn't quite help when we write $20/2x$ so for the sake of clarity, here is a figure:

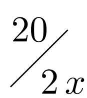

Figure 1.2.94 A picture of ambuiguous writing

In the figure, the placement of x is ambiguous: is the x part of the denominator, or is it separate from the fraction? If this is separate from the fraction, then this would be $\frac{20}{2} \cdot x$ which is not equal to $\frac{20}{2x}$. There are two ways to be convinced of this:

- First, we can take $\frac{20}{2} \cdot x$ and simplify $\frac{20}{2} \cdot x = 10x$, which is not equal to $\frac{20}{2x} = \frac{10}{x}$.
- Alternatively, we can pick a value for x and see what happens. It will be convenient here to pick $x = 5$. Then, $\frac{20}{2x} = \frac{20}{2\cdot 5} = \frac{20}{10} = 2$ while $\frac{20}{2} \cdot x = (20 \div 2) \cdot 5 = 10 \cdot 5 = 50$.

In the writing of $\frac{20}{2x}$ it is clear that $2x$ is the denominator, and thus it makes sense that the $2x$ is in a hidden set of parentheses. That is, $\frac{20}{2x}$ can be written $\frac{20}{(2x)}$ or even $20 \div (2 \cdot x)$ but none of these three expressions are equal to $(20 \div 2) \cdot x$.

Here is another example which also relies on our understanding of the hidden parentheses and how to simplify multiplication involving fractions:

Example 1.2.95

There comes a time when we will often need to write a fraction multiplied by a variable, like $\frac{3}{7}x$. When writing the fraction with a slanted fraction bar, a very typical way this gets written by hand is slightly different from the figure shown earlier, and instead tends to look like this:

$${}^{3}\!/_{7}\ x$$

Figure 1.2.96 Another picture of ambuiguous writing

There may be some variation in exactly where someone places the x. If the x were lower, perhaps this would mean $\frac{3}{7x}$. If the x were slightly higher, perhaps this would be interpreted as $\frac{3}{7^x}$ where x is an exponent. However, if we really need $\frac{3}{7}x$, then we should write this with a horizontal fraction bar.

To be clear, $\frac{3}{7}x$ and $\frac{3}{7x}$ and $\frac{3}{7^x}$ are all different expressions that are not equal to each other. In the case of $\frac{3}{7}x$, let's apply what we have learned about multiplication involving fractions:

$$\frac{3}{7}x = \frac{3}{7} \cdot x = \frac{3}{7} \cdot \frac{x}{1} = \frac{3x}{7}.$$

Note that in the last expression, the numerator contains x, while in $\frac{3}{7x}$ and $\frac{3}{7^x}$, the x appears somewhere in the denominator. Of course, $\frac{3}{7x}$ and $\frac{3}{7^x}$ are different because $7x$ and 7^x are different: for example $7 \cdot 2 = 14$ while $7^2 = 49$.

1.2.8 Applications Revisited

Let's revisit the applications we introduced at the beginning of this section.

Example 1.2.97

A baking recipe calls for $5\frac{1}{4}$ cups of flour. The recipe says that it serves 10. You are hosting a party where 35 people will attend, so you need to make $3\frac{1}{2}$ times the recipe. How many cups of flour do you need?

Solution. The original recipe needs $5\frac{1}{4}$ cups of flour, but we need to make $3\frac{1}{2}$ times the recipe. We have to multiply both of these numbers, which (at least through context) we can see are both mixed fractions. We convert the two mixed fractions into improper fractions: The mixed fraction $5\frac{1}{4}$ is $\frac{21}{4}$, and the mixed fraction $3\frac{1}{2}$ is $\frac{7}{2}$. Then, we multiply the two improper fractions:

$$\frac{21}{4} \cdot \frac{7}{2} = \frac{147}{8}.$$

Now, given that the problem is about baking, it makes a lot of sense to convert our final answer into a mixed fraction, since that's how measuring cups are typically used: We need $18\frac{3}{8}$ cups of flour, as a mixed fraction. As a reminder of what mixed fraction is, if we wanted, we could actually write this quantity unambigously by writing $18 + \frac{3}{8}$ cups of flour.

Example 1.2.98

You have a two-by-four piece of lumber that measures $2\frac{5}{8}$ feet long and another that is $1\frac{3}{4}$ feet long. You will glue the two pieces together to create one leg of a nightstand. But you'll need to go to the store to make three other legs of the same height. How long is each leg?

Solution. Because we are gluing one piece of lumber to another, we need to add their measurements. From context, both numbers provided are mixed fractions, so we convert them: The mixed fraction $2\frac{5}{8}$ is $\frac{21}{8}$, and the mixed fraction $1\frac{3}{4}$ is $\frac{7}{4}$. Then, we add the two improper fractions:

$$\frac{21}{8} + \frac{7}{4} = \frac{21}{8} + \frac{14}{8} = \frac{35}{8}.$$

Since we needed to simplify the addition of fractions, we first got both fractions to have a common denominator of 8, which we achieved by leaving the first fraction alone, and multiplying the numerator and denominator of the second fraction by 2. Each leg will be $\frac{35}{8}$ feet long. Now, given that the problem is about measuring lumber, it makes a lot of sense to convert our final answer into a mixed fraction, since that's how lumber is typically measured: Each leg will be $4\frac{3}{8}$ feet long, as a mixed fraction.

Example 1.2.99

A swimming pool is being filled at a rate of $8\frac{1}{4}$ gallons per hour. How many gallons of water will be in the pool after $2\frac{1}{4}$ hours?

Solution. Because we are filling the pool at a rate of $8\frac{1}{4}$ gallons per hour, and we are filling it for $2\frac{1}{4}$ hours, we need to multiply these two numbers. From context, both numbers provided are mixed fractions, so we convert them: The mixed fraction $8\frac{1}{4}$ is $\frac{33}{4}$, and the mixed fraction $2\frac{1}{4}$ is $\frac{9}{4}$. Then, we multiply the two improper fractions:

$$\frac{33}{4} \cdot \frac{9}{4} = \frac{297}{16}.$$

The pool will have $\frac{297}{16}$ gallons of water, or $18\frac{8}{16}$ gallons of water as a mixed fraction. Note that when we simiplified $\frac{33}{4} \cdot \frac{9}{4}$, we need to multiply the two denominators together (even though they matched).

Example 1.2.100

A car is traveling at a speed of 60 miles per hour. How far will the car travel in $4\frac{1}{2}$ hours?

Solution. Because we are traveling at a rate of 60 miles per hour, and we are traveling for $4\frac{1}{2}$ hours, we need to multiply these two numbers. From context, the first number is not a mixed fraction, but the second number is a mixed fraction, so we convert it: The mixed fraction $4\frac{1}{2}$ is $\frac{9}{2}$. Now, we are trying to compute $60 \cdot \frac{9}{2}$, and to resolve the awkwardness of multiplying a non-fraction by a fraction, we will turn 60 into $\frac{60}{1}$.

$$60 \cdot \frac{9}{2} = \frac{60}{1} \cdot \frac{9}{2} = \frac{540}{2} = 270.$$

The car will travel 270 miles.

Example 1.2.101

A group of 5 friends go out to dinner. The bill comes to \$48.75. If they split the bill evenly, how much does each person pay?

Solution. We can think of 48.75 as the mixed fraction $48\frac{75}{100}$ or by reducing the fraction, we can say this is the mixed fraction $48\frac{3}{4}$, and for the sake of nice computation, as an improper fraction, this is $\frac{195}{4}$. We need the value of $\frac{195}{4}$ to be a cost shared by five people equally, so we should divide this fraction by 5. We will actually make our very next step to change the 5 into $\frac{5}{1}$.

$$\frac{195}{4} \div 5 = \frac{195}{4} \div \frac{5}{1} = \frac{195}{4} \cdot \frac{1}{5} = \frac{195}{20} = \frac{39}{4}.$$

Each person's share is $\frac{39}{4}$ dollars, and this as a mixed fraction is $9\frac{3}{4}$ dollars and as a decimal is 9.75 dollars.

Example 1.2.102

A recipe calls for $\frac{3}{4}$ cup of sugar. You have a $\frac{1}{3}$ cup measuring cup. How many $\frac{1}{3}$ cups of sugar do you need to use to get the correct amount?

Solution. We need to find the value of $\frac{3}{4} \div \frac{1}{3}$, which is the same as $\frac{3}{4} \cdot \frac{3}{1}$.

$$\frac{3}{4} \div \frac{1}{3} = \frac{3}{4} \cdot \frac{3}{1} = \frac{9}{4}.$$

We need to fill the one-third cup measuring cup that we have $\frac{9}{4}$ times, or as a mixed fraction, $2\frac{1}{4}$ times. So, to fulfill what the recipe wants, we should fill our $\frac{1}{3}$ cup measuring cup full two times, and then do our best to estimate filling this measuring cup to one-fourth full.

Example 1.2.103

A recipe calls for $\frac{3}{4}$ cup of sugar. You put in $\frac{1}{3}$ cup of sugar by accident. How much more sugar do you need to put in to get back on track in following the recipe?

Solution. Since we already put in $\frac{1}{3}$ cup of sugar, we need to find out what's left of the $\frac{3}{4}$ cup of sugar that we haven't put in yet. That is, we need to find the value of $\frac{3}{4} - \frac{1}{3}$.

$$\frac{3}{4} - \frac{1}{3} = \frac{9}{12} - \frac{4}{12} = \frac{5}{12}.$$

We need to put in $\frac{5}{12}$ cup of sugar to get back on track.

1.2.9 Summary

1. We can take a fraction and create a fraction of the same value by multiplying by the same quantity on the top and bottom.

2. Undoing the multiplying of the same thing on top/bottom is the cancelling of a common factor. We can only cancel common factors, not common terms. (Recall that **factor** refers to the pieces of **multiplication**.)

3. To simplify adding or subtracting fractions, we need a common denominator first.

4. To multiply fractions, multiply straight across whether we have a common denominator or not.

5. To divide two fractions, note that dividing by a fraction is the same thing as multiplying by its reciprocal.

6. We stick to improper fractions as much as possible because it is much easier to perform operations on improper fractions, and only write a mixed fraction or

decimal when an application calls for it.

7. The entire numerator of a fraction is in a set of parentheses, which can be written in explicitly, or be "hidden." Similarly, the denominator of a fraction is also in a set of parentheses, which can be written in explicitly, or be "hidden." This is important to keep in mind when entering an answer that has fractions into a calculator: ignoring this may lead to *very* different decimal value than intended.

8. The fact that both the top and bottom can each be thought of as being in a hidden set of parentheses also enhances our understanding of how we can cancel in fractions: it still remains the case that we can only cancel common *factors*, but now we can see that the *entire* top and *entire* bottom can be considered factors through this understanding of hidden parentheses.

1.2.10 Exercises

1. Practice rewriting fractions using the Equal Fractions formula. Pay attention to unintended meanings of notation in your work, as described in Section .

   (a) What fraction is equal to $\frac{3}{4}$ but has 20 in the denominator?
   (b) What fraction is equal to $\frac{2}{5}$ but has 20 in the denominator?
   (c) What fraction is equal to $\frac{7}{10}$ but has 100 in the denominator?
   (d) What fraction is equal to $\frac{5}{6}$ but has 12 in the denominator?
   (e) What fraction is equal to $\frac{9}{8}$ but has 32 in the denominator?
   (f) What fraction is equal to $\frac{11}{12}$ but has 36 in the denominator?
   (g) What fraction is equal to $\frac{2}{x}$ but has $3x$ in the denominator?
   (h) What fraction is equal to $\frac{5}{y}$ but has $20y$ in the denominator?
   (i) What fraction is equal to $\frac{7}{2a}$ but has $10a$ in the denominator?
   (j) What fraction is equal to $\frac{x}{3}$ but has 12 in the denominator?
   (k) What fraction is equal to $\frac{3}{2b}$ but has $14b$ in the denominator?
   (l) What fraction is equal to $\frac{y}{4z}$ but has $12z$ in the denominator?
   (m) What fraction is equal to $\frac{3a}{10}$ but has 50 in the denominator?
   (n) What fraction is equal to $\frac{5m}{6n}$ but has $30n$ in the denominator?

2. Simplify the following expressions.

   (a) $\frac{3}{2} + \frac{2}{5}$.
   (b) $\frac{5}{6} + \frac{1}{3}$.
   (c) $\frac{7}{8} - \frac{3}{8}$.
   (d) $\frac{9}{10} - \frac{2}{5}$.
   (e) $\frac{2}{3} \cdot \frac{9}{4}$.

(f) $\frac{5}{7} \cdot \frac{14}{15}$.
(g) $\frac{3}{4} \div \frac{9}{16}$.
(h) $\frac{10}{21} \div \frac{5}{14}$.
(i) $\frac{1}{2} + \frac{2}{3} \cdot \frac{3}{4}$.
(j) $\frac{5}{6} - \frac{2}{5} \cdot \frac{3}{2}$.
(k) $\frac{3}{4} + \frac{5}{8} \div \frac{5}{16}$.
(l) $\frac{7}{9} - \frac{3}{10} \cdot \frac{5}{6}$.
(m) $(\frac{3}{4} + \frac{1}{8}) \cdot \frac{2}{3}$.
(n) $(\frac{5}{6} - \frac{1}{3}) \div \frac{1}{2}$.
(o) $(\frac{7}{10} + \frac{2}{5}) \cdot \frac{3}{7}$.
(p) $(\frac{9}{8} - \frac{5}{16}) \div \frac{3}{4}$.
(q) $\frac{3}{5} + \frac{2}{3} - \frac{4}{15}$.
(r) $\frac{7}{8} \cdot \frac{2}{5} + \frac{3}{10}$.
(s) $\frac{4}{9} + \frac{2}{3} \div \frac{8}{27}$.
(t) $\frac{5}{12} - \frac{3}{16} + \frac{7}{24}$.

3. Simplify the following expressions.

(a) $\frac{x}{2} + \frac{x}{3}$.
(b) $\frac{2a}{5} - \frac{a}{10}$.
(c) $\frac{3}{x} \cdot \frac{2}{5}$.
(d) $\frac{4}{y} \div \frac{2}{3y}$.
(e) $\frac{2}{3}x + \frac{x}{4} \cdot \frac{3}{2}$.
(f) $\frac{a}{6} + \frac{3a}{4} - \frac{a}{12}$.
(g) $\frac{y}{5} - \frac{2y}{15} \cdot \frac{3}{2}$.
(h) $\frac{m}{8} + \frac{2m}{3} \div \frac{4}{9}$.
(i) $(\frac{3}{4}x + \frac{x}{8}) \cdot \frac{2}{3}$.
(j) $(\frac{a}{5} - \frac{2a}{15}) \div \frac{a}{3}$.
(k) $(\frac{y}{6} + \frac{y}{4}) \cdot \frac{3}{2}$.
(l) $(\frac{2m}{7} - \frac{m}{14}) \div \frac{m}{21}$.
(m) $\frac{x}{2} + \frac{2x}{3} - \frac{x}{6}$.
(n) $\frac{3a}{4} \cdot \frac{2}{3a} + \frac{a}{2}$.
(o) $\frac{y}{3} + \frac{2y}{9} \div \frac{y}{6}$.
(p) $\frac{m}{5} - \frac{3m}{10} + \frac{7m}{20}$.
(q) $\frac{2x}{3} + \frac{x}{4} - \frac{3x}{8} \cdot \frac{2}{3}$.
(r) $\frac{5a}{6} - (\frac{a}{4} + \frac{a}{12}) \div \frac{a}{3}$.

(s) $\frac{3y}{8} + \frac{y}{6} - \frac{y}{4} \cdot \frac{9}{2}$.

(t) $(\frac{2m}{9} + \frac{m}{6} - \frac{m}{18}) \cdot \frac{3}{2}$.

4. A runner completed $\frac{2}{3}$ of a marathon in the morning and $\frac{1}{6}$ in the evening. How much of the marathon did the runner complete in total?

5. A carpenter cuts a piece of wood $3\frac{1}{2}$ feet long from a board that is $10\frac{1}{4}$ feet long. How much of the board remains?

6. You drink $\frac{3}{4}$ of a liter of water during a workout, and then another $\frac{2}{5}$ of a liter after. How much water did you drink altogether?

7. Your health routine requires drinking $1\frac{1}{4}$ liters of protein shake a day. You started the day by having $\frac{2}{3}$ liter of protein shake. How much should you have in the rest of the day?

8. A farmer harvested $\frac{7}{8}$ of a field of corn one week and $\frac{3}{10}$ of the field the next. How much of the field is harvested in total?

9. A rope is 12 meters long. You cut off a piece that is $\frac{5}{6}$ of a meter and another piece that is $\frac{7}{12}$ of a meter. How much rope is left?

10. You have $\frac{3}{4}$ of a pizza, and your friend has $\frac{5}{8}$ of the same size pizza. If you combine them, how much pizza do you have altogether?

11. A car uses $\frac{3}{5}$ of a tank of gas on one trip and $\frac{2}{7}$ of a tank on another trip. How much of a tank of gas is used altogether?

12. A cyclist rides $12\frac{1}{2}$ miles on Monday and $8\frac{3}{4}$ miles on Tuesday. How many miles did the cyclist ride in total?

13. A juice recipe requires $\frac{2}{3}$ of a cup of orange juice and $\frac{3}{5}$ of a cup of pineapple juice. How many cups of juice are needed in total?

14. A box of nails weighs $\frac{7}{8}$ of a pound. If you buy 4 boxes, how many pounds of nails do you have?

15. A swimming pool is being filled at a rate of $\frac{2}{3}$ of a gallon per minute. How many gallons of water will be added in 15 minutes?

16. A gardener uses $\frac{3}{4}$ of a bag of soil for one planter and $\frac{5}{6}$ of a bag for another. How many bags of soil does the gardener use altogether?

17. You buy $\frac{2}{3}$ of a pound of apples and $\frac{3}{4}$ of a pound of bananas. The apples cost \$1.20 per pound, and the bananas cost \$0.80 per pound. What is the total cost?

18. A skateboarder rides $\frac{5}{6}$ of a mile in the morning, then twice that distance in the afternoon. How far does the skateboarder ride in total?

19. A painter mixes $\frac{3}{5}$ of a gallon of red paint with $\frac{7}{10}$ of a gallon of white paint. How many gallons of paint are mixed altogether?

20. A scientist has a solution that contains $\frac{7}{8}$ of a liter of chemical A and adds another $\frac{3}{4}$ of a liter of chemical B. What is the total volume of the solution?

21. A movie lasts $2\frac{1}{4}$ hours, and a documentary lasts $1\frac{2}{3}$ hours. How long would it take to watch both back-to-back?

22. A train travels $\frac{2}{3}$ of a mile each minute. How far will it travel in 45 minutes?

23. A recipe calls for $\frac{3}{4}$ of a cup of sugar. If you want to make $\frac{2}{5}$ of the recipe, how much sugar do you need?

1.3 Solving Equations

Objectives

In this section, we learn how to:

1. Solve linear equations where the variable is written once.
2. Solve linear equations where the variable is written more than once.
3. Write equal signs appropriately in work for simplifying expressions and solving equations.
4. Use equations to model applied situations, solve the equation constructed, and interpret the solution in the context of the situation.

1.3.1 Applications

This section teaches us how to solve various equations, carefully introducing strategies. The mathematical skills introduced will allow us to answer these and similar questions:

1. A car rental costs $40 per day plus a one-time insurance fee of $15. If the total cost is $135, how many days was the car rented?
2. A chemist is cooling a liquid that starts at 90°C and decreases at a rate of 2°C per minute. After how many minutes will the liquid reach 30°C?
3. A train leaves a station traveling at 60 miles per hour. Another train leaves the same station two hours later traveling at 80 miles per hour. After how many hours will the second train catch up?
4. A landscaper buys bags of soil for $3 each and a one-time delivery charge of $12. If the total cost is $51, how many bags of soil were purchased?
5. An Australian internet service provider offers a base plan for $30 per month plus $5 for each additional gigabyte of data. If someone paid $65 one month, how many gigabytes of extra data did they use?
6. A taxi ride costs $3 plus $2.50 per mile. The taxi is cash only and you have $28.50 available. How many miles can you travel?
7. A gym charges a membership fee of $50 plus $15 per month. How many months can someone be a member if they paid $200 in total?

1.3.2 Introduction to Equations

We are going to introduce equations, which are not the same as expressions. Before introducing what an equation is, recall from Definition 1.1.3 that an expression is a mathematical notation representing a number. For example 23 is an expression and $\frac{7}{5}$ is an expression and $4x + 5$ is also an expression.

Definition 1.3.1

An **equation** is a mathematical statement that two expressions are equal. For example, $3x + 2 = 11$ is an equation stating that the expression $3x + 2$ is equal to 11

The key thing to note about *one* equation is that it states in its writing that *two* expressions are equal. An equation is saying to us "this" and "that" might not look the same, but "this" and "that" actually have the same value.

Example 1.3.2

Is $2x + 3 = 11$ an equation or an expression?

Solution. $2x + 3 = 11$ is an equation because it is stating that the expression $2x + 3$ is equal to the expression 11.

Example 1.3.3

Is $4y - 5$ an equation or an expression?

Solution. $4y - 5$ is an expression because it is not stating that two expressions are equal. It is just one expression: the value of $4y - 5$ is one number, based on whatever the mystery value of y is.

Try it 1.3.4

Both expressions and equations are mathematical writing involving notation. How would you explain to someone how to spot the difference between an expression and an equation?

Example 1.3.5

Which of the following are equations and which are expressions?

1. $5x + 7 = 3$
2. $6y - 4$
3. $8 = 2z + 1$
4. $12 = 4 + x$
5. $9a + 5b$

6. $7 - 3c$

Note 1.3.6 The reason that it is important to know the difference between expressions and equations is that we will be doing different things with them.

1. For an expression, our primary task will be to *simplify* the expression we are given.
2. For an equation, our primary task will be to *solve* the equation we are given.

Warning 1.3.7 An equation is not the same as an expression. We cannot "solve" an expression. We can only "simplify" an expression. The action of *solving* will be language we reserve for equations.

The primary tool we have for solving equations is to apply the same action to both sides. To be more specific:

Apply the same action to both sides.

Given an equation, we can:

1. Add the same expression to both sides of the equation.
2. Subtract the same expression from both sides of the equation.
3. Multiply by the same expression on both sides of the equation.
4. Divide by the same non-zero expression on both sides of the equation.

1.3.3 Equations when the variable is written once

What we just described tells us what the typical steps we can do are. (There are some other things we can do, but let's wait to explain those until we need them.) The key thing to remember is that whatever we do to one side of the equation, we must do to the other side of the equation. Saying "do the same thing to both sides" tells us what we can do, but it doesn't really give us a lot of guidace on how to do it effectively. There are many kinds of equations we will see in this algebra class. I encourage you to look for patterns and ask "why does this work in this equation, but not in another equation?" That said, I think it's helpful to be completely transparent, and admit that there is a bit of an art to solving equations. Because of this, I want to share specific situations and explain specific strategies for these situations.

We are going to start by looking at equations where the variable is written once. Examples include equations like $x - 9 = 5$ and $7x + 2 = 29$. There are equations where the where the variable is written more than once such as $2x + 3 = 7x - 8$, and these will be discussed after.

Note 1.3.8 To begin, it will be helpful to slow down by writing in the writing the action we are applying to both sides as its own step, then simplifying both sides as a separate step. While this is more time-consuming (and we'll eventually stop doing it this way), skipping this and working too fast makes it impossible to describe certain common errors. To be able to point out what's going on, we need to start slow. I ask for your patience in doing it this way for now. Once the important points have been made, we'll make the process a little smoother.

Example 1.3.9

Solve the equation $x + 3 = 30$.

Solution. To undo the $+3$ on the left side of the equation, we can subtract 3 from both sides of the equation:

$$\begin{aligned} x + 3 &= 30 \\ x + 3 - 3 &= 30 - 3 \\ x &= 27 \end{aligned}$$

On the left side, the $+3$ and the -3 together simplify to $+0$, so we are left with just x. On the right side, $30 - 3$ simplifies to 27.

In the answer to this first example, you might have felt annoyed that we wrote the equation $x + 3 - 3 = 30 - 3$ as its own step. Writing this equation is exactly what Note 1.3.8 is addressing. While you may feel like skipping this step in the future, let's just do a couple more where we write this step out.

Example 1.3.10

Solve the equation $x - 3 = 30$.

Solution. We can add 3 to both sides of the equation.

$$\begin{aligned} x - 3 &= 30 \\ x - 3 + 3 &= 30 + 3 \\ x &= 33 \end{aligned}$$

Warning 1.3.11 Now, let's examine a new equation where a certain type of error tends to first occur from learners. Given the equation

$$x + 3 = 30$$

a common error is to end up at

$$x = 10.$$

Let's analyze this thorougly:

1. First, does the answer obtained $x = 10$ make sense? If we substitue 10 for x in the original equation, we get $10 + 3 = 30$, which is not true. So, $x = 10$ cannot be the correct answer.

2. While we already see that there is a problem, we haven't discussed how the error occurred. Let's try to figure that out. Here, it will be very helpful to write the action taken on both sides without simplifying right away, just as Note 1.3.8 talked about:

   $$x + 3 = 30$$

   From the original equation, based on the final answer that someone wrote, I would guess that they thought about dividing by 3 on both sides. Writing this without simplifying first looks like this:

   $$\frac{x+3}{3} = \frac{30}{3}.$$

   Here, it is true that the right side simplifies to 10. However, the expression on the left side says $\frac{x+3}{3}$ I would guess the learner "canceled" the 3 on top and bottom, but the 3 appearing in the numerator is a term, not a factor. Recall that we can only cancel factors, and not terms. It would be hard to analyze this error if we hadn't slowed the process down.

3. To stick to principles, we can divide by 3 on both sides. What happens is this:

   $$\begin{aligned} x + 3 &= 30 \\ \frac{x+3}{3} &= \frac{30}{3} \\ \frac{x+3}{3} &= 10 \end{aligned}$$

   The steps shown above are all valid, but we haven't successfully solved for x, since x is not by itself on one side of the equation. This goes to show that while it is permissible to divide by 3 on both sides, it is not helpful in solving this equation. Let's show a couple more examples of sucessfully solving equations, and then summarize what kind of actions are helpful: we don't want to just "do" steps to do them; we want to find steps that "lead towards" our goal of isolating the variable on one side of the equation.

Example 1.3.12

Solve the equation $3x = 30$.

Solution. We can divide both sides of the equation by 3.

$$\begin{aligned} 3x &= 30 \\ \frac{3x}{3} &= \frac{30}{3} \\ x &= 10 \end{aligned}$$

In this example, dividing both sides by 3 was helpful. This is because in the fraction $\frac{3x}{3}$ there is a factor of 3 in both the numerator and denominator, so we can cancel the 3s. In slightly more detail, we can think of $\frac{3x}{3}$ as $\frac{3 \cdot x}{3 \cdot 1}$, which shows more clearly that 3 is a factor in both the numerator and denominator.

Note 1.3.13 In both of the last two examples, we divided by 3. In one example, this was helpful, and in the other example, this was not helpful. I want you to notice that when we divide both sides by a quantity, we end up creating a fraction even though we sometimes work so fast that we don't write the fraction that Note 1.3.8 encourages us to write. That fraction is there, whether we write it or are too lazy to write it, and we need to carefully apply what we learned in the last section about fractions: we can only cancel common factors.

Example 1.3.14

Solve the equation $\frac{x}{3} = 30$.

Solution. We can think of the left side as $x \div 3$. To undo the division by 3, we can multiply both sides by 3. We can multiply both sides of the equation by 3.

$$\begin{aligned} \frac{x}{3} &= 30 \\ \frac{x}{3} \cdot 3 &= 30 \cdot 3 \\ x &= 90 \end{aligned}$$

In this example, multiplying both sides by 3 was helpful. Why? Of the three equations that were one right after another, the left side of the second equation is the expression $\frac{x}{3} \cdot 3$. From the last section, we can treat the product of a fraction and a non-fraction in the usual way to get $\frac{x}{3} \cdot \frac{3}{1}$. Whether you multiply across first and cancel the factors of 3 on top and bottom after, or note that the 3s on top and bottom in the separate fractions would eventually become factors in the combined fraction $\frac{x \cdot 3}{3 \cdot 1}$ and cancel right away, we're left with $\frac{x}{1}$ which simplifies to x.

Solving Equations by Undoing Operations.

A general principle that is helpful to keep in mind is that we can think of solving equations as undoing operations. In more detail:

1. Subtraction undoes addition.
2. Addition undoes subtraction.
3. Division undoes multiplication.
4. Multiplication undoes division.

Warning 1.3.15 It is tempting to divide on both sides by 7 in an equation like $x - 7 = 31$ or $x - 7 = 2$. However, division does not undo subtraction. (Instead, addition undoes subtraction.)

Similarly, it is tempting to divide on both sides by 7 in an equation like $x + 7 = 40$ or $x + 7 = 5$. However, division does not undo addition. (Instead, subtraction undoes addition.)

In all four equations that have been mentioned in this box, dividing both sides by 7 is permissible, but not helpful. What ends up happening (if we slow things down the way Note 1.3.8 encourages us to) is that we have a fraction where the 7 in the numerator is a term, not a factor, so we cannot cancel it.

Example 1.3.16

Solve the equation $x + 30 = 2$.

Solution 1.

$$\begin{aligned} x + 30 &= 2 \\ x + 30 - 30 &= 2 - 30 \\ x &= -28 \end{aligned}$$

Solution 2. Since we made the point about writing the action that occurs on both sides without simplyfing in Note 1.3.8 so that we could analyze a situation like we did in Warning 1.3.11, we can now speed things up a little and show a perfectly acceptable answer to this question which doesn't show the middles step.

$$\begin{aligned} x + 30 &= 2 \\ x &= -28 \end{aligned}$$

While both answers are fine, we will still have situations later in the book where writing the action but not simplifying yet (as described in Note 1.3.8) makes it possible for us to discuss what went wrong. The overall point here is not to criticize skipping steps. Instead, if someone's work goes from Equation 1 to Equation 3 while skipping the writing of Equation 2, but there was an error going from Equation 2 to Equation 3, then we need to actually slow down by writing in all the steps. Skipping work is fine when the work is correct. But if there is an error hiding somewhere inside skipped work, then it is really helpful to include all the steps to be able to pinpoint exactly where the error occured.

Example 1.3.17

Solve the equation $2x = 25$.

Solution 1.

$$\begin{aligned} 2x &= 25 \\ \frac{2x}{2} &= \frac{25}{2} \\ x &= \frac{25}{2} \end{aligned}$$

Solution 2. Let's also present a solution which skips showing the action, and instead shows the result of simplifying where possible.

$$\begin{aligned} 2x &= 25 \\ x &= \frac{25}{2} \end{aligned}$$

While the two answers showed a different amount of work (by skipping the writing of one equation), we were able to cancel the 2 on top and on bottom (since 2 was a factor on top and a factor on bottom), even if we didn't explicitly write the fraction in the second answer. In both answers, $\frac{25}{2}$ didn't simplify further. If an application called for it, we might present this number as a mixed fraction or a decimal, but recall that leaving $\frac{25}{2}$ as written is what we usually prefer in algebra.

Example 1.3.18

Solve the equation $8x = 5$.

Solution.

$$\begin{aligned} 8x &= 5 \\ x &= \frac{5}{8} \end{aligned}$$

Example 1.3.19

Solve the equation $3 = x - 20$.

Solution.

$$\begin{aligned} 3 &= x - 20 \\ 23 &= x \end{aligned}$$

This last example shows that it's okay to have the variable on the right side of the final equation.

EQUATIONS WITH SEVERAL OPERATIONS

So far, we have only seen equations where a single operation is being applied to the variable, and therfore we applied an action to both sides and simplified, and we were done. We only needed to apply one action to both sides.

Let's now look at a more complicated equation. For example, consider the equation $3x + 2 = 17$. While it may seems strange to you, I want to first point out a common error that learners make when solving this equation. Sometimes, as a first step, people will divide both sides by 3 and might not write the step in between that is discussed in Note 1.3.8, but we will:

$$\frac{3x+2}{3} = \frac{17}{3}$$

What usually happens is that people will skip writing this step, and then they will "cancel" the 3s on top and bottom, getting $x + 2 = \frac{17}{3}$. However, this is an error, because the 3 in the numerator is not a factor of the numerator. (Instead, the numerator has a term of $3x$ and a term of 2.)

Another error that sometimes occurs in solving the equation $3x + 2 = 17$ comes from writing ambiguously. What happens is that someone will say they are taking $3x + 2 = 17$ and dividing both sides by 3 by writing

$$\frac{3x}{3} + 2 = \frac{17}{3}.$$

While it is true that the 3s cancel in $\frac{3x}{3}$, the problem is that we shouldn't be writing $\frac{3x}{3} + 2$ in the first place. We have to ensure that we divdide the entire left side by 3, not just part of the left side.

Apply action to the entirety of each side.

- When adding the same quantity to both sides, add that quantity to the entire left side and the entire right side.
- When subtracting the same quantity from both sides, subtract that quantity from the entire left side and the entire right side.

- When multiplying the same quantity on both sides, multiply the entire left side and the entire right side by that quantity.
- When dividing both sides by the same non-zero quantity, divide the entire left side and the entire right side by that quantity.

In the case of dividing on both sides, the easiest way to follow Summary is to ensure that the fraction bar drawn in is wide enough for the entire left side and the entire right side.

Example 1.3.20

Solve the equation $3x + 2 = 17$.

Solution. To undo the $+2$ on the left side of the equation, we can subtract 2 from both sides of the equation:

$$\begin{aligned} 3x + 2 &= 17 \\ (3x + 2) - 2 &= (17) - 2 \\ 3x + 2 - 2 &= 17 - 2 \\ 3x &= 15 \end{aligned}$$

From the second equation to the third equation, we dropped the parentheses due to the order of operations. It's more work than most of us want to show to write $(3x + 2) - 2 = (17) - 2$ in the first place, but those parentheses are shown to emphasize following the advice of Summary . Now, we have the equation $3x = 15$. To undo the multiplication by 3, we can divide both sides by 3.

$$\begin{aligned} 3x &= 15 \\ \frac{3x}{3} &= \frac{15}{3} \\ x &= 5 \end{aligned}$$

Example 1.3.21

Solve the equation $4 + 9x = 34$.

Solution. To solve the equation $4 + 9x = 34$, we can start by isolating the term with the variable x. We can do this by subtracting 4 from both sides of the equation:

$$\begin{aligned} (4 + 9x) - 4 &= (34) - 4 \\ 4 + 9x - 4 &= 34 - 4 \end{aligned}$$

$$9x = 30$$

Now, we have the equation $9x = 30$. To undo the multiplication by 9, we can divide both sides by 9:

$$\frac{9x}{9} = \frac{30}{9}$$
$$x = \frac{10}{3}$$

Note that a common error in starting this problem is to "divide" both sides by 9 first. While this is permissible, it is not helpful. We end up with

$$\frac{4 + 9x}{9} = \frac{34}{9}$$

and in this new equation, the 9s on the left side do not cancel because the 9 in the numerator is not a factor of the numerator. Instead of dividing both sides by 9 as the first action, the action that helped was to subtract 4 on both sides.

Also note that we cannot take the starting equation $4 + 9x = 34$ and "simplify" the left side by writing $13x = 34$. The error here in simplifying like this is that we would not be following the order of operations. Note that multiplication takes precedence over addition: to read $4 + 9x$ we should think something like "We would take 9 and multiply it by x, whatever that is, to have the valu of $9x$, and the value of this $9x$ would be added to 4."

Try it 1.3.22

In each equation below, identify several possible first actions that someone may try to perform on both sides. (Try to identify at least two different things that people might try to do first.) For each of these first actions, carefully write the resulting equation that has no simplification occurring, as discussed in Summary . Then, discuss whether or not that first action is helpful or not.

- $5x + 4 = 19$
- $22x - 3 = 40$
- $7 + 6x = 31$
- $3x - 5 = 16$
- $4 - 5x = 6$

I hope that you tried this actively conisdering what has the potential to work on each equation above, and what seems to not work on each equation above. I hope that the explanations in the previous answers have given you some useful ways to think about and analyze what could create success in solving equations, and what seems like it just increases the complexity of the problem, instead of making a smaller problem. You might feel like you don't have a complete answer on exactly how to solve each equation you'd ever encounter, and that's totally okay! All I really wanted you to get out of the exploration was to make some predictions of what might work and what might not work. Building a little bit of intuition on your own will help make things go smoother when we introduce specific strategies. We're about to introduce a strategy right now, but in case you didn't try exploring what might and might not work in the five equations above, I strongly encourage you to do so before reading any further! Actively thinking about these things will make the strategy we're about to introduce go smoother:

Strategy 1.3.23 Solving an equation when the variable is written once.

- ***When does this strategy apply?*** This strategy applies in any equation we are asked to solve when the variable is written once.
- ***How to apply the strategy***
  1. Identify the side of the equation which contains the variable.
  2. On the side which contains the variable, starting from the variable and working outwards, identify the order in which the operations occur.
  3. Apply an action on both sides of the equation to undo the last operation to occur. Continue in this way by undoing the second-to-last operation and the third-to-last operation and so on.

In short, the strategy tells us to "undo the last operation that occurs".

Example 1.3.24

Solve the equation $5x - 3 = 27$.

Solution. We are being asked to solve an equation where the variable x is written just once, so Strategy 1.3.23 applies. The variable x is written on the left side of the equation. Analyzing the left side according to the order of operations, we see that 5 multiplied by x occurs first, which gives us $5x$ and then from $5x$ subtracting 3 occurs next. Since this finishes our analysis of reading the left side, we note that subtracitng 3 occured last. The strategy tells us to undo subtracting 3 first, so we add 3 to both sides.

$$\begin{aligned} 5x - 3 &= 27 \\ 5x &= 30 \end{aligned}$$

Now there is only one operation remaining. We undo the multiplication by 5 by dividing both sides by 5 leading to

$$x = 6.$$

Example 1.3.25

Solve the equation $4x - 11 = 2$.

Solution. We are being asked to solve an equation where the variable x is written just once, so Strategy 1.3.23 applies. The variable x is written on the left side of the equation. Analyzing the left side according to the order of operations, we see that 4 multiplied by x occurs first, which gives us $4x$ and then from $4x$ subtracting 11 occurs next. Since this finishes our analysis of reading the left side, we note that subtracitng 11 occured last. The strategy tells us to undo subtracting 11 first, so we add 11 to both sides.

$$\begin{aligned} 4x - 11 &= 2 \\ 4x &= 13 \end{aligned}$$

Now there is only one operation remaining. We undo the multiplication by 4 by dividing both sides by 4 leading to

$$x = \frac{13}{4}$$

Example 1.3.26

Solve the equation $12 = 10 + 3y$.

Solution. We are being asked to solve an equation where the variable y is written just once, so Strategy 1.3.23 applies. The variable appears on the right side. Analyzing the right side using the Order of Operations, the multiplication between the 3 and the y occurs first leading to $3y$, whatever the value of that is. Then, we add this value $3y$ to 10. Since adding $3y$ and 10 together happens last (we can switch the order of addition if it's convenient for our minds), we will undo adding 10 by making our first step to subtract 10 on both sides, leading to

$$2 = 3y.$$

The only remaining operation on the right is multiplication of 3 and y. We can undo the multiplication by 3 if we divide both sides by 3:

$$\frac{2}{3} = y.$$

Example 1.3.27

Solve the equation

$$3 + \frac{5x - 2}{10} = 7.$$

Solution. We are being asked to solve an equation where the variable x is written just once, so Strategy 1.3.23 applies. The variable x is written on the left side of the equation. Since the variable is on the left side, we analyze the left side according to the Order of Operations. The multiplication between 5 and x occurs first, represented as $5x$. Then from $5x$ we subtract 2 to get $5x - 2$. (Recall the entire numerator of a fraction is in a hidden set of) Then from $5x - 2$ we divide by 10 to get $\frac{5x-2}{10}$. Finally, the addition between 3 and $\frac{5x-2}{10}$ occurs last, so we will undo this addition by subtracting 3 from both sides first.

$$3 + \frac{5x-2}{10} = 7$$
$$\frac{5x-2}{10} = 4$$

Now the final operation on the left $\frac{5x-2}{10}$ is the division by 10, so we multiply both sides by 10:

$$5x - 2 = 40.$$

The final operation on the left is the subtraction of 2, so we add 2 to both sides:

$$5x = 42.$$

Finally, we divide both sides by 5 to isolate x:

$$x = \frac{42}{5}.$$

Example 1.3.28

Solve the equation

$$30 = \frac{3+4x}{5} + 20.$$

Solution. Strategy 1.3.23 applies. On the side with the variable, adding 20 occurs last, so we undo this first: we'll subtract 20 on both sides leading to the new equation

$$10 = \frac{3+4x}{5}.$$

In the new equation the last operation on the right side is the division by 5 which we undo by multiplying both sides by 5. We get

$$50 = 3 + 4x.$$

The final operation on the right is the addition of 3, so we subtract 3 from both sides:

$$47 = 4x.$$

Finally, we divide both sides by 4 to isolate x:

$$\frac{47}{4} = x.$$

Note, it is perfectly okay for the final equation to display the variable on the right side instead of the left side.

1.3.4 Comments and clarifications

Before moving on to equations where the variable is written more than once, let's make some comments and clarifications about what we've done so far.

Warning 1.3.29 Instead of applying an action on one side and the opposite (or "undoing") action on the other side, we must always apply the same action to both sides.

Example 1.3.30

In solving the equation $x - 10 = 25$, it would not be correct to make the next equation $x = 15$. How does someone get to this second equation? I would say that they added 10 to the left side (to undo the subtraction of 10) but then they subtracted 10 from the right side. Adding 10 on the left and subtracting 10 on the right is not the same action on both sides. The correct next step would be to add 10 to both sides, leading to $x = 35$.

Similar situations can occur when someone multiplies on one side and divides on the other side.

When solving equations, we are not simplifying an expression. We have seen two main tasks: simplifying an expression and solving an equation. Properly writing an answer for simplifying an expression involves writing equal signs, and properly writing an answer for solving an equation requires equal signs, but the exact locations of the equal signs "feels" really different, and definitely looks really different. As we dig into this explanation right now, I encourage you to keep in mind that the equal sign $=$ truly means that one expression is equal in value to another expression. We write an equal sign in between one expression and another to indicate that the two expressions have the same value.

When we are asked to simplify an expression, we can present our work "horizontally" or "vertically". Either way that we do it, we start with the given expression. Then, after writing an equal sign, we write a new expression that is equal in value to the previous expression. We can continue in this way, writing a new expression that is equal in value to the previous expression after each equal sign. We stop when we have an expression that cannot be simplified any further. It can feel like the equal sign means "and the next step is", but the equal sign doesn't mean "next step".

Instead, the equal sign means "the expression on the left has the same value as the expression on the right". As a coincidence, for simplifying expressions, the next step is to write an expression that is equal to the previous expression. Let's look at an example, and show the work both horizontally and vertically, then discuss.

Example 1.3.31

Simplify the expression $3+\frac{4}{5}\cdot 2-\frac{1}{3}$.

Solution 1. $3+\frac{4}{5}\cdot 2-\frac{1}{3}=3+\frac{4}{5}\cdot\frac{2}{1}-\frac{1}{3}=3+\frac{8}{5}-\frac{1}{3}=\frac{15}{5}+\frac{8}{5}-\frac{1}{3}=\frac{23}{5}-\frac{1}{3}=\frac{69}{15}-\frac{5}{15}=\frac{64}{15}$

Solution 2.

$$\begin{aligned} 3+\frac{4}{5}\cdot 2-\frac{1}{3} &= 3+\frac{4}{5}\cdot\frac{2}{1}-\frac{1}{3} \\ &= 3+\frac{8}{5}-\frac{1}{3} \\ &= \frac{15}{5}+\frac{8}{5}-\frac{1}{3} \\ &= \frac{23}{5}-\frac{1}{3} \\ &= \frac{69}{15}-\frac{5}{15} \\ &= \frac{64}{15} \end{aligned}$$

Both answers are really the same, but they have different visual shapes (horizontal versus vertical). Note, we followed the order of operations (so multiplication occurred before addition/subtraction). In viewing either answer, I encourage you to think about what an equal sign really means instead of thinking of it as "next step". To say start saying it in words: $3+\frac{4}{5}\cdot 2-\frac{1}{3}$ is equal in value to $3+\frac{4}{5}\cdot\frac{2}{1}-\frac{1}{3}$ which is equal in value to $3+\frac{8}{5}-\frac{1}{3}$ which is equal to $\frac{15}{5}+\frac{8}{5}-\frac{1}{3}$ which is equal to $\frac{23}{5}-\frac{1}{3}$ which is equal $\frac{69}{15}-\frac{5}{15}$ which is finally equal to $\frac{64}{15}$.

Note 1.3.32 When simplifying an expression, we can write the equal sign between equal expressions. Be sure that the the expressions are equal.

Note that this focus on equal signs helps us create good habits. We really want the next expression we write to be equal to the previous expression. For example, in both answers above, the first four expressions all included $\frac{1}{3}$ even though this part wasn't changing at all. It would be incorrect to write $3+\frac{4}{5}\cdot 2-\frac{1}{3}=3+\frac{4}{5}\cdot\frac{2}{1}$ because the two sides are not equal. It can be tempting to drop parts that aren't changing, but we have to be careful to ensure that the two sides of an equal sign are truly equal. Building this habit (even if it feels really tedious) helps to prevent losing track of parts of very large expressions, and writing with good habits (with equal signs) helps to reinforce what an equal sign actually means.

The way that we organize the work for equations looks different. When solving an equation, we start with the given equation. When we have performed an action on both sides, we write the next equation one line below. The equation after that is written one line below that, and so on. Note that an equation (a statement that two expressions are equal) has an equal sign in the middle of its writing.

Example 1.3.33

Solve the equation $\frac{2x-1}{3}+4=10$.

Solution.

$$\begin{aligned}\frac{2x-1}{3}+4&=10\\ \frac{2x-1}{3}&=6\\ 2x-1&=18\\ 2x&=19\\ x&=\frac{19}{2}\end{aligned}$$

Notice where equal signs are written, and where they are not written. It is helpful for us to say in words what is really happening: Starting from the fact that $\frac{2x-1}{3}+4$ is equal to 10, we learn that $\frac{2x-1}{3}$ is equal to 6. From the fact that $\frac{2x-1}{3}$ is equal to 6, we learn that $2x-1$ is equal to 18. From the fact that $2x-1$ is equal to 18, we learn that $2x$ is equal to 19. Finally, from the fact that $2x$ is equal to 19, we learn that x is equal to $\frac{19}{2}$.

Warning 1.3.34 When solving an equation, do not write an equal sign to mean "and the next step is". Here are two examples of incorrect writing for the previous question:

- $$\begin{aligned}\frac{2x-1}{3}+4&=10\\ &=\frac{2x-1}{3}=6\\ &=2x-1=18\\ &=2x=19\\ &=x=\frac{19}{2}\end{aligned}$$

- $\frac{2x-1}{3}+4=10=\frac{2x-1}{3}=6=2x-1=18=2x=19=x=\frac{19}{2}$

Both of these "answers" are essentially the same, but just with different shapes: the first "answer" is presented vertically, while the second "answer" is presented horizontally. Every other equal sign is the equal sign that appears in the middle of an equation, but then the other equal signs are being written because someone is thinking “my next step is” which coincidentally may have worked for simplifying expressions, writing like this is reinforcing an incorrect idea of what an equal sign means. If we read either version out lout, at face value, the writing in this incorrect answer is saying: $\frac{2x-1}{3} + 4$ is equal to 10 which is equal to $\frac{2x-1}{3}$ which is equal to 6 which is equal to $2x - 1$ which is equal to 18 which is equal to $2x$ which is equal to 19 which is equal to x which is equal to $\frac{19}{2}$. Please note how strange this is, especially when saying this aloud: among other things, we have said that the number 6 is equal to something which is then eventually equal to the number 18, but these numbers are not equal!

For equations, to prevent the common presentation error, it is good practice to write vertically instead of horizontally. Write an equation. Then write the next equation one line below, being sure that there is an equal sign in the middle (since an equation is a statement that two expressions are equal), but not an equal sign before the new equation. Then write the next equation one line below that, again ensuring that we don’t write an equal sign before the new equation. Continue in this way until the variable is isolated.

Summary of expectations for presenting work.

1. ***Simplifying expressions:*** Start with the given expression. After each equal sign, write a new expression that is equal in value to the previous expression. (To ensure that the new expression is equal, do not ignore the parts of the expression that are not changing.) Stop when we have an expression that cannot be simplified any further using valid mathematical techniques. When simplifying an expression, we can write horizontally or vertically: in either case, an equal sign is written from one expression to the next equal expression.

2. ***Solving equations:*** Start with the given equation. Write the next equation one line below. While the equation itself contains an equal sign, we should not write an equal sign before the new equation. Continue in this way until the variable is isolated. We write vertically because when the next equation is written to the side of the previous equation, it is really tempting to write an equal sign in that empty space before the new equation, which is incorrect.

While we often see the final equation having the variable on the left and the constant on the right, there is no requirement to have the variable on the left side. We also have a perfectly acceptable final equation if the variable is on the right side. Let’s see an example, and show two answer presentations:

Example 1.3.35

Solve the equation $19 = 7 + 2x$.

Solution 1.

$$\begin{aligned} 19 &= 7 + 2x \\ 19 - 2x &= 7 \\ -2x &= -12 \\ x &= \frac{-12}{-2} \\ x &= 6 \end{aligned}$$

Solution 2.

$$\begin{aligned} 19 &= 7 + 2x \\ 12 &= 2x \\ 6 &= x \end{aligned}$$

In the first answer, we isolated x on the left side, while in the second answer we isolated x on the right side. Both answers are correct. The first answer ended up being a little longer due to the minus signs that appear. The second answer is a little shorter. This is the benefit of not insisting that the variable must be on the left side in the final equation: by having the flexibility to isolate the variable on either side, we can reap the benefit of a possibly shorter answer.

Isolated variable on either side.

When solving an equation, it is just as acceptable to isolate the variable on the right side as it is to isolate the variable on the left side. You can choose the side that makes your work easier or shorter!

We are about to introduce strategies for solving equations where the variable is written more than once. Before we do that, I think it's important for us to clarify exactly what is the expectation when solving equations.

Solving equations: expectation for final step.

When solving an equation, the final equation should have exactly one copy of the variable isolated (meaning with nothing else) on one side of the equation, and the other side of the equation should be a constant (an expression with no variables).

Example 1.3.36

We have not yet developed any strategies for solving equations like $7x - 2 = 3x + 14$. However, we will use this example to illustrate the point being made above. Suppose someone takes $7x - 2 = 3x + 14$ and does the following steps, all of which are valid:

$$\begin{aligned} 7x - 2 &= 3x + 14 \\ 7x &= 3x + 16 \\ x &= \frac{3x + 16}{7} \end{aligned}$$

This doesn't meet the expectation that we just discussed: though the left side just has the variable x on its own, the right side is not a constant because the variable x was written.

So far, regarding solving equations, we have discussed:

- ***Primary tool:*** Apply the same action on both sides of an equation.
- ***When we stop:*** When the variable is isolated on one side of the equation, and the other side is a constant.

We've discussed our primary tool which is how we solve an equation. We've also discussed when we stop. We haven't really discussed what this signifies. In other words, what we haven't discussed is what it means to solve an equation.

Principle 1.3.37 Solving equations: what it means.

To solve an equation means to find all values of the variable that make the left side of the original equation equal to the right side of the original equation.

Example 1.3.38

Let's revisit the equation $7x - 2 = 3x + 14$. We will discuss strategies for solving this equation soon, but for now, say that someone claims to have solved this equation and wrote $x = 4$ as their final line of work. Let's see if using 4 for x makes the origial left side and the original right side equal:

1. The left side of the original equation is the expression $7x - 2$. Substituting 4 for x gives $7(4) - 2 = 28 - 2 = 26$.
2. The right side of the original equation is the expression $3x + 14$. Substituting 4 for x gives $3(4) + 14 = 12 + 14 = 26$.

Substiting $x = 4$ made both the left and right side equal to 26. This means that $x = 4$ is a solution to the equation $7x - 2 = 3x + 14$. Note that this focuses on what it means to solve an equation. We have actually avoided discussing how someone got $x = 4$, which we will do soon. In addition, note that substituing any value of x other than 4 will make both sides unequal.

Example 1.3.39

When asked to solve the equation $x^2 - 3x = 35 - 5x$, using techinques we'll introduce in the next chapter, someone presented $x = -7, 5$ as their final answer. When substituting -7 for x in the original equation, both the left and the right side equal 70. When substituting 5 for x in the original equation, both the left and the right side equal 10. When substituting any other value for x, the left and right sides are not equal. (For example, if we tried substituting 10 for every x in the original equation, the left side would equal 70 while the right side would equal -15, which are not equal. So $x = 10$ is not a solution to the equation.) It turns out that $x = -7$ and $x = 5$ are the only two solutions to the equation are $x = -7$ and $x = 5$.

Understanding that solving an equation means finding all values of the variable that make the original left and right sides equal, this gives us a useful tool that we can use to check our work after solving any equation!

Principle 1.3.40 Checking solutions to equations.

After solving an equation, we can check our work by substituting our solution(s) into the original equation. If substituting a solution into the original equation makes both sides equal, then it is indeed a solution. If substituting a solution into the original equation does not make both sides equal, then there was an error in the work. If there are multiple solutions, we should check each one.

Example 1.3.41

In Example 1.3.24 we started with the equation $5x - 3 = 27$ and we found that $x = 6$ is a solution. We didn't check our work then, so let's check it now, following the advice of Principle 1.3.40. (If you have leftover time in a quiz or exam, it's a good idea to check your work on solving equations by substituting your solution(s) into the original equation.) Whenever we check our work, we always start with the original equation. Let's substitute 6 for x in the original equation:

1. The right side of the original equation is the expression 27. There is no x written, so we don't even have to work on substituting 6 for every x. The right side has value 27.

2. The left side of the original equation is the expression $5x-3$. Substituting 6 for x gives $5(6) - 3 = 30 - 3 = 27$.

Since both sides are equal, we have verified that $x = 6$ is indeed a solution to the equation $5x - 3 = 27$.

Example 1.3.42

Consider the equation $x^3 - x + 8 = 3x^2 + 5x$. For this equation:

- Is $x = 2$ a solution?
- Is $x = 1$ a solution?
- Is $x = -2$ a solution?
- Is $x = 3$ a solution?
- Is $x = 4$ a solution?

Solution.

- Substituting $x = 2$, the left side simplifies to 14 and the right side simplifies to 22. Since these two values do not match, $x = 2$ is not a solution to $x^3 - x + 8 = 3x^2 + 5x$.
- Substituting $x = 1$, the left side simplifies to 8 and the right side simplifies to 8. Since these two values match, $x = 1$ is a solution to $x^3 - x + 8 = 3x^2 + 5x$.
- Substituting $x = -2$, the left side simplifies to -2 and the right side simplifies to -2. Since these two values match, $x = -2$ is a solution to $x^3 - x + 8 = 3x^2 + 5x$.
- Substituting $x = 3$, the left side simplifies to 26 and the right side simplifies to 30. Since these two values do not match, $x = 3$ is not a solution to $x^3 - x + 8 = 3x^2 + 5x$.
- Substituting $x = 4$, the left side simplifies to 56 and the right side simplifies to 56. Since these two values match, $x = 4$ is a solution to $x^3 - x + 8 = 3x^2 + 5x$.

We have found three solutions $(-2, 1, 4)$ to this equation and two non-solutions $(2, 3)$. Actually, the three solutions we have verified are the only solutions to this equation, though we have not explained how we know this to be true. We have only verified that 2 and 3 are not solutions, but just about any number we pick (including 5 or π) will not be a solution. To be clear, we did not give any practical strategy for solving the equation $x^3 - x + 8 = 3x^2 + 5x$. In fact, the specific reason we picked an equation like $x^3 - x + 8 = 3x^2 + 5x$ where we haven't yet introduced a strategy that discusses things like what we should do on both sides is that it helps focus our conversation: instead of focusing on how to solve this equation, we focused on what it means to solve an equation. We get to practice Principle 1.3.40 which is a useful tool to check our work after solving any equation. (Say that someone else claimed that $x = 3$ is a solution to this equation. We could use the same process to verify that $x = 3$ is not a solution.) Practicing this kind of question also encourages us to think about what it means to solve an equation as introduced in Principle 1.3.37.

1.3.5 Equations when the variable is written more than once

We will now discuss solving some types of equations where the variable is written more than once, but not all types of equations where the variable is written more than once.

Strategy 1.3.43 Solving an equation when the variable is written more than once, with

- ***When does this strategy apply?*** This strategy applies in any equation where the variable is written more than once. In addition, the left side and right side of the equation must be sums where each term is either a constant or a constant times the variable.
- ***How to apply the strategy***
  1. On each side, identify the terms that contain the variable, and the term that don't.
  2. Choose which side you'd like the variable to appear on.
  3. Add on both sides and/or subtract on both sides to reach the goal of having all terms with the variable on the side you chose in the previous step, and having all terms without the variable on the other side.
  4. Independently simplify each side. On the side where each term contains the variable, we can collect like terms (or in more complicated situations, apply the more general technique of factoring out the variable). Once this step is complete, the variable will only be written once.
  5. Apply a division on both sides (or a multiplication on both sides) to isolate the variable.

In Example 1.3.36 we brought up the equation $7x - 2 = 3x + 14$ to discuss details about solving equations, but we left it to know to discuss an actual strategy for solving this equation. Let's go through the steps, based on the strategy we just introduced.

Example 1.3.44

Solve the equation $7x - 2 = 3x + 14$.

Solution. First, let's check that the strategy applies. We were given an equation to solve. Each side is a sum (or difference) of terms. Since each side consists of terms, the strategy applies. We need to pick a side to have the variable on. It doesn't matter which side we pick, so let's pick the left side. Currently, the terms on the left side are $7x$ and -2, while the terms on the right side are $3x$ and 14. Since we want terms with x to be on the left side, we should keep the $7x$ term on the left side. However, the next term on the left side is -2. This term does not contain x, so to get rid of it validly, let's add 2 to both sides, obtaining:

$$7x = 3x + 14 + 2.$$

Now, the left side has only terms with x, but the right side still has a term with x. To get rid of the $3x$ term on the right side, let's subtract $3x$ from both sides, obtaining:

$$7x - 3x = 14 + 2.$$

Now, the left side has only terms with x, while the right side has only terms without x, which is a key goal in the process of this strategy. We can simplify both sides:

$$4x = 16.$$

The left side can be thought of as collecting like terms, or we can factor x from the left side to get $(7 - 3)x$, which simplifies to $4x$. Finally, we can isolate x by dividing both sides by 4:

$$x = 4.$$

There are many other answers which perform the actions on both sides in different orders.

Example 1.3.45

Solve the equation $7x + 10 = 12x - 2$.

Solution 1. The variable x is written more than once, and each side is a sum of terms, so the strategy applies. Let's add and subtract both sides with the goal of getting all terms with x on the left side, and all terms without x on the right side. We can start by subtracting 10 from both sides:

$$7x = 12x - 2 - 10.$$

Now, the left side has only terms with x. Let's examine the right side. We wanted all terms with x on the left side, so we have to do something about the $12x$ term on the right side. Let's subtract $12x$ from both sides:

$$7x - 12x = -2 - 10.$$

Now, the left side has only terms with x, while the right side has only terms without x. This was a key achievement described in this strategy. We can simplify each side independently:

$$-5x = -12.$$

Finally, divide both sides by -5 to isolate x:

$$x = \frac{-12}{-5}$$

$$x = \frac{12}{5}$$

Solution 2. As a second solution, we can apply the same startegy, but getting the terms with x on the right side and the terms without x on the left side. Starting from

$$7x + 10 = 12x - 2$$

we can subtract $7x$ from both sides:

$$10 = 12x - 7x - 2$$

The right side has a term without x, so we add 2 to both sides:

$$10 + 2 = 12x - 7x$$

Now, the left side has only terms without x, while the right side has only terms with x. We can simplify each side independently:

$$12 = 5x$$

Finally, divide both sides by 5 to isolate x:

$$x = \frac{12}{5}$$

The second solution is slightly faster because we didn't need to cancel negative signs. In the original equation, note there was a $12x$ on the right and a $7x$ on the left. Strategically, to avoid minus signs, since 12 is larger than 7 and the coefficient of 12 appears on the right, this is why we might choose to keep all terms with the variable on the right side.

Example 1.3.46

Solve the equation $3(x + 2) = 5x - 11$.

Solution. We are asked to solve an equation in which the variable x is written more than once. However, the left side is a product, not a sum. So the strategy doesn't directly apply. However, we can rewrite the right side by distributing. The original left side $3(x+2)$ can be replaced with $3x+6$. With this replacement, our equation becomes

$$3x + 6 = 5x - 11$$

Now, the strategy applies. Since 5 is bigger than 3, let's choose to have all terms with x on the right, and all terms without x on the left. (You could make the opposite choice, and still achieve a perfectly valid answer, but this choice will minimize the number of minus signs that appear.) Starting from $3x+6 = 5x-11$, to get all terms with x on the right and all terms without x on the left, we can subtract $3x$ from both sides and also add 11 to both sides, obtaining:

$$6 + 11 = 5x - 3x$$

You might choose to skip the equation we just wrote above to directly write:

$$17 = 2x$$

Finally, divide both sides by 2 to isolate x:

$$\frac{17}{2} = x$$

Example 1.3.47

Solve the equation $10(x+6) = 3x$.

Solution. We are asked to solve an equation in which the variable x is written more than once. However, the left side is a product, not a sum. So the strategy doesn't directly apply. However, we can rewrite the left side by distributing.

$$10x + 60 = 3x$$

Now, the strategy applies. Since 10 is bigger than 3, let's choose to have all terms with x on the left, and all terms without x on the right. (You could make the opposite choice, and still achieve a perfectly valid answer, but this choice will minimize the number of minus signs that appear.) Starting from $10x+60 = 3x$, we subtract $3x$ from both sides and also subtract 60 from both sides, obtaining:

$$10x - 3x = -60$$

You might choose to skip the equation we just wrote above to directly write:

$$7x = -60$$

Finally, divide both sides by 7 to isolate x:

$$x = \frac{-60}{7}$$

Example 1.3.48

Solve the equation $2(3x - 4) + 5 = 6(x + 1) + 4x$.

Solution. After distributing, we have:

$$6x - 8 + 5 = 6x + 6 + 4x$$

Now, the strategy applies. You can continue work noticing that there are already like terms on the right, so we can combine those like terms. (A successful solution can be made without combining like terms at this moment.) Let's combine like terms on the right, and as long as we're doing this, simplify the constants on the left:

$$6x - 3 = 10x + 6$$

Since 10 is bigger than 6, let's choose to have all terms with x on the right, and all terms without x on the left. Subtract $6x$ on both sides and also subtract 6 from both sides:

$$-9 = 4x$$

Finally, divide both sides by 4 to isolate x:

$$x = \frac{-9}{4}$$

As a friendly reminder, collecting like terms is really skipping a two-step process of factoring out the variable. This is important to keep in mind, because there are examples (like the next one) where only thinking about collecting like terms usually gets people stuck.

Example 1.3.49

Solve the equation $\sqrt{7}x - 5 = 2x + 30$.

Solution. The strategy directly applies. Let's get all terms that have x on the left side, and all terms without x on the right side. We subtract $2x$ from both sides and also add 5 to both sides:

$$\sqrt{7}x - 2x = 30 + 5.$$

Of course, the new right side $30 + 5$ is easy to simplify, but to make a point, I didn't simplify here yet, because the left side simplification is not obvious (because how are we supposed to "collect like terms" here?). While we simplify the right side in the next equation, let's factor out x on the left side:

$$(\sqrt{7}-2)x = 35.$$

Now the content in parentheses is a constant (the variable doesn't appear), so this is a number. It's a funky number -- it take some time and some space to write it -- but it is just a number/constant. Let's divide both sides by the number $\sqrt{8}-2$. In fact, let's slow this down by showing the division on both sides without simplifying yet:

$$\frac{(\sqrt{7}-2)x}{\sqrt{7}-2} = \frac{35}{\sqrt{7}-2}$$

On the left side, the factor of $\sqrt{7}-2$ in the numerator cancel with the the entire denominator $(\sqrt{7}-2)$, recalling that the entire denominator of any fracition is a factor, since the entire denominator is in a hidden set of parentheses. After this cancellation, we have x by itself on one side, and no xs on the other side, so we're done:

$$x = \frac{35}{\sqrt{7}-2}$$

In the appendix, we talk about a technique called rationalizing the denominator, which can be used to rewrite this answer in a different form.

AAAA NOTE ON WRITING: dividing on both sides in a LAZY way: making the existing equation part of a fraction.

AAAA NOTE ON WRITING: indicating action on only ONE side.

1.3.6 Applications Revisited

Let's revisit the applications we introduced at the beginning of this section. Before going through the answers, there is one important point to make:

Warning 1.3.50 When solving application problems, it is important to write an equation that models the situation. It is tempting to skip the explanation, and try to coach yourself to look for a pattern and boil everything in the question down to an arithmetic problem. However, we introduced these application problems as applications of solving equations. It would be easy to say "but I see a pattern, so I don't need to set up an equation". This view is limiting in multiple ways: first off, any pattern devloped on coached arithmetic actually comes from the equation. Second, the skill of writing an equation to model a situation is a critical skill in algebra. It is a really powerful tool to write a variable to stand in for a quantity that we don't know yet. Third, the pattern you see may not be correct for a different problem that is similar but not exactly the same. More generally, attempting to do arithmetic over algebra will eventually limit the kinds of problems you can solve.

The process of modeling will feel awkward, but with practice it becomes more natural. Everyday uses of mathematics later in life (from things as mundane as creating an approximate budget for a kitchen remodel job) will require modeling. Setting up a relevant equation which contains a variable is a core idea of algebra, so I strongly encourage you to give it a try! Run towards this, instead of running away! You can do it!

Example 1.3.51

A car rental costs \$40 per day plus a one-time insurance fee of \$15. If the total cost is \$135, how many days was the car rented?

Solution. We introduce the variable d to represent the number of days the car was rented. So, if $d = 7$ then the car was rented for 7 days. The total cost of renting the car for d days is $15 + 40d$. We were told the total cost of renting the car was 135. Since we have two different-looking ways to represent the total cost, namely $15 + 40d$ and 135, we can set them equal to each other:

$$\begin{aligned} 15 + 40d &= 135 \\ 40d &= 120 \\ d &= 3 \end{aligned}$$

The car was rented for 3 days.

Example 1.3.52

A chemist is cooling a liquid that starts at 90°C and decreases at a rate of 2°C per minute. After how many minutes will the liquid reach 30°C?

Solution. We introduce the variable t to represent the time in minutes. (Hypothetically, if $t = 5$ then the liquid has been cooling for 5 minutes.) The temperature of the liquid after t minutes is $90 - 2t$. We were told that we want to find when the temperature reaches 30. Since we have two different-looking ways to represent the temperature, namely $90 - 2t$ and 30, we can set them equal to each other:

$$\begin{aligned} 90 - 2t &= 30 \\ -2t &= -60 \\ t &= 30 \end{aligned}$$

The liquid will reach 30°C after 30 minutes.

Example 1.3.53

A train leaves a station traveling at 60 miles per hour. Another train leaves the same station two hours later traveling at 80 miles per hour. After how many hours will the second train catch up?

Solution. Let h represent the number of hours traveled by the faster train. (Hypothetically, if $h = 5$ then the faster train has been traveling for 5 hours.)

The slower train leaves earlier, meaning it travels for more time. Since the fast train travels for h hours, the slow train travels for $h + 2$ hours. The distance of the slow train is $60(h + 2)$.

The distance of the fast train is $80h$.

Both trains travel the same distance. This means that the distances $60(h+2)$ and $80h$ must be equal, even though these expressions look different. We create an equation by setting this different-looking expressions equal to each other:

$$\begin{aligned} 60(h + 2) &= 80h \\ 60h + 120 &= 80h \\ 120 &= 20h \\ 6 &= h \end{aligned}$$

The faster train will catch up after 6 hours.

Example 1.3.54

A landscaper buys bags of soil for \$3 each and a one-time delivery charge of \$12. If the total cost is \$51, how many bags of soil were purchased?

Solution. Let b represent the number of bags of soil purchased. (Hypothetically, if $b = 4$ then the landscaper bought 4 bags of soil.) The total cost of buying b bags of soil is $3b + 12$. We were told that the total cost was 51. Since we have two different-looking ways to represent the total cost, namely $3b + 12$ and 51, we can set them equal to each other:

$$\begin{aligned} 3b + 12 &= 51 \\ 3b &= 39 \\ b &= 13 \end{aligned}$$

The landscaper bought 13 bags of soil.

Example 1.3.55

An Australian internet service provider offers a base plan for \$30 per month plus \$5 for each additional gigabyte of data. If someone paid \$65 one month, how many gigabytes of extra data did they use?

Solution. Let d represent the amount of data used by the customer in gigabytes. (Hypothetically, if $d = 4$ then the customer used 4 gigabytes of data.) The total cost of using d gigabytes of data is $30 + 5d$ for the month, but the total cost is also 65 for the month, so we set these equal to create an equation:

$$\begin{aligned} 30 + 5d &= 65 \\ 5d &= 35 \\ d &= 7 \end{aligned}$$

The customer used 7 gigabytes of extra data.

Example 1.3.56

A taxi ride costs \$3 plus \$2.50 per mile. The taxi is cash only and you have \$28.50 available. How many miles can you travel?

Solution. Let d represent the distance traveled in miles. (Hypothetically, if $d = 2.75$ then the customer traveled 2.75 miles.)

The total cost of the ride can be expressed as $3 + 2.5d$. We know the total cost is 28.50, so we can set up the equation:

$$\begin{aligned} 3 + 2.5d &= 28.5 \\ 2.5d &= 25.5 \\ d &= 10.2 \end{aligned}$$

The customer can travel 10.2 miles.

Example 1.3.57

A gym charges a membership fee of \$50 plus \$15 per month. How many months can someone be a member if they paid \$200 in total?

Solution. Let m represent the number of months the person was a member. (Hypothetically, if $m = 4$ then the person was a member for 4 months.) The total cost of being a member for m months is $50 + 15m$. We were told that the total cost was 200. Since we have two different-looking ways to represent the total cost, namely $50 + 15m$ and 200, we can set them equal to each other:

$$50 + 15m = 200$$

$$15m = 150$$
$$m = 10$$

The person was a member for 10 months.

1.3.7 Summary

1. Expressions and equations are different mathematical writing. An expression is notation representing a number, while an equation is a statement that two expressions are equal. Generally, an expression is simplified, while an equation is solved.

2. The individual steps of solving an equation involve applying the same action to the entire left side and the entire right side of a fraction.

3. To solve an equation where the variable is written once, undo the final operation which occured on the side that has the variable.

4. To solve an equation where the variable is written more than once, if each side is (or can be, after algebra) written as a sum, apply additions and subtractions to both sides to get all terms with the variable on one side and all terms without the variable on the other side.

5. The placement of equal signs can be confusing at first, but recall that an equal sign that is written can always be intrerpreted to mean equal but does not always mean "and the next step is".

6. While we often concern ourselves with the process of how to solve an equation (performing the same action to both sides), the meaning is to provide a list of all values for the variable which make the original equation's left side equal to the original equation's right side. For this reason, we can always check our answer(s) by substituting our solution(s) into the original equation.

1.3.8 Exercises

1. Review the strategies for solving equations that we have seen, paying attention to when each strategy applies. Then solve each equation.

   (a) $x + 5 = 12$
   (b) $5x = 12$
   (c) $x - 7 = 4$
   (d) $4 = 7x$
   (e) $2x + 7 = 15$
   (f) $10 - x = 4$
   (g) $8 - 5x = 2$

(h) $6x + 9 = 3$

(i) $4x - 3 = 2x + 7$

(j) $7x + 5 = 2x + 20$

(k) $12 - 3x = 4x - 2$

(l) $\dfrac{2x-1}{6} + 3 = 55$

(m) $\dfrac{3x+4}{5} - 2 = 6$

(n) $2(x + 3) = 10$

(o) $3(x - 2) = 15$

(p) $4(x - 1) + 2 = 10$

(q) $5(x + 2) - 3 = 2x + 10$

(r) $\dfrac{x+5}{2} = 7$

(s) $4 + \dfrac{x-2}{3} = 10$

(t) $7(x - 4) = 2(x + 1) + 9$

(u) $3(x - 2) = 7(x + 3) + 2x + 1$

(v) $12 - \sqrt{3}x = 4x - 2$

(w) $\sqrt{7}x + 5 = 2x + 20$

(x) $\dfrac{5x+2}{7} = 8$

(y) $12 = \dfrac{3x-4}{2}$

(z) $\dfrac{15-2x}{5} = 1$

2. Is the given value a solution to the given equation? Justify your answer.

(a) Is $x = 4$ a solution to $x^3 - 3x^2 - 6x + 8 = 0$?

(b) Is $x = 3$ a solution to $x + 5 = 8$?

(c) Is $x = -2$ a solution to $x^3 = -8$?

(d) Is $x = 5$ a solution to $x^2 = 20$?

(e) Is $x = -1$ a solution to $(x + 1)(x - 4) = 0$?

(f) Is $x = 6$ a solution to $5x = 30$?

(g) Is $x = 1$ a solution to $x^3 - 3x^2 = 6x - 8$?

(h) Is $x = -3$ a solution to $x^2 = 9$?

(i) Is $x = 2$ a solution to $(x - 2)(x - 3) = 0$?

(j) Is $x = -4$ a solution to $x^3 + 64 = 0$?

(k) Is $x = 4$ a solution to $x^2 - 5x + 6 = 0$?

(l) Is $x = 3$ a solution to $x^3 - 27 = 0$?

(m) Is $x = 0$ a solution to $x(x - 5) = 0$?

(n) Is $x = -2$ a solution to $2x + 7 = 3$?

(o) Is $x = 7$ a solution to $(x - 2)(x - 7) = 0$?

(p) Is $x = 2$ a solution to $x^2 - 3x + 2 = 0$?

(q) Is $x = 1$ a solution to $x^3 = 1$?

(r) Is $x = 5$ a solution to $x^2 - 25 = 0$?

(s) Is $x = 4$ a solution to $3x - 2 = 10$?

(t) Is $x = -2$ a solution to $x^3 + 8 = 3x^2 + 6x$?

3. Model each situation by first clearly defining a variable to represent the unknown quantity, write an equation that models the situation (by equating two different-looking expressions which are equal due to the situation described), and then solve the equation.

   (a) A taxi ride costs a flat fee of \$4 plus \$2.50 per mile. If the total cost is \$29.50, how many miles was the trip?

   (b) A recipe requires $3\frac{1}{2}$ cups of sugar to make one batch of cookies. If you want to make 5 batches, how many cups of sugar do you need?

   (c) A water tank is being filled at a rate of 12 liters per minute. How long will it take to fill 300 liters?

   (d) You earn \$15 per hour babysitting. If you earned \$180 last month, how many hours did you babysit?

   (e) A scientist has a sample cooling at a steady rate of 1.5°C per minute. If it started at 95°C, after how many minutes will it reach 35°C?

   (f) A gym charges \$40 per month plus a one-time registration fee of \$20. If someone paid \$140 total, how many months were they a member?

   (g) A concert ticket costs \$45. If you save \$270 by skipping other expenses, how many tickets can you buy?

   (h) You drive at an average speed of 60 miles per hour. How long does it take to drive 210 miles?

   (i) A factory produces 250 parts per hour. How many hours does it take to produce 6,000 parts?

   (j) A cell phone plan charges \$30 plus \$0.05 per text message. If the monthly bill is \$47.50, how many text messages were sent?

   (k) A bicycle rental costs \$12 per hour. If the total cost was \$66, for how many hours was the bike rented?

   (l) A sports drink container has 1.5 liters. How many containers are needed to hold 21 liters?

   (m) A printing company charges \$50 setup fee plus \$0.20 per page. If the total charge is \$110, how many pages were printed?

(n) A runner maintains a pace of 8 minutes per mile. How long does it take to complete a 13-mile run?

(o) A streaming service charges $12 per month. If you paid $96 in total, how many months were covered?

(p) A car uses 3.5 gallons of gas to drive 100 miles. How many gallons are needed to drive 350 miles?

(q) A plumber charges $75 for a service call plus $50 per hour of labor. If the bill was $275, how many hours of labor were included?

(r) A hiking trail is 6.4 kilometers long. If you walk at a steady pace of 4 kilometers per hour, how long does it take to finish the trail?

(s) A landscaper charges $30 per hour plus a one-time fee of $50. If the total cost was $200, how many hours did the landscaper work?

1.4 Equations with Fractions

Objectives

In this section, we learn how to:

1. Solve equations with fractions present.

1.4.1 A new strategy

In Section 1.2, we learned how to simplify expressions with fractions. In this section, we look beyond simplifying expressions with fractions, and instead look at solving equations with fractions. The last two sections discussed expressions with fractions followed by solving equations, so now it's time to see efficient ways to solve equations with fractions. Recall from Warning 1.2.66 that simplifying expressions with fractions does not always require a common denominator. (More specifically, when asked to simplify the product of two fractions, we do not need a common denominator.) So, I invite you -- as we go through this section -- to be willing to question the idea that we'd always need a common denominator when solving equations with fractions. After all, it wasn't every fractional expression that required common denominators: only some of them did.

Strategy 1.4.1 Solving an equation with fractions.

- ***When does this strategy apply?*** This strategy applies in any equation we are asked to solve when there is one or more fractions.
- ***How to apply the strategy***
  1. Inventory all of the fractions, whether they appear on the left side, the right side, or both. In particular, identify all of the denominators present.
  2. Off to the side, compute the least common denominator (LCD) of all of the fractions. This phrase **least common denominator** is shorthand to mean the **least common multiple** of all of the denominators. (See APPENDIX for details on least common multiples.)
  3. Take the least common multiple computed in the previous step and multiply both sides of the equation by this expression. (Because we have to multiply the entire left side and the entire right side, it may be necessary to use parentheses due to the Order of Operations.)
  4. On each side, distribute if applicable.
  5. In each term on each side, cancel where applicable. This always results in an equation which has no fractions.

Example 1.4.2

Solve $\frac{x}{3}+\frac{2}{5}=\frac{3}{10}$.

Solution. We are being asked to solve an equation with fractions, so Strategy 1.4.1 applies. First, we inventory all of the fractions and identify the denominators, which are 3 and 5 and 10. The least common multiple of 3 and 5 and 10 is 30.

We multiply both sides of the equation by 30:

$$30\left(\frac{x}{3}+\frac{2}{5}\right)=30\cdot\frac{3}{10}.$$

Note that on the left side of the equation, we used parentheses to ensure that the entire left side is multiplied by 30: the Order of Operations tells us that multiplication is performed before addition, so without parentheses, only the first fraction would be multiplied by 30. On the right side, we could have drawn parentheses around the entire right side, but because there is only one term on the right side, it was not necessary.

Now, distribute on the left side:

$$30\cdot\frac{x}{3}+30\cdot\frac{2}{5}=30\cdot\frac{3}{10}.$$

Next, we cancel where applicable. In the first term (all content before the first plus sign), the 30 in the numerator partially cancels with the 3 in the denominator, or said differently, after cancelling factor of 3 on top and bottom, we are left with a factor of 10 on top. In the second term (all content between the plus sign and the equal sign), the 30 in the numerator partially cancels with the 5 in the denominator, leaving a factor of 6 on top. On the right side, the 30 in the numerator partially cancels with the 10 in the denominator, leaving a factor of 3 on top. Here's what it looks like with the cancellations written in:

$$\cancel{3}\cdot 10\cdot\frac{x}{\cancel{3}}+\cancel{5}\cdot 6\cdot\frac{2}{\cancel{5}}=\cancel{10}\cdot 3\cdot\frac{3}{\cancel{10}}.$$

After performing the cancellations, we are left with:

$$10x+12=9.$$

This is an equation with no fractions, so we can solve it using the strategies we have learned previously. Subtracting 12 from both sides gives:

$$10x=-3.$$

Dividing both sides by 10 gives:

$$x=-\frac{3}{10}.$$

Note 1.4.3 Before seeing more examples, let's discuss why this strategy works:

- We multiplied by the appropriate quantity (in our last example 30) which ensured that, after distributing where needed, all factors in the denominator would cancel. Using the least common multiple of the denominators as the number to multiply by on both sides ended up leading to this value appearing as a factor in the numerator of each term, allowing each copy to create all the necessary cancellations so that each denominator ends up being 1 after cancellations occur. In other words, this ends up creating an equation without fractions.

- For a fraction expression, there are not two sides, so to simplify adding fractions, a common denominator is needed. However, when you have an equation there are two sides. We can take avantage of the two sides by multiplying by the same thing on both sides! In the case of an expression, there aren't "sides" in the first place.

- In an equation with fractions, multiplying both sides by the least common multilple of the denominators is the fastest way to get rid of fractions.

Example 1.4.4

Solve the equation $\frac{x}{6}+\frac{2}{3}=\frac{3x}{4}$.

Solution. We are being asked to solve an equation with fractions, so Strategy 1.4.1 applies. The least common multiple of the denominators 6 and 3 and 4 is 12. So we multiply both sides of the equation by 12:

$$12\left(\frac{x}{6}+\frac{2}{3}\right)=12\cdot\frac{3x}{4}.$$

Distributing on the left side gives:

$$12\cdot\frac{x}{6}+12\cdot\frac{2}{3}=12\cdot\frac{3x}{4}.$$

Now we cancel where applicable.

$$\cancel{6}\cdot 2\cdot\frac{x}{\cancel{6}}+\cancel{3}\cdot 4\cdot\frac{2}{\cancel{3}}=\cancel{4}\cdot 3\cdot\frac{3x}{\cancel{4}}.$$

After performing the cancellations, we are left with:

$$2x+8=9x.$$

This is an equation with no fractions. Subtracting $2x$ from both sides gives:

$$8=7x.$$

Dividing both sides by 7 gives:

$$\frac{8}{7} = x.$$

Example 1.4.5

Solve the equation $\frac{x}{6} + \frac{1}{2} = \frac{2}{3}$.

Solution. The least common multiple of the denominators 6 and 2 and 3 is 6. So we multiply both sides of the equation by 6, though Order of Operations requires us to wrap the left side in parentheses:

$$6\left(\frac{x}{6} + \frac{1}{2}\right) = 6 \cdot \frac{2}{3}.$$

Distributing on the left side gives:

$$6 \cdot \frac{x}{6} + 6 \cdot \frac{1}{2} = 6 \cdot \frac{2}{3}.$$

Now we cancel where applicable.

$$\cancel{6} \cdot 1 \cdot \frac{x}{\cancel{6}} + \cancel{2} \cdot 3 \cdot \frac{1}{\cancel{2}} = \cancel{3} \cdot 2 \cdot \frac{2}{\cancel{3}}.$$

After performing the cancellations, we are left with:

$$x + 9 = 12.$$

This is an equation with no fractions. Subtracting 9 from both sides gives:

$$x = 3.$$

You may have never heard of Strategy 1.4.1 before, but it is the fastest way to solve any equation that has fractions. You may be used to another method that works involving common denominators, but as a friendly reminder, Warning 1.2.66 told us that even when we looked at fractional expressions, we didn't always need common denominators. Let's compare the two methods in the next example.

Example 1.4.6

Solve $\frac{x}{6} + \frac{x-3}{10} = \frac{4}{3}$

Solution 1. Here's what it looks like to solve $\frac{x}{6} + \frac{x-3}{10} = \frac{4}{3}$ by ignoring Strategy 1.4.1 and doing everything with common denominators instead:

$$\frac{x}{6} + \frac{x-3}{10} = \frac{4}{3}$$

$$\begin{aligned} \frac{5x}{30} + \frac{3(x-3)}{30} &= \frac{40}{30} \\ \frac{5x}{30} + \frac{3x-9}{30} &= \frac{40}{30} \\ \frac{5x+3x-9}{30} &= \frac{40}{30} \end{aligned}$$

Multiply both sides by 30.

$$5x + 3x - 9 = 40$$

$$\begin{aligned} 8x - 9 &= 40 \\ 8x &= 49 \\ x &= \frac{49}{8} \end{aligned}$$

Solution 2. Here's what it looks like to solve $\frac{x}{6} + \frac{x-3}{10} = \frac{4}{3}$ by using Strategy 1.4.1. First, the least common multiple of the denominators 6 and 10 and 3 is 30, so we make our first step to multiply both sides by 30.

$$30\left(\frac{x}{6} + \frac{x-3}{10}\right) = 30 \cdot \frac{4}{3}$$

On the left side, distribute 30.

$$30\frac{x}{6} + 30\frac{x-3}{10} = 30 \cdot \frac{4}{3}$$

After cancelling, we have:

$$5x + 3(x-3) = 10 \cdot 4$$

$$\begin{aligned} 5x + 3x - 9 &= 40 \\ 8x - 9 &= 40 \\ 8x &= 49 \\ x &= \frac{49}{8} \end{aligned}$$

Let's be clear that both answers to the previous question are completely valid. In both answers, it took multiplying by 30 to get rid of the fractions. (I ask you to pause for a second and think about this: if we didn't multiply by 30, we would not have been able to get rid of the fractions.) Since we have to multiply by 30 anyway, why not do it right away and save ourselves some time? This saves us time because we don't have to keep writing fractions. Fractions take longer to write than non-fractions (because fractions have a top and a bottom). To be clear, there are ways to skip some of the steps in both answers above (but for a fair comparison, both answers stated the original equation at the start of the answer).

Strategy 1.4.1 works because an equation has two sides. If we only had an expression, then we don't have two "sides" to work with.

Example 1.4.7

Simplify the expression $\frac{x}{6}+\frac{x-3}{10}$.

Solution.

$$\begin{aligned}&\frac{x}{6}+\frac{x-3}{10}\\ =&\frac{5x}{30}+\frac{3(x-3)}{30}\\ =&\frac{5x}{30}+\frac{3x-9}{30}\\ =&\frac{5x+3x-9}{30}\\ =&\frac{8x-9}{30}\end{aligned}$$

In the previous example, we could not use Strategy 1.4.1 because there is no second "side" to multiply by the least common multiple. Instead, we had to find a common denominator in order to simplify the addition of fractions. This is why Strategy 1.4.1 only applies to equations with fractions. However, when we have an equation with fractions, even if there is a plus sign between two fractions, there is no requirement to simplify the addition of fractions. Instead, we can use Strategy 1.4.1 to get rid of the fractions entirely.

Example 1.4.8

Solve the equation $\frac{2x}{9}+\frac{3x+1}{10}=\frac{5x}{6}$.

Solution 1. An answer that ignores (((xref without ref, first/last, or provisional attribute (check spelling)))) and uses common denominators instead looks like this:

$$\begin{aligned}\frac{2x}{9}+\frac{3x+1}{10}&=\frac{5x}{6}\\ \frac{20x}{90}+\frac{9(3x+1)}{90}&=\frac{75x}{90}\\ \frac{47x+9}{90}&=\frac{75x}{90}\end{aligned}$$

Multiply both sides by 90.

$$47x+9=75x$$

$$-28x=-9$$

$$x = \frac{9}{28}$$

To try to speed things up, we skipped writing several steps.

Solution 2. An answer that uses (((xref without ref, first/last, or provisional attribute (check spelling)))) first computes that the least common multiple of 9 and 10 and 6 is 90. To create a fair comparison with the previous answer, we start by writing the original equation:

$$\frac{2x}{9} + \frac{3x+1}{10} = \frac{5x}{6}$$

Now, if we were in the setting of a quiz or exam, this equation is probably already typed for you. For the multiplying by 90 on both sides, you could actually handwrite the 90 on both sides, surrounding the typed left side in parentheses. That is, by handwriting on top of the original equation, you could see this next equation without having to write it in full:

$$90\left(\frac{2x}{9} + \frac{3x+1}{10}\right) = 90 \cdot \frac{5x}{6}$$

In fact, the step that would appear after this involves distributing 90 on the left, as we saw in previous examples using (((xref without ref, first/last, or provisional attribute (check spelling)))). Instead of writing the equation

$$90 \cdot \frac{2x}{9} + 90 \cdot \frac{3x+1}{10} = 90 \cdot \frac{5x}{6}$$

we might just stare at each term to figure out what results after cancellation. In other words, by looking at $\frac{2x}{9}$ we could see that result of multiplying by 90 is $10 \cdot 2x$. Similarly, looking at the second term $\frac{3x+1}{10}$, multiplying by 90 gives $9(3x+1)$. For the term on the right side, multiplying $\frac{5x}{6}$ by 90 simplifies to $15 \cdot 5x$. Thus, our first full equation to write could be this equation, which has no fractions:

$$20x + 9(3x+1) = 75x$$

$$20x + 27x + 9 = 75x$$

$$47x + 9 = 75x$$

$$-28x = -9$$

$$x = \frac{9}{28}$$

Example 1.4.9

Solve $\frac{2x+1}{3} - \frac{x-1}{4} = \frac{x+2}{6}$.

Solution 1. An answer that ignores (((xref without ref, first/last, or provisional attribute (check spelling)))) and uses common denominators looks like this:

$$\frac{8(2x+1)}{24} - \frac{6(x-1)}{24} = \frac{4(x+2)}{24}$$
$$\frac{16x+8-6x+6}{24} = \frac{4x+8}{24}$$
$$\frac{10x+14}{24} = \frac{4x+8}{24}$$

Multiply both sides by 24.

$$10x + 14 = 4x + 8$$
$$6x = -6$$
$$x = -1$$

To try to minimize the amount of writing, some steps were skipped. Even with skipped steps, there were three equations written after the original equation which had fractions, and fractions take longer to write than non-fractions. (One common point of confusion is why the left side of the second equation has $16x + 8 - 6x + 6$ instead of $16x + 8 - 6x - 6$. The reason for this is that the numerator on the left is really $8(2x + 1) - 6(x - 1)$ before any distributing occured. Some people would prefer to distribute before combining fractions, in which case one numerator is $16x + 8$ in a hidden set of parentheses and the second numerator is $(6x - 6)$ also in a hidden set of parentheses.)

Solution 2. Using (((xref without ref, first/last, or provisional attribute (check spelling)))), the least common multiple of 3 and 4 and 6 is 12. We could handwrite a 12 on each side of the original typed equation, surrounding the left side in parentheses. Knowing that we will distribute 12 on the left side anyway, we can really just think of what it means to multiply each term on each side by 12. Multiplying $\frac{2x+1}{3}$ by 12 means that a factor of 4 remains in the numerator from the 12 so we'd have $4(2x + 1)$. Multiplying $\frac{x-1}{4}$ by 12 means that a factor of 3 remains in the numerator from the 12 so we'd have $3(x - 1)$ after the minus sign. Multiplying $\frac{x+2}{6}$ by 12 means that a factor of 2 remains in the numerator from the 12 so we'd have $2(x + 2)$. So, the first full equation we write doesn't have fractions:

$$4(2x + 1) - 3(x - 1) = 2(x + 2)$$
$$8x + 4 - 3x + 3 = 2x + 4$$
$$5x + 7 = 2x + 4$$
$$3x = -3$$
$$x = -1$$

In an attempt to make a fair comparison and skip as much as people might eventually feel comfortable in writing (just to compare how much writing there is, and how much time it takes), a certain aspect of the first answer may have been confusing. For that reason, outside of the comparison, I think it would be good for us to see the first several steps of the first answer without skipping any work. There are actually two subvariants of the first answer, and we'll go through the steps of both subvariants. Spending the time to do this isn't to persuade you to move away from Strategy 1.4.1. Far from it. In fact, the subtle things that we're about to dig into aren't a problem by following Strategy 1.4.1. That said, going into the details (as annoying as it might be) I hope will demystify some of the steps that are too often glossed over quickly (and I don't blame people who are confused)! The throughout explanations we're about to dig into are meant to ensure we don't create confusion by shoving things under the rug.

- Starting from $\frac{2x+1}{3} - \frac{x-1}{4} = \frac{x+2}{6}$ the usual work to create a common denominator leads to
$$\frac{8(2x+1)}{24} - \frac{6(x-1)}{24} = \frac{4(x+2)}{24}.$$
At this moment, we can either distribute 8 and distribute 6, or since we have common denominators on the left, we can combine the fractions. I don't think it's wise at this moment to try to do both at the same time. One thing at a time. So, let's see what happens if we rewrite each denominator by distributing:
$$\frac{16x+8}{24} - \frac{6x-6}{24} = \frac{4x+8}{24}$$
Now, before doing the next step, something that tiny that can get easily ignored can lead to problems. Recall that the entire numerator and entire denominator of every fraction is in a set of parentheses. For this problem, we need to really think about this for the numerators (and don't need to really think about it for the denominators). While we really actually need to think about these parentheses for the second numerator, let's just draw in thoe hidden parentheses for all numerators, making visible those once-hidden parentheses:
$$\frac{(16x+8)}{24} - \frac{(6x-6)}{24} = \frac{(4x+8)}{24}$$
Now, on the left side, since we are subtracting fractions of a common denominator, we can combine this into one fraction: we leave in the parentheses for now, just to process one thing at a time.
$$\frac{(16x+8)-(6x-6)}{24} = \frac{(4x+8)}{24}$$
The first set of parentheses can be dropped. The parentheses on the right side can also be dropped. However, the second set of parentheses on the left side cannot be dropped because of the minus sign in front of it. In fact, we

addressed the expression $(16x + 8) - (6x - 6)$ in Example 1.1.87, so we can apply what we learned there:

$$\frac{16x + 8 - 6x + 6}{24} = \frac{4x + 8}{24}$$

We'll stop working on the problem here, because the previous equation appeared in the first answer we already provided.

- Starting over, once we got common denominators which gave the equation

$$\frac{8(2x + 1)}{24} - \frac{6(x - 1)}{24} = \frac{4(x + 2)}{24},$$

instead of distributing first, we could have combined the fractions first:

$$\frac{8(2x + 1) - 6(x - 1)}{24} = \frac{4(x + 2)}{24},$$

Now, distributing 8 and -6 and 4 gives us the equation:

$$\frac{16x + 8 - 6x + 6}{24} = \frac{4x + 8}{24},$$

Again, we stop working on the problem here.

1.4.2 Clarifications, Cautions, and Connections

We've seen a technique for solving equations that have fractions, but there are some things that we should clarify. I invite you to read through this section carefully. My hope is to really make sure everything that you've seen about equations with fractions is honored: we'll discuss it all! Moreover, I hope that what you read is empowering, because we've worked hard together to lay a foundation for what's happening step-by-step!

Warning 1.4.10 When being asked to solve an equation with fractions, a common error occurs among those who insist on a common denominator approach. What usually happens is that there is adding and/or subtracting on both sides to reach the goal of having one side of the equation becoming just zero. After achieving this, often the work shown will involve only looking at the busier side and simplifying the expression.

Let's illustrate exactly what this looks like with an example. When asked to solve $\frac{x}{6} + \frac{x-3}{10} = \frac{4}{3}$, someone might first subtract $\frac{4}{3}$, and their work in full might look like this:

$$\frac{x}{6} + \frac{x - 3}{10} = \frac{4}{3}$$

$$\frac{x}{6} + \frac{x - 3}{10} - \frac{4}{3} = 0$$

$$= \frac{5x}{30} + \frac{3x-9}{30} - \frac{40}{30}$$
$$= \frac{5x+3x-9-40}{30}$$
$$= \frac{8x-49}{30}$$

Doesn't the end of this work seem a little strange? Normally, when we solve an equation, we expect the last line of work to look something like $x = 19$ or perhaps $x = \frac{31}{7}$. The end of the work didn't tell us what number x should be. Instead, our final work was an expression. How did this happen? Often, when people see a zero on one side (as you see in the work above), people feel like zero is insignificant, and feel no need to keep copying this from one line to the next. This mistake does not happen when using Strategy 1.4.1, which gives another compelling reason (besides speed) to use that strategy. The mistake made earlier never seems to happen in an equation without fractions, but just to demonstrate and to intentionally try to create that error, if someone is asked to solve $5x + 6 = 3x - 10$ and someone shows as their steps

$$5x + 6 = 3x - 10$$
$$5x + 6 - 3x + 10 = 0$$
$$= 2x + 16$$

and declares that their final answer is $2x + 16$, that seems strange.

Example 1.4.11

Solve $\frac{x}{6} + \frac{x-3}{10} = \frac{4}{3}$.

Solution. After multiplying both sides by 30, effectively, we can think about what cancellations are created by multiplying each term by 30. The new first term would be $5x$, the new second term would be $3(x-3)$, and the new third term would be $10 \cdot 4$. So, we have:

$$5x + 3(x-3) = 10 \cdot 4$$
$$5x + 3x - 9 = 40$$
$$8x - 9 = 40$$
$$8x = 49$$
$$x = \frac{49}{8}$$

Just to be thorough, the steps that we didn't show were right at the beginning, namely,

$$30\left(\frac{x}{6}+\frac{x-3}{10}\right)=30\cdot\frac{4}{3}$$
$$30\cdot\frac{x}{6}+30\cdot\frac{x-3}{10}=30\cdot\frac{4}{3}$$

Example 1.4.12

Solve $\frac{2x+1}{3}-\frac{x-1}{4}=\frac{x+2}{6}+\frac{x}{3}$.

Solution. The least common multiple of 3 and 4 and 6 and 3 is 12. Multiplying both sides by 12 gives:

$$12\left(\frac{2x+1}{3}-\frac{x-1}{4}\right)=12\left(\frac{x+2}{6}+\frac{x}{3}\right)$$

Distributing on both sides gives:

$$12\cdot\frac{2x+1}{3}-12\cdot\frac{x-1}{4}=12\cdot\frac{x+2}{6}+12\cdot\frac{x}{3}$$

After cancelling, we have:

$$4(2x+1)-3(x-1)=2(x+2)+4x$$
$$8x+4-3x+3=2x+4+4x$$
$$5x+7=6x+4$$
$$3=x$$

The original equation was slightly different from our earlier examples in that the original equation had multiple fractions on each side. However, Strategy 1.4.1 still applies: the strategy applies to any equation with fractions, no matter how many fractions are on each side.

Principle 1.4.13 Expectation: equations where the variable appears in the denominator

*If the original equation had variables in the denominator, check to ensure there are no false solutions. (Any solution which causes division by zero to occur is a **false solution**.)*

Example 1.4.14

Solve $\frac{3}{2x-4}-\frac{5}{x+3}=\frac{2}{x-2}$

Solution. Because the equation we are asked to solve has fractions, we use Strategy 1.4.1. It will be easier to compute the least common multiple of the denominators if we factor the first denominator: $2x-4=2(x-2)$. The least common multiple of $2(x-2)$ and $x+3$ and $x-2$ is $2(x-2)(x+3)$. So we will multiply both sides by $2(x-2)(x+3)$.

$$2(x-2)(x+3)\left(\frac{3}{2(x-2)}-\frac{5}{x+3}\right)=2(x-2)(x+3)\cdot\frac{2}{x-2}$$

Just a friendly reminder: the original left side had two terms, so due to Order of Operations, to indicate that we are multiplying the entire left side, we wrapped the original left side in parentheses. Now we distribute $2(x-2)(x+3)$. The language being used here can be confusing: I don't mean that we look at $2(x-2)(x+3)$ and expand this in any way, which is not only valid, this is a completely reasonable interpretation of distributing in this context. I mean that we place a copy of $2(x-2)(x+3)$ to the side of each of the original terms on the left.

$$2(x-2)(x+3)\cdot\frac{3}{2(x-2)}-2(x-2)(x+3)\cdot\frac{5}{x+3}=2(x-2)(x+3)\cdot\frac{2}{x-2}$$

In the first term (which is everything before the minus sign), a factor of 2 and a factor of $(x-2)$ cancel on top and bottom. In the second term, a factor of $(x+3)$ cancels. On the right side, a factor of $(x-2)$ cancels. We are left with

$$(x+3)\cdot 3-2\cdot 5(x-2)=2\cdot 2\cdot(x+3)$$

$$\begin{aligned}3(x+3)-10(x-2)&=4(x+3)\\3x+9-10x+20&=4x+12\\-7x+29&=4x+12\\17&=11x\\\frac{17}{11}&=x\end{aligned}$$

Since the original equation had the variable written in the denominator, Principle 1.4.13 applies, and we need to check that the solution that we got doesn't cause any denominator to be zero.

- The first denominator is $2x-4$. When we put in $x=\frac{17}{11}$, we see $2\cdot\frac{17}{11}-4\neq 0$.
- When we put in $x=\frac{17}{11}$ into the second denominator $x+3$, we see $\frac{17}{11}+3\neq 0$.
- When we put in $x=\frac{17}{11}$ into the third denominator $x-2$, we see $\frac{17}{11}-2\neq 0$.

We checked that $x=\frac{17}{11}$ does not cause any of the three denominators to simplify to zero. Therefore, $x=\frac{17}{11}$ is a solution.

Example 1.4.15

Solve $\frac{3}{x+1} - \frac{1}{2} = \frac{1}{3x+3}$

Solution. The third denominator in factored form is $3(x+1)$. The least common multiple of $x+1$ and 2 and $3(x+1)$ is $2 \cdot 3 \cdot (x+1)$, or in other words, $6(x+1)$. We multiply both sides by $6(x+1)$, noting that we have to wrap the left side in parentheses:

$$6(x+1)\left(\frac{3}{x+1} - \frac{1}{2}\right) = 6(x+1) \cdot \frac{1}{3(x+1)}$$

Now we distribute $6(x+1)$. By this, I'm not saying that we write $6x+6$. Instead, I mean that a copy of $6(x+1)$ is written to the side of both fractinos on the left:

$$6(x+1) \cdot \frac{3}{x+1} - 6(x+1) \cdot \frac{1}{2} = 6(x+1) \cdot \frac{1}{3(x+1)}$$

After processing cancellations,

$$18 - 3(x+1) = 2$$

$$18 - 3x - 3 = 2$$

Now, there are multiple ways to proceed, so don't fret if your next steps look different from my next steps:

$$15 - 3x = 2$$

$$12 = 3x$$

$$4 = x$$

Since the original equation had the variable written in the denominator, Principle 1.4.13 applies, and we need to check that the solution that we got doesn't cause any denominator to be zero.

- When we put in $x = 4$ into the first denominator $x+1$, we see $4+1 \neq 0$.
- The second denominator is 2 without needing to substitute for x, since there was no x in the second denominator. Note $2 \neq 0$.
- When we put in $x = 4$ into the third denominator $3x+3$, we see $3 \cdot 4 + 3 = 15 \neq 0$.

We checked that $x = 4$ does not cause any of the three denominators to become zero. Therefore, $x = 4$ is a solution.

Example 1.4.16

Solve $\frac{10x}{x-2} - \frac{13}{x-2} = \frac{14}{2x-4}$

Solution. The third denominator in factored form is $2(x-2)$. The least common multiple of $x-2$ and $x-2$ and $2x-4$ is $2(x-2)$. We multiply both sides by $2(x-2)$, noting that we have to wrap the left side in parentheses:

$$2(x-2)\left(\frac{10x}{x-2} - \frac{13}{x-2}\right) = 2(x-2) \cdot \frac{14}{2(x-2)}$$

Now we distribute $2(x-2)$. By this, I'm not saying that we write $2x-4$. Instead, I mean that a copy of $2(x-2)$ is written to the side of both fractions on the left:

$$2(x-2) \cdot \frac{10x}{x-2} - 2(x-2) \cdot \frac{13}{x-2} = 2(x-2) \cdot \frac{14}{2(x-2)}$$

After processing cancellations,

$$2 \cdot 10x - 2 \cdot 13 = 14$$

$$20x - 26 = 14$$

$$20x = 40$$

$$x = 2$$

Since the original equation had the variable written in the denominator, Principle 1.4.13 applies, and we need to check that the solution that we got doesn't cause any denominator to be zero. When we put in $x = 2$ into the first denominator $x-2$, we see $2-2 = 0$. Therefore, $x = 2$ is a false solution, and the equation has no solution. (We don't even need to bother checking the other denominators: any time one or more denominators become zero, the solution is false.) It turns out that the equation $\frac{10x}{x-2} - \frac{13}{x-2} = \frac{14}{2x-4}$ has no solution.

Note 1.4.17 At this point, we have two different concepts of checking an equation. To be transparent with you, it's time for some clarification. Let's compare:

- In Principle 1.4.13, we gave an expectation, or really a requirement: after solving an equation where the variable appears in the denominator, we are required to check that the value(s) obtanied for x do not cause any denominator to equal zero.

- In Principle 1.3.40, we mentioned that you can always see if the value(s) of x make the left and right side of the original equation to be equal.

This check is never required, but it is recommended. If we encounter a time when the left and right side do not end up being equal, we should scan our work for any errors. After all, we're all human!

The purpose of discussing this is to spell out that the two checks are checking for different things. In the first case, we're just checking to see that denominators are never zero, and this check is required whenever the variable appears in the denominator. In the second, we're checking for something else (do the left and right sides of the original equation equal each other), and only because getting different values for the left and right hints to us that there's an error in our work, and we should go over it. In fact, we only completed the first kind of check in Example 1.4.14, and we didn't do the second kind of check mentioned in Principle 1.3.40. To illustrate that point more fully, we can check that putting in $\frac{17}{11}$ in for x into both the left side $\frac{3}{2x-4} - \frac{5}{x+3}$ and the right side $\frac{2}{x-2}$ shows that both sides equal $\frac{22}{-5}$ or from a calculator, -4.4.

In summary, when the variable appears in the denominator, we must check that we never have zero in the denominator due to the value(s) of x. In any equation, we can always check to see if the original left and right sides are equal for the sake of quality control.

Warning 1.4.18 In an equation with fractions where all fractions have the same denominator, we cannot say that we are "deleting" all the denominators.

Example 1.4.6(((xref without ref, first/last, or provisional attribute (check spelling))))

$$\frac{5x}{30} + \frac{3x-9}{30} = \frac{40}{30}.$$

$5x + 3x - 9 = 40$"just got rid of"looksmultiplied both sides by 30"got rid of denominators"

Example 1.4.19

When asked to solve

$$\frac{x}{5} \cdot \frac{3}{5} = \frac{2}{5},$$

a well-meaning but incorrect thought is to say that since all three fractions have the same denominator, we can "get rid of" the denominators. This is incorrect, but let's follow through with this. This would lead to the equation $x \cdot 3 = 2$, which eventually gets us to $x = \frac{2}{3}$. Now we can follow up by checking the original equation like Principle 1.3.40 encourages us to do. Putting in $\frac{2}{3}$ for x into the left side gives $\frac{\frac{2}{3}}{5} \cdot \frac{3}{5} = \frac{2}{15}$. Putting in $\frac{2}{3}$ for x into the right side gives $\frac{2}{5}$. Since $\frac{2}{15} \neq \frac{2}{5}$, we know that $x = \frac{2}{3}$ is not a solution.

Example 1.4.20

When asked to solve the equation

$$\frac{x}{2} \cdot \frac{5}{2} + \frac{7}{2} = \frac{3x}{2},$$

a well-meaning but incorrect thought is to say that since all four fractions have the same denominator, we can "get rid of" the denominators. This is would lead to $x \cdot 5 + 7 = 3x$ which gives us $5x + 7 = 3x$ so $2x = -7$ and $x = -\frac{7}{2}$. Let's following Principle 1.3.40. Putting in $-\frac{7}{2}$ for x into the left side gives $\frac{-\frac{7}{2}}{2} \cdot \frac{5}{2} + \frac{7}{2} = -\frac{7}{4} \cdot \frac{5}{2} + \frac{7}{2} = -\frac{35}{8} + \frac{28}{8} = -\frac{7}{8}$. Putting in $-\frac{7}{2}$ for x into the right side gives $\frac{3 \cdot -\frac{7}{2}}{2} = -\frac{21}{4}$. Because the left and right sides simplified to different values, $x = -\frac{7}{2}$ is not a solution. The problem was created by the incorrect idea that we could "get rid of" the denominators.

The two examples above illustrate that we cannot say that we "get rid of" denominators when all fractions have the same denominator. (Our last two examples show that we can't just "delete" denominators when they are all the same, and we certainly can't delete denominators when fractions have different denominators). We're never erasing denominators. Instead, we stick to the only tool we really have for equations: which is to apply the same action to both sides. The action of multiplying on both sides helps create the possibility that cancellations can occur: the newly introduced factor in the numerator (the number we multiply both sides by) works to cancel with factors in the denominator. In fact, the reason that Strategy 1.4.1 works is that we are multiplying both sides by the least common multiple of all denominators, and this quantity is guaranteed to have all the factors needed to create cancellations in every denominator.

Note 1.4.21 Up until now, we have avoided talking about cross multiplying. The reason for waiting for so long to talk about this is that there is a common setting in which learners try to apply cross multiplying where the techinque is not valid. Specifically, given the task of simplifying the multiplying of two fractions, learners sometimes try to cross multiply. Let's dig into an example:

1. When asked to simplify $\frac{2}{3} \cdot \frac{5}{7}$, a well-meaning but incorrect thought is to say that we can cross multiply. This is incorrect, but let's follow through with this. This would lead to the expression $\frac{2 \cdot 7}{3 \cdot 5} = \frac{14}{15}$.

2. Before immediately dismissing this answer, let's talk about it. I believe that you have the background to follow the discussion we're about to make, and taking the time to actively read this discussion will be empowering for building your confidence in math, and moreover, can help you avoid mistakes in the future.

Say we take the same problem about simplifying the product of fractions, but just switch the order of the fractions:

$$\frac{5}{7} \cdot \frac{2}{3}.$$

Then, applying cross multiplying would lead to $\frac{5 \cdot 3}{7 \cdot 2} = \frac{15}{14}$. This is a different answer from the previous answer of $\frac{14}{15}$. This seems strange, because we know the multiplication is commutative, either from our everyday experience with examples like $2 \cdot 3 = 3 \cdot 2$, or from giving a geometric argument that the Commutative Property of Multiplication in Principle 1.1.57 is true in general. Because we got different results from cross multiplying, it's reasonable for us to be suspicious of our two answers, and the technique in general. In a moment, we will in fact see that neither $\frac{14}{15}$ nor $\frac{15}{14}$ is the correct final answer. Cross multiplying is not a valid technique for simplifying the product of two fractions.

3. To simplify $\frac{2}{3} \cdot \frac{5}{7}$, recall that Principle 1.2.56 tells us to multiply straight arcoss:

$$\frac{2}{3} \cdot \frac{5}{7} = \frac{2 \cdot 5}{3 \cdot 7} = \frac{10}{21}.$$

Cross multiplying is really just a special case of Strategy 1.4.1. Cross multiplying only applies when there are two fractions in an equation, one on each side of the equal sign. In this case, the least common multiple of the two denominators is simply the product of the two denominators. So, cross multiplying is just multiplying both sides by the product of the two denominators. This is exactly what Strategy 1.4.1 tells us to do. So, cross multiplying is not a different technique, but rather a special case of Strategy 1.4.1. Because cross multiplying is so limited in its application, and because it is just a special case of Strategy 1.4.1, we have avoided talking about it. More importantly, we have avoided talking about it because of the common mistake of trying to apply it to simplify the multiplication of two fractions.

Example 1.4.22

Solve the equation $\frac{2x-1}{3} = \frac{5x+6}{4}$.

Solution 1. Because we have an equation with a single fraction on each side, we can use the cross multiplying technique that we have avoided talking about until now. This leads to $4(2x-1) = 3(5x+6)$. So:

$$\begin{aligned} 8x - 4 &= 15x + 18 \\ -22 &= 7x \\ x &= -\frac{22}{7} \end{aligned}$$

Solution 2. Alternatively, we can use Strategy 1.4.1. The least common multiple of 3 and 4 is 12. Multiplying both sides by $3 \cdot 4$ gives:

$$3 \cdot 4 \cdot \frac{2x-1}{3} = 3 \cdot 4 \cdot \frac{5x+6}{4}$$

After cancelling, we have:

$$4(2x-1) = 3(5x+6)$$

$$8x - 4 = 15x + 18$$

$$-22 = 7x$$

$$x = -\frac{22}{7}$$

In the last example, I admit that cross multiplying got us slightly faster to the equation $4(2x-1) = 3(5x+6)$ than Strategy 1.4.1 did. However, the difference in speed is very small, and cross multiplying only applies in a very limited setting. People, both students and instructors alike, will differ in their opinions about cross multiplying. If you wish to use this techinque, it's important to "read the fine print". This would be like knowing the side effects of a medication: it's just good to be informed. In the case of cross multiplying, it cannot be used to simplify the multiplication of two fractions, and it only applies to solving an equation when there are exactly two fractions in an equation, one on each side of the equal sign, and each side consists only of a fraction. Furthermore, cross multiplying is just a special case of Strategy 1.4.1, which applies to any equation with fractions.

A word on writing.

When we worked on solving $\frac{x}{6} + \frac{1}{2} = \frac{2}{3}$ in Example 1.4.5, one of our first steps was:

$$6\left(\frac{x}{6} + \frac{1}{2}\right) = 6 \cdot \frac{2}{3}.$$

We then distributed (which ended up removing the parentheses) then canceled. It would be strange to not yet distribute and write in cancellations like this:

$$\cancel{6}\left(\frac{x}{\cancel{6}} + \frac{1}{2}\right) = 6 \cdot \frac{2}{3}.$$

Work that looks like this gets dangerous, since we need to distribute 6 first. Distributing 6 creates two copies of 6 on the left, and it's actually one of the copies from the distribution that cancels with the 6 in the denominator of the first fraction. Let's think of the parentheses for what they really are: we can say that these grouping symbols are protecting the content inside, and preventing this kind of cancellation.

1.4.3 SUMMARY

1. A technique that always works for solving equations with fractions is to first compute the least common multiple of all denominators, then multiply both sides by this least common multiple. (Insisting on common denominators is always more time consuming, and there are several types of errors that can occur when insisting on common denominators. When we discussed simplifying fractional expressions, we even saw back then that multiplying fractions does not require a common denominator.)

2. Cross multiplying is a special case of multiplying both sides by the least common multiple of the denominators. This technique only applies when each side of the equation has exactly one fraction and nothing else. This technique is what sometimes people erroneously think of when working simplifying the product of fractions, which is why some people avoid talking about cross multiplying.

1.4.4 EXERCISES

1. Solve $\frac{x}{2} = \frac{5}{6}$.
2. Solve $\frac{x}{3} + \frac{1}{6} = \frac{5}{6}$.
3. Solve $\frac{x+1}{4} + \frac{x-2}{3} = \frac{5}{6}$.
4. Solve $\frac{2x+1}{5} - \frac{x-1}{4} = \frac{x+2}{10} + \frac{x}{5}$.
5. Solve $\frac{4x-1}{3} = \frac{2x+5}{4}$.
6. Solve $\frac{x}{5} + \frac{2}{3} = \frac{19}{15}$.
7. Solve $\frac{3}{x} = \frac{6}{5}$.
8. Solve $\frac{2}{x-1} = \frac{5}{x+2}$.
9. Solve $\frac{x+3}{7} = \frac{2x-1}{5}$.
10. Solve $\frac{x}{4} - \frac{3}{8} = \frac{5}{8}$.
11. Solve $\frac{5}{x} + \frac{2}{3} = \frac{19}{6}$.
12. Solve $\frac{3}{x+2} - \frac{5}{x-1} = \frac{2}{x-4}$.
13. Solve $\frac{x-4}{6} = \frac{2x+1}{9}$.
14. Solve $\frac{7}{2x} = \frac{3}{x+1}$.
15. Solve $\frac{2}{x} - \frac{3}{x+1} = \frac{1}{2}$.
16. Solve $\frac{x+5}{x-2} = \frac{3}{2}$.

17. Solve $\frac{2}{x-3} + \frac{1}{x+1} = \frac{3}{4}$.

18. Solve $\frac{x+1}{x-4} = \frac{x-2}{x+3}$.

19. Solve $\frac{2x}{x-1} = \frac{3}{2}$.

20. Solve $\frac{1}{x-2} + \frac{1}{x+2} = \frac{1}{3}$.

1.5 Exponents and Radicals

Objectives

In this section, we learn how to:

1. Simplify and rewrite expressions containing exponents.
2. Simplify and rewrite expressions containing radicals.

1.5.1 Applications

After having covered the content of this section, we will be able to answer these questions:

1. You deposit 100 dollars into an account earning 5% interest compounded continuously. How much money is in the account at the end of 3 years, and what is the factor by which the amount in the account grows on an annual basis?
2. As an immunologist, you start with a culture of 1000 bacteria. The population doubles every 3 hours. How many bacteria are there after 12 hours, and by what factor does the population grow in a single hour?
3. You are filling small and large boxes to move apartments. Each small box is a cube with side length s. If you double the side length, by what factor does the volume of the cube increase? This will tell us how much more can fit into the larger box over the smaller box.
4. Now that you're in your new apartment, you want to build in a bookshelf as large as you can. You will bring in a slab of wood "diagonally" through a door that has a width of 3 feet and a height of 6 feet. What is the diagonal length of the door, and thus what is the longest piece of wood you can bring in through the door?
5. Continuing with the bookshelf, you have identical slabs of wood coming from a previous project. These slabs have their width determined by taking the hypotenuse of a right triangle with side lengths 1 foot and 2 feet. (The measurements of 1 foot and 2 feet came from the height and width of a storage space in your previous apartment.) How many of these slabs should be glued together side-by-side to create the back of the bookshelf in the previous question?

1.5.2 Exponents

Writing expressions involving exponents provides a powerful way to describe certain quantifiable phenomena in a variety of applications. We will need some exponent rules that help us rewrite expressions involving exponents. These rules will also help us simplify expressions involving exponents. Before introducing the rules, let's make sure we're on the same page about what an exponent means.

Principle 1.5.1 What does an exponent mean?

Exponentiation can be interpreted to mean repeated multiplication. An exponent is a way to represent repeated multiplication of a number by itself: a^n means a is multiplied by itself a total of n times, where n is a positive integer.

This principle only really makes sense when n is a positive integer such as $1, 2, 3, \ldots$. For example, a^4 means $a \cdot a \cdot a \cdot a$, and a^6 means $a \cdot a \cdot a \cdot a \cdot a \cdot a$. We will later talk about situations where the exponent is zero, a negative number, or a fraction. In those stranger situations, Principle 1.5.1 doesn't apply.

Note 1.5.2 In the desire to make sure we have really said everything (no math secrets here!), let's also say that $a^1 = a$. It makes sense that if $a^3 = a \cdot a \cdot a$ and $a^2 = a \cdot a$, then to follow the pattern a^1 must equal a. So $u^1 = u$ and $8^1 = 8$ and $(x + 7)^1 = x + 7$, for example.

There are times that we'll need to simplify an expression like p^1 as p. There will actually be other times we need to go in reverse, turning p into p^1. Informally, we can think of it this way: whenever we see an expression, if there is no exponent explicitly written in, we can always write in an exponent of 1 explicitly. I like to call this the "hidden exponent of 1", analogous to the "hidden denominator of 1" that we discussed in Note 1.2.4.

Principle 1.5.3 Exponent rules, part 1.

- $a^m a^n = a^{m+n}$
- $(a^m)^n = a^{mn}$

Reading from left to right, the first formula says that when multiplying two expressions with the same base, we can add the exponents (in which case we write the base once). This is why the left side has $a^m a^n$ where the same base is a appears twice, while the right side has a^{m+n} where the base a appears only once. The first formula's requirement requiring that the two expressions being multiplied together have the same base is analogous to the requirement that two fractions being added or subtracted must have the same denominator. Reading from right to left, an expression of the form a^{m+n} can be rewritten as $a^m a^n$.

Reading from left to right, the second formula says that when taking an exponential expression such as a^m and raise it to a power, we can multiply the exponents. Read from right to left, an expression of the form a^{mn} can be rewritten as $(a^m)^n$, meaning that when we see an expression with an exponent that is a product, we can rewrite it as an expression where the base is raised to one of the factors, and then the entire expression wrapped in parentheses is raised to the other factor.

Example 1.5.4

Simplify $(a^3)^5$.

Solution. Using the exponent rule $(a^m)^n = a^{mn}$, we have $(a^3)^5 = a^{3\cdot 5} = a^{15}$.

Example 1.5.5

Simplify a^3a^5.

Solution. Using the exponent rule $a^m a^n = a^{m+n}$, we have $a^3a^5 = a^{3+5} = a^8$.

Example 1.5.6

Simplify $2(x^4)^2 + 5x^4x^2 + 10x^6 + 20x^8$.

Solution. Using the exponent rule $(a^m)^n = a^{mn}$, we have $(x^4)^2 = x^{4\cdot 2} = x^8$. Thus, $2(x^4)^2 = 2x^8$.

Next, using the exponent rule $a^m a^n = a^{m+n}$, we have $5x^4x^2 = 5x^{4+2} = 5x^6$.

The third term and fourth term are already simplified: $10x^6$ and $20x^8$.

Now we can combine like terms: $2x^8 + 5x^6 + 10x^6 + 20x^8 = (2+20)x^8 + (5+10)x^6 = 22x^8 + 15x^6$.

Putting all of this together, from the initial expression to our final answer, with equal signs where they belong, we have $2(x^4)^2 + 5x^4x^2 + 10x^6 + 20x^8 = 2x^8 + 5x^6 + 10x^6 + 20x^8 = 22x^8 + 15x^6$.

An example like the one we just did is meant to provide a bit of caution. After applying exponent rules, we get to the expression $2x^8 + 5x^6 + 10x^6 + 20x^8$. It is tempting to say that the final expression is $37x^{14}$. Combine only what we can (which is just up to like terms).

Example 1.5.7

Simplify $c^7c^{1/2}$.

Solution. Using the exponent rule $a^m a^n = a^{m+n}$, we have $c^7c^{1/2} = c^{7+\frac{1}{2}} = c^{\frac{14}{2}+\frac{1}{2}} = c^{\frac{15}{2}}$.

In a simpler example, namely Example 1.5.5, we wrote $a^3a^5 = a^{3+5} = a^8$, though many prefer to skip the "middle step" and directly go from a^3a^5 to a^8. This is fine to do when we can do the simplification in our heads, but in the example we just did, it might be harder to see how the sum of the exponents simplifies. That's why even in the easier problem, we wrote out the step. In a situation like this that's more complicated, it is practical to write the addition (without having simplified yet), and then in the next step(s), we can "zoom in" focusing only in the exponent area, and simplify the exponent from $7+\frac{1}{2}$ to $\frac{14}{2}+\frac{1}{2}$ to $\frac{15}{2}$. Even if you see it in this example without writing all the steps, there's definitely eventually going to be a challenging enough example where we can't just picture it all. Being willing to write that two things are added together without immediately simplifying it is a power tool to have.

Note 1.5.8 In this example, we wrote some fractions using a slash notation. Back in Principle 1.2.92, we discussed that writing slanted fraction bars can be ambiguous. So, it's not exactly a rule that we can never write fractions using a slanted fraction bar, but that we should just be cautious. In the case of $c^{\frac{1}{2}}$, because the font for the exponent is tiny, to avoid having someone think that this was c times $\frac{1}{2}$, some people prefer to write and/or type $c^{1/2}$.

However, writing slanted fractions for exponents isn't always the right choice. See, for instance, the writing below and compare this to the writing in the answer: $c^7c^{1/2}=c^{7+1/2}=c^{14/2+1/2}=c^{15/2}$

The point is that writing fractions with a slanted bar is sometimes used to compensate for the small space in an exponent area, making the expression easier to read. However, based on what we learned in Principle 1.2.92, we should be mindful of what we write, especially the types of common ambiguities that were mentioned in the examples after Principle 1.2.92.

Example 1.5.9

Simplify $m^4m^{\frac{3}{10}}m^{\frac{2}{5}}$.

Solution.

$$\begin{aligned}
&m^4m^{\frac{3}{10}}m^{\frac{2}{5}}\\
&=m^{4+\frac{3}{10}+\frac{2}{5}}\\
&=m^{\frac{40}{10}+\frac{3}{10}+\frac{4}{10}}\\
&=m^{\frac{47}{10}}
\end{aligned}$$

Example 1.5.10

Simplify $m^4(m^{\frac{3}{10}})^{\frac{2}{5}}$.

Solution.

$$\begin{aligned}
&m^4(m^{\frac{3}{10}})^{\frac{2}{5}}\\
&=m^4m^{\frac{3}{10}\cdot\frac{2}{5}}
\end{aligned}$$

$$
\begin{aligned}
&= m^4 m^{\frac{6}{50}} \\
&= m^{4+\frac{6}{50}} \\
&= m^{\frac{200}{50}+\frac{6}{50}} \\
&= m^{\frac{206}{50}}
\end{aligned}
$$

Unlike the previous question, notice that we had to use both exponent rules in Principle 1.5.3 to simplify this expression.

We showed two standard formulas in Principle 1.5.3 how they usually look: $a^m a^n = a^{m+n}$ and $(a^m)^n = a^{mn}$. However, these two are often confused for each other, and it's easier to see why people confuse these two formulas when we see all operations written in, with none hidden:

1. $a^m \cdot a^n = a^{m+n}$
2. $(a^m)^n = a^{m \cdot n}$

These are actually the same formulas, and I admit they look bulkier with all the operations explicitly shown. But this helps us see why there is confusion! Notice that the formula that has multiplication on the left has addition on the right, while the second formula has multiplication the right. So, it is incorrect to turn $c^2 \cdot c^3$ into $c^{2\cdot 3}$ which would eventually be c^6, but I can understand why people would make that error! (Instead, the correct simplification is $c^2 \cdot c^3 = c^{2+3} = c^5$.)

This leads us to ask a more general question: how are we supposed to know what to do when we see an expression of the form $a^m \cdot a^n$ versus what we are supposed to do when we see an expression of the form $(a^m)^n$? The short answer (just memorize the formulas) is really unsatisfying. It's easy to "cross the wires" here. Fortunately, there is a tool that we have that can help us remember how each of these formulas end, and more importantly, if we keep practicing the process that we're about to suggest, it actually provides a reason why these formulas are true in the first place! That's a win-win!

Principle 1.5.11 How to keep track of exponent formulas.

For the two formulas we presented in Principle 1.5.3, the left sides said $a^m \cdot a^n$ and $(a^m)^n$. How can know what is on the right side of each formula? What can we do when we've forgotten what's on the right side of each formula? And finally, how can we know why the right sides that get presented are actually legitimate in the first place? The answer to all of these questions is to "expand" the expression after selecting your own constants for the exponents (but not for the base), and expand based on Principle 1.5.1.

Example 1.5.12

Suppose we recall that there is a formula that starts with $a^m \cdot a^n$, but we can't remember what is on the right side of the formula. Say we also recall there is a formula that starts with $(a^m)^n$, but we can't remember what is on the right side of that formula either. Which of these two expressions involves adding the exponents, and which involes multiplying the exponents? That is, which has a power of $m+n$ and which has a power of $m \cdot n$? To figure this out, we can pick our own values for m and n, and then expand each expression based on Principle 1.5.1. While we can pick m and n to be as large as we want, let's pick small values to make the expansion easier. That said, we can't pick m and n to be too small: for example if we picked both $m = 2$ and $n = 2$, both $m + n$ and $m \cdot n$ would be equal to 4, which wouldn't help us distinguish between the two formulas.

Let's pick $m = 2$ and $n = 3$, which is about as minimal as we can pick. Based on what exponents mean in Principle 1.5.1:

1. We can expand $a^m \cdot a^n$ as $a^2 \cdot a^3 = (a \cdot a) \cdot (a \cdot a \cdot a) = a \cdot a \cdot a \cdot a \cdot a = a^5$.
2. We can expand $(a^m)^n$ as $(a^2)^3 = (a \cdot a)^3 = (a \cdot a) \cdot (a \cdot a) \cdot (a \cdot a) = a \cdot a \cdot a \cdot a \cdot a \cdot a = a^6$.

Noting that $5 = 2 + 3$ and $6 = 2 \cdot 3$, we see that $a^m \cdot a^n$ involves adding the exponents, while $(a^m)^n$ involves multiplying the exponents.

Note 1.5.13 Note that we never picked a specific value for a. It's not that we can't. We can! But, if we did (say we picked $a = 5$), then we'd be looking at an expression like $5^2 \cdot 5^3$, and with constants in all places, it's too tempting to work on simplyfing arithmetic, which is both time-consuming, and hides what it is we're trying to get out of this. The point is that we just want to count how many times the base a appears as a factor, and record that number of appearances as an exponent.

Example 1.5.14

Say we need to simplify $x^{\frac{1}{3}} x^{\frac{2}{5}}$, but forgot how the formulas go. Go through a process to recover the formula, then apply it it to simplify this expression.

Solution. Suppose we forgot how the formulas go. The shape of what we have is multiplying to expressions that have the same base: this is addressed by $a^m \cdot a^n$. The question becomes whether we add or multiply the exponents. To figure this out, we can work on the side where we temporarily pick our own values for m and n. Let's say we pick $m = 4$ and $n = 3$, which are relatively small numbers. Then, based on Principle 1.5.1, we can expand $a^m \cdot a^n$ as

$a^4 \cdot a^3 = (a \cdot a \cdot a \cdot a) \cdot (a \cdot a \cdot a) = a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a$ and counting how many times the base is a factor, this is a^7. So, in the case of $a^m \cdot a^n$, we add the exponents, which is why the full formula (which we might forget, but we just recovered through this process) is $a^m \cdot a^n = a^{m+n}$.

Now we can apply this formula: $x^{\frac{1}{3}} x^{\frac{2}{5}} = x^{\frac{1}{3}+\frac{2}{5}} = x^{\frac{5}{15}+\frac{6}{15}} = x^{\frac{11}{15}}$. Because it is challenging to simplify the addition of these fractions right away, we just first wrote the addition without simplifying it by writing $x^{\frac{1}{3}+\frac{2}{5}}$, though we still needed to write x as the base in order to ensure that the equal sign truly means equal. In the next step, we focused on simplifying the exponent area (by mentally zooming in) and found a common denominator.

The natural question arises: it's easy to remember what to do to get these formulas when the author or teacher hints to me what to do, but how am I supposed to recall? In the end, the two formulas we introduced (and several more!) all can be recovered using the same tool: select values for exponents (but not for the base), and expand based on asking yourself what exponents mean (Principle 1.5.1), then count the number of times the base appears. This works for the two formulas we introduced, and it also works for several more formulas that we will introduce next. Before showing the new formulas, it would be good to put this useful tool into a box!

Principle 1.5.15

Based on what exponents mean in Principle 1.5.1

- *Select values for exponents. (To avoid the temptation to do unnecessary arithmetic, do not select a value for base(s).)*

  Expand based on what exponents mean.

- *Count the number of times the base appears, and rewrite based on what exponents mean.*

Let's introduce a couple new formulas where the tool in Principle 1.5.15 applies. Let's then demonstrate full examples of how to practice this tool, then do full examples of computations based on the new formulas.

Principle 1.5.16 Exponent rules, part 2.

- $(a \cdot b)^n = a^n \cdot b^n$
- $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$

It would be easy to fall into the temptation of saying "Well, if I don't remember what's on the other side of one of these exponent formulas, I can just look it up." Let me be the first to admit that, yes, of course you can look it up in certain situations.

However, as this algebra course progresses, we will be looking at more involved problems, and in these problems, applyning exponent rules will need to happen as one pieces of the process. It will really slow us down if we don't gain a little bit of familiarity with these formulas. (Imagine if a professional chef who has to keep inventing new recipes works from the assumption of saying "I don't want to spend the brain space to think about the differences of the taste of basil, cinnamon, rosemary, and anise." Sure, anyone -- including the chef -- can go online and find descriptions of all four of these spices. But intentionally lacking familiarity is going to slow this cook down.) Just like in speaking a foreign language, learning these exponent formulas is going to take some practice through repetition. It won't stick just looking at them once or studying them once. Before we dig into what's next, in the hopes that you'll really read what's next with effort, I also just want to say that it's hard for anyone to memorize a thing when there's no meaning behind it. It would be hard to go in a room full of thirty strangers and memorize each person's name, address, and phone number if each person told you their info once. The reason is that all the data is just disconnected: the phone number you learn for a stranger will just be a bunch of digits. These exponent formulas can seem the same way if there's no meaning attached to it. And people try doing that! People try to memorize these formulas, and struggle to do so. Instead, let's practice a technique that helps us memorize these formulas better each time we practice what we're about to do, and more importantly, it does the job not only by having us stare (like we often passively do with flashcards), but instead shows us the meaning behind the notation which convinces us why the formula is true in the first place!

Example 1.5.17

Suppose that in the course of working on an algebra problem, we encounter the expression $\left(\frac{c}{2}\right)^5$. Based on the shape of that expression, we think about a formula where one side says $(ab)^n$. If we have forgotten the other side of that formula, how do we recover it? Then, how do we apply the recovered formula to our expression?

Solution. The original expression $\left(\frac{c}{2}\right)^5$ is in the format of $\left(\frac{a}{b}\right)^n$ because the original expression is a fraction (numerator of denominator) raised to a power.

We want to figure out what goes on the other side of the formula that has $\left(\frac{a}{b}\right)^n$ on one side. To apply Principle 1.5.15, we should only pick values for exponent(s), but not for base(s), so we will select something for n but not for a or b. Let's pick $n = 3$, which is a small positive integer. Then, Principle 1.5.1 reminds us what an exponent means, so we can expand $\left(\frac{a}{b}\right)^n$ as

$$\left(\frac{a}{b}\right)^3 = \frac{a}{b} \cdot \frac{a}{b} \cdot \frac{a}{b} = \frac{a \cdot a \cdot a}{b \cdot b \cdot b} = \frac{a^3}{b^3}.$$

Therefore, we have recovered the formula

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

We can apply this formula to $\left(\frac{c}{2}\right)^5$ and we see that

$$\left(\frac{c}{2}\right)^5 = \frac{c^5}{2^5}.$$

We just illustrated using the formula, but because the denominator has no variables, we can simplify slightly, getting $\frac{c^5}{32}$.

Stick with it! This takes practice. You may never have had to do this in algebra before, but I encourage you to try this. This kind of activity actually gives us a real reason to believe the formulas are true in the first place, and it is a lot easier to remember and know a thing that you believe is true with evidence, rather than a thing that someone just told you to believe is true.

Example 1.5.18

A major algebra formula has $(a \cdot b)^n$ on one side. How can we recover what goes on the other side, and in doing so, help reveal why the formula is true in the first place?

Solution. To apply Principle 1.5.15, we should only pick values for exponent(s), but not for base(s), so we will select something for n but not for a or b. Let's pick $n = 4$, which is a small positive integer. Then, Principle 1.5.1 reminds us what an exponent means, so we can expand $(a \cdot b)^n$ as

$$(a \cdot b)^4 = (a \cdot b) \cdot (a \cdot b) \cdot (a \cdot b) \cdot (a \cdot b) = a \cdot a \cdot a \cdot a \cdot b \cdot b \cdot b \cdot b = a^4 b^4.$$

Therefore, we have recovered the formula

$$(a \cdot b)^n = a^n \cdot b^n.$$

Example 1.5.19

Simplify $(rs)^3$.

Solution. Using the exponent rule $(a \cdot b)^n = a^n \cdot b^n$, we have $(rs)^3 = r^3 s^3$.

Example 1.5.20

Simplify $(gh^2)^5$.

Solution. Using the exponent rule $(a \cdot b)^n = a^n \cdot b^n$, we have $(gh^2)^5 = (g)^5(h^2)^5$. Note, in our use of this exponent rule, we treated g as a and h^2 as b, since this assignment of values truly captures the idea that the value of a and the value of b are multiplied together, and that result is taken to the power n. When we wrote $(g)^5(h^2)^5$ we could have skipped the parentheses around the g and write $g^5(h^2)^5$ instead. However, we kept parentheses around h^2. Otherwise, there's too much going on, since in the next step, we use $(a^m)^n = a^{mn}$ to turn $(h^2)^5$ into $h^{2\cdot5}$. It doesn't make sense for us to skip writing the work the very first time we ever need to combine the use of $(a \cdot b)^n = a^n \cdot b^n$ and the the use of $(a^m)^n = a^{mn}$. Then, using the exponent rule $(a^m)^n = a^{mn}$, we have $(h^2)^5 = h^{2\cdot5} = h^{10}$. Based on the discussion above, a thorough answer would be:

$$(gh^2)^5 = g^5(h^2)^5 = g^5h^{2\cdot5} = g^5h^{10}$$

while an answer (that is still based on the discussion above) that skips a little bit more would be:

$$(gh^2)^5 = g^5h^{10}.$$

Example 1.5.21

Simplify $(x^2y^3)^4$.

Solution. First, we use the exponent rule $(a \cdot b)^n = a^n \cdot b^n$ to rewrite $(x^2y^3)^4$ as $(x^2)^4(y^3)^4$. We wrapped both the x^2 and the y^3 in parentheses, for the same reason we discussed in the previosu example. Then, we use the exponent rule $(a^m)^n = a^{mn}$ to rewrite each factor $(x^2)^4 = x^{2\cdot4} = x^8$ and also to rewrite $(y^3)^4 = y^{3\cdot4} = y^{12}$. Showing all of what was explained above, writing with good notation (see where the equal signs are), we have:

$$(x^2y^3)^4 = (x^2)^4(y^3)^4 = x^{2\cdot4}y^{3\cdot4} = x^8y^{12}.$$

Example 1.5.22

Simplify $(3x^2y^3)^4$.

Solution. First, we use the exponent rule $(a \cdot b)^n = a^n \cdot b^n$ to rewrite $(3x^2y^3)^4$ as $(3)^4(x^2)^4(y^3)^4$. In the new expression, $(3)^4 = 3^4 = 81$ by arithmetic. We use the exponent rule $(a^m)^n = a^{mn}$ to rewrite the other two factors: $(x^2)^4 = x^{2\cdot4} = x^8$, and also to rewrite $(y^3)^4 = y^{3\cdot4} = y^{12}$. Showing all of what was explained above, writing with good notation (see where the equal signs are), we have:

$$(3x^2y^3)^4 = (3)^4(x^2)^4(y^3)^4 = 81x^8y^{12}.$$

Example 1.5.23

Simplify $(xy^4)^3x^8(x^3z)^{10}$.

Solution. Using the exponent rule $(a \cdot b)^n = a^n \cdot b^n$, we have $(xy^4)^3 = x^3(y^4)^3 = x^3y^{12}$. Please note that inside the first parentheses where the text is xy^4, we must read this using the Order of Operations, and recognize that the exponent 4 applies only to y, and not to x. Thus, $(xy^4)^3x^8(x^3z)^{10} = (x^3y^{12})x^8(x^3z)^{10}$. Next, using the exponent rule $(a \cdot b)^n = a^n \cdot b^n$ again, we have $(x^3z)^{10} = (x^3)^{10}z^{10} = x^{30}z^{10}$. Thus, $(xy^4)^3x^8(x^3z)^{10} = (x^3y^{12})x^8(x^{30}z^{10})$. Now, using the exponent rule $a^ma^n = a^{m+n}$, we have

$$x^3x^8x^{30} = x^{3+8+30} = x^{41}.$$

Putting all our work together, from starting expression to final answer, with equal signs where they belong, we have: $(xy^4)^3x^8(x^3z)^{10} = (x^3y^{12})x^8(x^{30}z^{10}) = x^{3+8+30}y^{12}z^{10} = x^{41}y^{12}z^{10}$.

Example 1.5.24

Simplify $\left(\frac{b}{c^2d^4}\right)^3$.

Solution. Using the exponent rule $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$, we have $\left(\frac{b}{c^2d^4}\right)^3 = \frac{b^3}{(c^2d^4)^3}$. Then, using the exponent rule $(a \cdot b)^n = a^n \cdot b^n$, we have $(c^2d^4)^3 = (c^2)^3(d^4)^3$. Then, using the exponent rule $(a^m)^n = a^{mn}$, we have $(c^2)^3 = c^{2 \cdot 3} = c^6$ and also $(d^4)^3 = d^{4 \cdot 3} = d^{12}$. Therefore, $\left(\frac{b}{c^2d^4}\right)^3 = \frac{b^3}{c^6d^{12}}$.

Putting all our work together, with equal signs where they belong, we have: $\left(\frac{b}{c^2d^4}\right)^3 = \frac{b^3}{(c^2d^4)^3} = \frac{b^3}{(c^2)^3(d^4)^3} = \frac{b^3}{c^6d^{12}}$.

Note 1.5.25 When a fraction is raised to a power, surrounding the fraction in parentheses is required to avoid ambiguous writing. For example, if we mean that we want to cube the fraction $\frac{x}{y}$, it would be ambiguous to write as in this figure:

$$\frac{x}{y}^{\,3}$$

Figure 1.5.26 A picture of ambuiguous writing

Depending on how long the horizontal fraction bar is drawn and the exact placement of the 3 this might get interpreted as $\frac{x^3}{y}$. However, this is different than $\left(\frac{x}{y}\right)^3$ which is $\frac{x}{y} \cdot \frac{x}{y} \cdot \frac{x}{y}$ which is ultimately $\frac{x^3}{y^3}$, and we note that we could have obtained $\frac{x^3}{y^3}$ directly from $\left(\frac{x}{y}\right)^3$ using the exponent rule $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$. The point here is that $\frac{x^3}{y} \neq \left(\frac{x}{y}\right)^3$, calling attention to the fact that the fraction on the right is equal to $\frac{x^3}{y^3}$.

We can also see how this would affect the expression we examined in the previous example: the expression we were given $\left(\frac{b}{c^2d^4}\right)^3$ is different than $\frac{b^3}{c^2d^4}$. In the former, the entire fraction is raised to the power of 3, and $(\frac{a}{b})^n = \frac{a^n}{b^n}$ tells us we will eventually apply the exponent to both numerator and denominator, while in the latter, only the numerator is raised to the power of 3.

In Principle 1.5.3, we saw $a^m a^n = a^{m+n}$, which in symbols says that multiplying two expressions that have the same base can be rewrtten as a single expression with that same base, with the exponent being the sum of the two starting exponents. Now, we show a formula for dividing two expressions that have the same base, and we see on the other sid eof the formula that we subtract the two starting exponents:

Principle 1.5.27

$\frac{a^m}{a^n} = a^{m-n}$

Example 1.5.28

The algebra formula we just presented has $\frac{a^m}{a^n}$ on one side. How can we recover what goes on the other side, and in doing so, help reveal why the formula is true in the first place?

Solution. To apply Principle 1.5.15, we should only pick values for exponent(s), but not for base(s), so we will select something for m and n but not for a. Let's pick $m = 5$ and $n = 2$. Then, Principle 1.5.1 reminds us what an exponent means, so we can expand $\frac{a^m}{a^n}$ as

$$\frac{a^5}{a^2} = \frac{a \cdot a \cdot a \cdot a \cdot a}{a \cdot a}.$$

Then, we can cancel two factors of a from the numerator and denominator, getting $\frac{a \cdot a \cdot a}{1} = a^3$. We see that in the example where $m = 5$ and $n = 2$, this simplified to an expression where, with the base a written once, the exponent is $5 - 2$. Therefore, we have recovered the formula

$$\frac{a^m}{a^n} = a^{m-n}.$$

Example 1.5.29

Simplify $\frac{c^7}{c^3}$.

Solution 1. While we can use our new formula in Principle 1.5.27, we can skip this formula and instead apply Principle 1.5.1. Based on Principle 1.5.1, we can expand $\frac{c^7}{c^3} = \frac{c \cdot c \cdot c \cdot c \cdot c \cdot c \cdot c}{c \cdot c \cdot c}$. Then, we can cancel three factors of c from the numerator and denominator, getting $\frac{c \cdot c \cdot c \cdot c}{1} = c^4$.

Solution 2. Using the exponent rule $\frac{a^m}{a^n} = a^{m-n}$ in Principle 1.5.27, we have $\frac{c^7}{c^3} = c^{7-3} = c^4$, which leads to the same final answer.

We found two ways to answer the last example. One answer used the formula given in Principle 1.5.27, while the other answer didn't (and only used what exponents mean from Principle 1.5.1). We bring this up for several reasons:

- One natural reaction to hearing "we didn't need the formula" in Principle 1.5.27 is to ask why bother showing this formula in the first place. In a small enough example, it's fine to expand based on Principle 1.5.1, but if we had a larger example like
$$\frac{u^{30000}}{u^{41}}$$
it would be impractical to expand based on Principle 1.5.1, since nobody wants to write out u as a factor 30,000 times in the numerator and 41 times in the denominator. In this case, the formula in Principle 1.5.27 is very useful, which lets us quickly write
$$\frac{u^{30000}}{u^{41}} = u^{30000-41} = u^{29959}.$$

- It is also good to see that we got the same final answer of c^4 both using the new formula and not using the new formula. Getting the same final answer through two different routes (an old route and a new route) gives us more confidence that the new formula is correct.

- Finally, it's just good to keep in mind that there are often multiple ways to answer a question. This is true in math, as it is in life. It is good to be flexible and adaptable. There is often more than one way to approach a problem. Especially in the area of exponents (and radicals, which we will see next), there are often multiple ways to answer a question. I bring this up because in several of the practice problems that are to follow, I will be providing several different ways to answer the same question. In fact, for many of the questions we'll see, there are many possible different first steps! This is not to confuse you, but rather to show you that there are often multiple ways to approach a problem. Moreover, if a question naturally has several different answers (with different starts) that are all equally reasonable as answers (none much longer than the other), then I think it does a disservice to you to not show multiple answers: this might lead you to think that the way you're thinking about the problem doesn't work. My experience working with students has shown me that, especially in the area of exponents and radicals, multiple students start the same problem differently with approaches that all work!

Before digging into practice problems that have different answers, let's do just a little more practice with the formula in Principle 1.5.27.

Example 1.5.30

Simplify $\frac{u^{2/3}}{u^{1/6}}$.

Solution. Since we are dividing two exponential expressions that have the same base, the formula $\frac{a^m}{a^n} = a^{m-n}$ in Principle 1.5.27 applies.

$$\frac{u^{2/3}}{u^{1/6}} = u^{\frac{2}{3}-\frac{1}{6}} = u^{\frac{4}{6}-\frac{1}{6}} = u^{\frac{3}{6}} = u^{\frac{1}{2}}.$$

Note that in the second step, we found a common denominator of 6 to subtract the two fractions. In the last step, we simplified the fraction $\frac{3}{6}$. It was very useful to write the step $u^{\frac{2}{3}-\frac{1}{6}}$ even though we didn't simplify the exponent right away: we worked on getting a common denominator for the subtraction of fractions in the step after that. I can understand when someone doesn't do this in an easier problem like $\frac{u^7}{u^2}$, skipping the step of writing u^{7-2} and going directly to u^5, but that's because it's easier to simplify $7 - 2$ than it is to simplify $\frac{2}{3} - \frac{1}{6}$ in your head. When the simplification of the exponent is not immediate, it's just nicer to take it a little slower and write what the subtraction is first before simplifying the subtraction.

Warning 1.5.31 Note $\frac{a^m}{a^n}$ turns into a^{m-n}, not $a^{m \div n}$. (In other words, to simplify $\frac{a^m}{a^n}$, we must subtract the exponents instead of dividing the exponents.)

Example 1.5.32

A common error is to take an expression such as $\frac{p^{12}}{p^4}$ and due to seeing the exponents of 12 and 4 incorrectly conclude that the result is p^3, which someone likely obtains by thinking about $12 \div 4$. However, $\frac{p^{12}}{p^4} = p^{12-4} = p^8$.

Example 1.5.33

Simplify $\left(\frac{(x^4y^5)^2}{x^2y^3}\right)^{10}$.

Solution 1. First, we will zoom in to the portion of the expression that says $(x^4y^5)^2$, where we use the exponent rule $(a \cdot b)^n = a^n \cdot b^n$ to rewrite $(x^4y^5)^2$ as $(x^4)^2(y^5)^2 = x^{4 \cdot 2}y^{5 \cdot 2} = x^8y^{10}$. Thus, $\left(\frac{(x^4y^5)^2}{x^2y^3}\right)^{10} = \left(\frac{x^8y^{10}}{x^2y^3}\right)^{10}$.

Next, we use the exponent rule $\frac{a^m}{a^n} = a^{m-n}$ to rewrite $\frac{x^8}{x^2} = x^{8-2} = x^6$, and also to rewrite $\frac{y^{10}}{y^3} = y^{10-3} = y^7$. Thus, $\left(\frac{x^8y^{10}}{x^2y^3}\right)^{10} = (x^6y^7)^{10}$.

Finally, we use the exponent rule $(a \cdot b)^n = a^n \cdot b^n$ to rewrite $(x^6y^7)^{10} = (x^6)^{10}(y^7)^{10}$, and then use the exponent rule $(a^m)^n = a^{mn}$ to rewrite $(x^6)^{10} = x^{6\cdot 10} = x^{60}$ and also to rewrite $(y^7)^{10} = y^{7\cdot 10} = y^{70}$.

Therefore, $(x^6y^7)^{10} = x^{60}y^{70}$.

In summary,

$$\left(\frac{(x^4y^5)^2}{x^2y^3}\right)^{10} = \left(\frac{x^8y^{10}}{x^2y^3}\right)^{10} = (x^6y^7)^{10} = (x^6)^{10}(y^7)^{10} = x^{6\cdot 10}y^{7\cdot 10} = x^{60}y^{70}.$$

Solution 2. We could have started by using $(\frac{a}{b})^n = \frac{a^n}{b^n}$ to rewrite $\left(\frac{(x^4y^5)^2}{x^2y^3}\right)^{10} = \frac{((x^4y^5)^2)^{10}}{(x^2y^3)^{10}}$.

Then, we use the exponent rule $(a^m)^n = a^{mn}$ to rewrite $((x^4y^5)^2)^{10} = (x^4y^5)^{2\cdot 10} = (x^4y^5)^{20}$. Thus, $\frac{((x^4y^5)^2)^{10}}{(x^2y^3)^{10}} = \frac{(x^4y^5)^{20}}{(x^2y^3)^{10}}$.

Next, we use the exponent rule $(a \cdot b)^n = a^n \cdot b^n$ to rewrite $(x^4y^5)^{20} = (x^4)^{20}(y^5)^{20}$, and then use the exponent rule $(a^m)^n = a^{mn}$ to rewrite $(x^4)^{20} = x^{4\cdot 20} = x^{80}$ and also to rewrite $(y^5)^{20} = y^{5\cdot 20} = y^{100}$. Thus, $\frac{(x^4y^5)^{20}}{(x^2y^3)^{10}} = \frac{x^{80}y^{100}}{(x^2y^3)^{10}}$.

Next, we do similar work in the denominator: we use the exponent rule $(a \cdot b)^n = a^n \cdot b^n$ to rewrite $(x^2y^3)^{10} = (x^2)^{10}(y^3)^{10}$, and then use the exponent rule $(a^m)^n = a^{mn}$ to rewrite $(x^2)^{10} = x^{2\cdot 10} = x^{20}$ and also to rewrite $(y^3)^{10} = y^{3\cdot 10} = y^{30}$. Thus, $\frac{x^{80}y^{100}}{(x^2y^3)^{10}} = \frac{x^{80}y^{100}}{x^{20}y^{30}}$.

The next step we discuss is often glossed over, but we wanted to be thorough here. Recall that when simplifying the product of two fractions, we can just multiply the numerators and multiply the denominators. We can do this in reverse, too: when we see a single fraction, we can turn this into a product of two fractions: $\frac{x^{80}y^{100}}{x^{20}y^{30}} = \frac{x^{80}}{x^{20}} \cdot \frac{y^{100}}{y^{30}}$

Next, we use the exponent rule $\frac{a^m}{a^n} = a^{m-n}$ to rewrite $\frac{x^{80}}{x^{20}} = x^{80-20} = x^{60}$, and also to rewrite $\frac{y^{100}}{y^{30}} = y^{100-30} = y^{70}$. Therefore,

$$\frac{x^{80}y^{100}}{x^{20}y^{30}} = x^{60}y^{70}.$$

In summary,

$$\left(\frac{(x^4y^5)^2}{x^2y^3}\right)^{10} = \frac{((x^4y^5)^2)^{10}}{(x^2y^3)^{10}} = \frac{(x^4y^5)^{20}}{(x^2y^3)^{10}} = \frac{(x^4)^{20}(y^5)^{20}}{(x^2)^{10}(y^3)^{10}} = \frac{x^{80}y^{100}}{x^{20}y^{30}} = x^{60}y^{70}.$$

Solution 3. We could have started by using $(\frac{a}{b})^n = \frac{a^n}{b^n}$ to rewrite $(\frac{(x^4y^5)^2}{x^2y^3})^{10} = \frac{((x^4y^5)^2)^{10}}{(x^2y^3)^{10}}$.

Then, we use the exponent rule $(a \cdot b)^n = a^n \cdot b^n$ to rewrite $(x^4y^5)^2$ as $(x^4)^2(y^5)^2 = x^{4\cdot 2}y^{5\cdot 2} = x^8y^{10}$. Thus, $\frac{((x^4y^5)^2)^{10}}{(x^2y^3)^{10}} = \frac{(x^8y^{10})^{10}}{(x^2y^3)^{10}}$.

Next, we use the exponent rule $(a \cdot b)^n = a^n \cdot b^n$ to rewrite $(x^8y^{10})^{10} = (x^8)^{10}(y^{10})^{10}$, and then use the exponent rule $(a^m)^n = a^{mn}$ to rewrite $(x^8)^{10} = x^{8\cdot 10} = x^{80}$ and also to rewrite $(y^{10})^{10} = y^{10\cdot 10} = y^{100}$. Thus, $\frac{(x^8y^{10})^{10}}{(x^2y^3)^{10}} = \frac{x^{80}y^{100}}{(x^2y^3)^{10}}$.

Similarly, in the denominator, we use the exponent rule $(a \cdot b)^n = a^n \cdot b^n$ to rewrite $(x^2y^3)^{10} = (x^2)^{10}(y^3)^{10}$, and then use the exponent rule $(a^m)^n = a^{mn}$ to rewrite $(x^2)^{10} = x^{2\cdot 10} = x^{20}$ and also to rewrite $(y^3)^{10} = y^{3\cdot 10} = y^{30}$. Thus, $\frac{x^{80}y^{100}}{(x^2y^3)^{10}} = \frac{x^{80}y^{100}}{x^{20}y^{30}}$.

Applying the multiplication of fractions "backwards" just as we did in the previous answer, $\frac{x^{80}y^{100}}{x^{20}y^{30}} = \frac{x^{80}}{x^{20}} \cdot \frac{y^{100}}{y^{30}}$

Next, we use the exponent rule $\frac{a^m}{a^n} = a^{m-n}$ to rewrite $\frac{x^{80}}{x^{20}} = x^{80-20} = x^{60}$, and also to rewrite $\frac{y^{100}}{y^{30}} = y^{100-30} = y^{70}$.

Therefore, $\frac{x^{80}y^{100}}{x^{20}y^{30}} = x^{60}y^{70}$.

In summary,

$$\left(\frac{(x^4y^5)^2}{x^2y^3}\right)^{10} = \frac{((x^4y^5)^2)^{10}}{(x^2y^3)^{10}} = \frac{(x^4y^5)^{20}}{(x^2y^3)^{10}} = \frac{(x^4)^{20}(y^5)^{20}}{(x^2)^{10}(y^3)^{10}} = \frac{x^{80}y^{100}}{x^{20}y^{30}} = x^{60}y^{70}.$$

Note that this third solution is very similar to the second solution, except that in the second solution, we first simplified $(x^4y^5)^2$ before simplifying $(x^2y^3)^{10}$, while in this third solution, we first simplified $(x^2y^3)^{10}$ before simplifying $(x^4y^5)^2$. Both approaches work equally well.

An example like this one shows that there are often multiple ways to approach a problem, and that is okay. We just showed three different ways to answer the problem, all leading to the same final answer. These didn't even all start exactly the same way! The first solution started by zooming quite far in, simplifying $(x^4y^5)^2$, while the second and third solutions started by rewriting the entire expression using $(\frac{a}{b})^n = \frac{a^n}{b^n}$. All three answers (and in fact, additional answers that we didn't provide) are all great, and each takes about the same amount of time.

Example 1.5.34

Simplify $((x^2)^3(xy^4)^5)^{10}$.

Solution 1. $((x^2)^3(xy^4)^5)^{10} = ((x^6)(x^5(y^4)^5))^{10} = (x^6x^5y^{20})^{10} = (x^{11}y^{20})^{10} = (x^{11})^{10}(y^{20})^{10} = x^{110}y^{200}$

Solution 2. $((x^2)^3(xy^4)^5)^{10} = ((x^2)^3)^{10}((xy^4)^5)^{10} = (x^2)^{30}(xy^4)^{50} = x^{60}x^{50}(y^4)^{50} = x^{110}y^{200}$

Solution 3. $((x^2)^3(xy^4)^5)^{10} = (x^2)^{30}(xy^4)^{50} = x^{60}(x^5(y^4)^5)^{10} = x^{60}(x^5)^{10}(y^4)^{50} = x^{110}y^{200}$

Example 1.5.35

Simplify $\frac{h^{11}}{h^{-2}}$

Solution. Using the exponent rule $\frac{a^m}{a^n} = a^{m-n}$ in Principle 1.5.27, we have $\frac{h^{11}}{h^{-2}} = h^{11-(-2)} = h^{11+2} = h^{13}$.

A common error for this problem is to write h^{11-2} instead of $h^{11-(-2)}$. This is likely because of the negative exponent in the denominator. Note that we need to subtract exponents, even when the exponent in the denominator is negative. To break things down, the way that we can see this is to identify the roles of a and m and n when we view $\frac{h^{11}}{h^{-2}}$ as an expression in the form $\frac{a^m}{a^n}$. Then based on the position of things, $a = h$ and $m = 11$ and $n = -2$. Then putting these values into a^{m-n}, we get $h^{11-(-2)}$. Our expression $h^{11-(-2)}$ is correct, and has two minus signs. Here are the reasons:

- The first minus sign (the minus sign right after writing 11) is there because the right side of the formula in Principle 1.5.27 has a minus sign.
- The second minus sign is actually part of writing the number -2. This is because in the formula's right side, we see the exponent $m - n$, so we need to subtract the value of n, which we already identified was $n = -2$.

As one other way to try to see this, let us think about which expression would lead us to h^{11-2} using the formula $\frac{a^m}{a^n} = a^{m-n}$. If we wanted to use the formula to get h^{11-2}, we had to have started with $\frac{h^{11}}{h^2}$, which is not the same as starting with $\frac{h^{11}}{h^{-2}}$.

There are three last exponent rules that we discuss:

Principle 1.5.36

- $a^0 = 1$
- $a^{-n} = \dfrac{1}{a^n}$
- $a^n = \dfrac{1}{a^{-n}}$

Example 1.5.37

Simplify c^{5-5}.

Solution 1. $c^{5-5} = \frac{c^5}{c^5}$ by the exponent rule $\frac{a^m}{a^n} = a^{m-n}$ in Principle 1.5.27.

The fraction we wrote is saying that we are dividing a quantity by itself, so this equals 1. Therefore, $c^{5-5} = 1$.

Solution 2. Since $5 - 5 = 0$, we have $c^{5-5} = c^0$. Using the exponent rule $a^0 = 1$ in Principle 1.5.36, we have $c^0 = 1$. Therefore, $c^{5-5} = 1$.

Our first answer did not use the new exponent formula $a^0 = 1$, while our second answer did. The fact that both answers led to the same final answer of 1 gives us more confidence that the new formula is correct.

Example 1.5.38

Simplify 5^0.

Solution. Using the exponent rule $a^0 = 1$ in Principle 1.5.36, we have $5^0 = 1$.

Example 1.5.39

Rewrite $\frac{u^3}{u^6}$ in two different ways. In one writing, the only exponent fact/concept to use is $\frac{a^m}{a^n} = a^{m-n}$ in Principle 1.5.27. In the other writing, the only exponent fact/concept to use is what an exponent means from Principle 1.5.1.

Solution.

- Using the exponent rule $\frac{a^m}{a^n} = a^{m-n}$ in Principle 1.5.27, we have $\frac{u^3}{u^6} = u^{3-6} = u^{-3}$. The only exponent fact/concept we used was $\frac{a^m}{a^n} = a^{m-n}$.

  Using Principle 1.5.1, we can expand $\frac{u^3}{u^6} = \frac{u \cdot u \cdot u}{u \cdot u \cdot u \cdot u \cdot u \cdot u}$. Then, we can cancel three factors of u from the numerator and denominator, getting $\frac{1}{u \cdot u \cdot u} = \frac{1}{u^3}$. While we applied cancellation rules for fractions, the only exponent fact/concept we used was what an exponent means from Principle 1.5.1.

Following the instructions, we saw that $\frac{u^3}{u^6}$ became either u^{-3} or became $\frac{1}{u^3}$. All of our work was valid. Therefore, $u^{-3} = \frac{1}{u^3}$. In fact, our conclusion of $u^{-3} = \frac{1}{u^3}$ is an example of the exponent rule $a^{-n} = \frac{1}{a^n}$ in Principle 1.5.36 at play, and this activity gives us reason to believe $a^{-n} = \frac{1}{a^n}$.

Warning 1.5.40 Just a note that $\frac{u^3}{u^6}$ does not become $u^{1/2}$. For more info, see Warning 1.5.31.

Try it 1.5.41

Rewrite $\frac{c^2}{c^7}$ in two different ways. In one writing, the only exponent fact/concept to use is $\frac{a^m}{a^n} = a^{m-n}$ in Principle 1.5.27. In the other writing, the only exponent fact/concept to use is what an exponent means from Principle 1.5.1.

Example 1.5.42

Simplify $\frac{u^{14}}{u^{-2}}$.

Solution 1. One option is to use $a^{-n} = \frac{1}{a^n}$ to turn u^{-2} into $\frac{1}{u^2}$. Then the original expression $\frac{u^{14}}{u^{-2}}$ is equal to $\frac{u^{14}}{\frac{1}{u^2}}$.

If we wish to avoid the look of the fraction inside the fraction (the overall expression is a fraction whose denominator is itself a fraction), we can instead write the expression $u^{14} \div \frac{1}{u^2}$.

Instead of dividing a non-fraction and a fraction, we can rewrite this $\frac{u^{14}}{1} \div \frac{1}{u^2}$, which is a division of two fractions, which then becomes $\frac{u^{14}}{1} \cdot \frac{u^2}{1}$, where we multiply across to get $\frac{u^{14} \cdot u^2}{1 \cdot 1} = \frac{u^{16}}{1} = u^{16}$.

If we present all the work using equal signs (and skipping a reasonable number of steps from above), we have $\frac{u^{14}}{u^{-2}} = \frac{u^{14}}{\frac{1}{u^2}} = \frac{u^{14}}{1} \div \frac{1}{u^2} = \frac{u^{14}}{1} \cdot \frac{u^2}{1} = u^{16}$

Solution 2. Instead of zooming in to replace u^{-2} with $\frac{1}{u^2}$, we can instead completely zoom out, looking at the whole expression, and use the exponent rule $\frac{a^m}{a^n} = a^{m-n}$ to replace the entire expression $\frac{u^{14}}{u^{-2}}$ with $u^{14-(-2)} = u^{14+2} = u^{16}$. Note that we wrote the exponent first as $14 - (-2)$ before simplifying to $14 + 2$. Why are there two minus signs? One minus sign is the minus sign that's already built into the formula $\frac{a^m}{a^n} = a^{m-n}$, and the other minus sign is the one that is part of the exponent -2.

Warning 1.5.43 Just a note that $\frac{u^{14}}{u^{-2}}$ does not become u^{14-2} using $\frac{a^m}{a^n} = a^{m-n}$. Instead, a different expression, namely $\frac{u^{14}}{u^2}$, becomes u^{14-2} using $\frac{a^m}{a^n} = a^{m-n}$.q

Note 1.5.44 In Principle 1.5.3 we saw the formula $a^m a^n = a^{m+n}$. In Principle 1.5.16 we saw the formula $(ab)^n = a^n b^n$. The first formula tells us that when we have an expression of the form $a^m a^n$ we can replace it with the equal expression a^{m+n}. However, we can also go in reverse: if we see an expression of the form a^{m+n}, we can replace it with the equal expression $a^m a^n$. The second formula tells us that when we have an expression of the form $(ab)^n$ we can replace it with the equal expression $a^n b^n$. However, we can also go in reverse: if we see an expression of the form $a^n b^n$, we can replace it with the equal expression $(ab)^n$.

If we compare the left side of the first formula and the right side of the second formula, they have the same "shape" to them. What I mean is that both expressions $a^m a^n$ and $a^n b^n$ happen to be a base raised to an exponent, multiplied by a base raised to an exponent. So, we really have to pay attention to which variable(s) are repeated and which variable(s) are not repeated. For example, in $a^m a^n$, the base a is repeated, while in $a^n b^n$, the exponent n is repeated.

Thus, in the expression $3^c 10^c$, both exponents are the same, meaning that the formula which has $a^n b^n$ on one side applies, since $a^n b^n$ also has exponents which match. In the expression $\sqrt{7}^k \sqrt{7}^4$, the bases are the same, so the formula which has $a^m a^n$ on one side applies, since $a^m a^n$ also has bases which match.

Example 1.5.45

Rewrite $3^c 10^c$ using a relevant exponent formula.

Solution. Since both exponents are the same, we can use the formula $(ab)^n = a^n b^n$ in Principle 1.5.16 to rewrite $3^c 10^c = (3 \cdot 10)^c = 30^c$. While we have practiced using the formula $(ab)^n = a^n b^n$ to turn an expression of the form $(ab)^n$ into an expression of the form $a^n b^n$ (see for example Example 1.5.19), we can also use the formula in reverse, turning an expression of the form $a^n b^n$ into an expression of the form $(ab)^n$, like we just did in this example.

Example 1.5.46

Rewrite $\sqrt{7}^k \sqrt{7}^4$ using a relevant exponent formula.

Solution. Since both bases are the same (they happen to be $\sqrt{7}$), we can use the formula $a^m a^n = a^{m+n}$ in Principle 1.5.3 to rewrite $\sqrt{7}^k \sqrt{7}^4 = \sqrt{7}^{k+4}$. In this example, the new exponent $k+4$ does not simplify further. In another example like $u^3 u^5$, we can apply the formula to get u^{3+5}, or even skip writing this step, and in either case we could simplify the new exponent $3+5$ to 8.

Note 1.5.47 In Principle 1.5.16 we saw the formula $(\frac{a}{b})^n = \frac{a^n}{b^n}$. In Principle 1.5.27 we saw the formula $\frac{a^m}{a^n} = a^{m-n}$. The right side of the first formula and the left side of the second formula have the same "shape" to them. However, in $\frac{a^n}{b^n}$ the exponents match, while in $\frac{a^m}{a^n}$ the bases match.

Thus, in the expression $\frac{20^c}{5^c}$, both exponents are the same, meaning that the formula which has $\frac{a^n}{b^n}$ on one side applies, since $\frac{a^n}{b^n}$ also has exponents which match. In the expression $\frac{p^8}{p^2}$, the bases are the same, so the formula which has $\frac{a^m}{a^n}$ on one side applies, since $\frac{a^m}{a^n}$ also has bases which match.

Example 1.5.48

Rewrite $\frac{20^c}{5^c}$ using a relevant exponent formula.

Solution. Since both exponents are the same, we can use the formula $(\frac{a}{b})^n = \frac{a^n}{b^n}$ in Principle 1.5.16 to rewrite $\frac{20^c}{5^c} = (\frac{20}{5})^c = 4^c$. While we have practiced using the formula $(\frac{a}{b})^n = \frac{a^n}{b^n}$ to turn an expression of the form $(\frac{a}{b})^n$ into an expression of the form $\frac{a^n}{b^n}$ (see for example AAAAA), we can also use the formula in reverse, turning an expression of the form $\frac{a^n}{b^n}$ into an expression of the form $(\frac{a}{b})^n$, like we just did in this example.

Example 1.5.49

Rewrite $\frac{p^8}{p^2}$ using a relevant exponent formula.

Solution. Since both bases are the same (they happen to be p), we can use the formula $\frac{a^m}{a^n} = a^{m-n}$ in Principle 1.5.27 to rewrite $\frac{p^8}{p^2} = p^{8-2} = p^6$. While we have practiced using the formula $\frac{a^m}{a^n} = a^{m-n}$ to turn an expression of the form $\frac{a^m}{a^n}$ into an expression of the form a^{m-n} (see for example AAAA), we can also use the formula in reverse, turning an expression of the form a^{m-n} into an expression of the form $\frac{a^m}{a^n}$, like we just did in this example.

Note that a common mistake is to turn $\frac{p^8}{p^2}$ into p^4. However, there is no exponent rule that justifies this. In fact, among all the exponent rules we have seen up to this point, there is never a time that one exponent gets divided by another exponent.

Warning 1.5.50 It is a very common error to turn an expression of the form $(x+y)^n$ into an expression of the from $x^n + y^n$.

Here are specific examples of what this warning is trying to address:

- We cannot turn $(x+y)^2$ into x^2+y^2.
- We cannot turn $(x+8)^2$ into x^2+8^2.
- We cannot turn $(u+7)^9$ into u^9+7^9.
- We cannot turn $(2+k)^c$ into 2^c+k^c.

We have seen various versions of the distributive property in AAAAAA. For example, specific versions included $a(b+c) = ab+ac$ and $(b+c)a = ba+ca$. It is true that in our examples where we said that we couldn't "distribute", we had addition inside the parentheses. So, what's the problem? Why doesn't the distributive rule apply?

The issue is that in the format of $a(b+c)$, we have to be very careful to pay attention the operation between a and the content in parentheses. In $a(b+c)$, the operation between a and $(b+c)$ is multiplication. In other words, a is a factor that is multiplied. The Distributive Law only addresses the situation where a factor is multiplied by a sum or a difference. The point is that it is not enough to just think that anything outside of parentheses can be "distributed" to everything inside the parentheses. In all three of our examples above, the thing outside of parentheses is an exponent, not a factor.

The Distributive Law only applies to the situation that it is describing through its formula, and nothing else. This is why we have to pay close attention to the formula as written: we examine each formula for what operations are occuring, where these operations are occuring, and any repeated variables. If we don't pay close attention to these details, we may misapply a formula.

We have introduced the exponent rule $(ab)^n = a^n b^n$ in Principle 1.5.16. In this formula, the operation between a and b is multiplication, and the role of the thing outside of parentheses *is* an exponent. So, this formula tells us that if we have multiplication inside the parentheses and an exponent outside, the exponent can apply to each of the factors.

We have also introduced the exponent rule $(\frac{a}{b})^n = \frac{a^n}{b^n}$ in Principle 1.5.16. In this formula, the operation between a and b is division, and the role of the thing outside of parentheses *is* an exponent. So, this formula tells us that if we have division inside the parentheses and an exponent outside, the exponent can apply to both the numerator and denominator.

1.5.3 Radicals

Definition 1.5.51 What does a radical mean?

$\sqrt[n]{a}$ asks for the value that fills in the blank:

$$(\underline{\quad\quad})^n = a.$$

If two numbers can fill in the blank (positive and negative), then by definition, the positive number is the answer. The answer to the radical question identifies the value of the base.

Example 1.5.52

Find $\sqrt[2]{64}$.

Solution. This asks for the value that fills in the blank:

$$(\underline{\quad\quad})^2 = 64.$$

Both 8 and -8 fill in the blank, but by definition, the positive number is the answer. Therefore, $\sqrt[2]{64} = 8$.

Example 1.5.53

Find $\sqrt[3]{64}$.

Solution. This asks for the value that fills in the blank:

$$(\underline{\quad})^3 = 64.$$

The only number that fills in the blank is 4. (The reason that -4 does not fill in the blank is that $(-4)^3 = -64$, not 64.) Therefore, $\sqrt[3]{64} = 4$.

Example 1.5.54

Find $\sqrt[2]{25}$.

Solution. When we raise the base 5 to the power 2, the result is 25. So the base 5 answers the question. Written as notation with an equal sign properly, $\sqrt[2]{25} = 5$.

Warning 1.5.55 It would be incorrect to write

$$\sqrt[2]{25} = \sqrt[2]{5}.$$

Of course, the completion to the right of the equal sign should have a 5, so when someone writes $\sqrt[2]{5}$, it is good to see the 5 present, since they are thinking about the correct base. However, the radical should not surround the 5 because we have already processed what the radical does.

Principle 1.5.56 Radical missing small number.

$\sqrt{a} = \sqrt[2]{a}$

If the small number outside of the radical is missing, it is a hidden 2.

We do this because the small number outside of the radical is often (but not always) a 2. In other words, we often need to find the base when the exponent is 2, but there are also times we need to find the base when the exponent is something else, like 3 or 4 or 5, etc.

Example 1.5.57

Find $\sqrt{25}$.

Solution. This question is asking $\sqrt[2]{25}$, so $\sqrt{25} = \sqrt[2]{25} = 5$.

Warning 1.5.58 Like in Warning 1.5.55, it would be incorrect to write

$$\sqrt{25} = \sqrt{5}.$$

Instead, we should write

$$\sqrt{25} = 5.$$

Example 1.5.59

Simplify $\sqrt{81}$.

Solution 1. $\sqrt{81} = \sqrt[2]{81} = 9$

Solution 2. $\sqrt{81} = 9$

Note, in the second answer, we skipped writing the middle expression $\sqrt[2]{81}$. This is fine to do, but it's good to remember that when the little number is missing outside of the radical, it is really a hidden 2. The reason we're taking time to say this is that sometimes people breeze through then the hidden number is a 2, but then struggle when there's another number there such as a 3. The point is that Definition 1.5.51 gives us a uniform and consistent way to think about all radicals, whether the small number outside of the radical is a 2 (missing or written) or another number is written there such as a 3 or 4. In all cases, a radical is asking us to identify the base while the little number outside the radical is telling us the exponent (and the content inside the radical is telling us the result of taking the base raised to the exponent).

Language 1.5.60 How to say radicals.

$\sqrt[n]{a}$ should be said "n-th root of a". When the value of n is small, there are also special names for these:

- *$\sqrt[2]{a}$ or $\sqrt{a}$ can be said "2nd root of a" or "square root of a".*
- *$\sqrt[3]{a}$ is spoken "3rd root of a" or "cube root of a", but not "3 square root of a".*

For other values of n, we just say "n-th root of a". For example:

- *$\sqrt[4]{a}$ is spoken "4th root of a" but not "4 square root of a".*
- *$\sqrt[7]{a}$ is spoken "7th root of a" but not "7 square root of a".*

Warning 1.5.61 It is tempting to look at $\sqrt[7]{5}$ and want to say "seven square root of five", but saying "seven square root of five" actually refers to $7\sqrt{5}$ which can also be written $7\sqrt[2]{5}$. The point is that $\sqrt[7]{5}$ and $7\sqrt{5}$ are different values:

- $\sqrt[7]{5}$ asks for what is the base that when raised to the exponent 7 results in the value of 5. Using a calculator, I found this to be approximately 1.258.
- $7\sqrt{5}$ asks for what is the base that when raised to the exponent 2 results in the value of 5, and then multiplies that result by 7. Using a calculator, I found $\sqrt{5}$ is approximately 2.236. I then had to multiply this by 7. Thus, the value of $7\sqrt{5}$ is approximately 15.652.

Because of what we're about to bring up next, let's revisit some of the problems we've done using a new formula. I admit that most people won't prefer thinking of the problems we've done through this new lens, but it will be helpful for what we're about to do right after this!

Principle 1.5.62

$\sqrt[n]{a^n} = a$

In words, this formula says that an nth root and an nth power undo each other.

Example 1.5.63

Find $\sqrt{81}$.

Solution. Inside the radical is 81, which we can replace with 9^2, since 9^2 and 81 are equal. That is,

$$\sqrt{81} = \sqrt[2]{81} = \sqrt[2]{9^2}.$$

Note that in the above, we still had to write the second root: we have not processed the root here yet (we have only replaced 81 with something that is equal to 81). Now, we can use the formula $\sqrt[n]{a^n} = a$ to process the radical: $\sqrt[2]{9^2} = 9$. Presenting all the work with equal signs, we have

$$\sqrt{81} = \sqrt[2]{81} = \sqrt[2]{9^2} = 9.$$

It may seem strange to include the step of writing $\sqrt[2]{9^2}$ when we already provided two perfectly good solutions in Example 1.5.59, but there are two reasons we are doing this:

1. Including this step makes it easier to illustrate the use of the formula $\sqrt[n]{a^n} = a$ in Principle 1.5.62.
2. This gives us yet another way to simplify an expression containing a radical, and while this isn't most people's preferred way of thinking about $\sqrt{81}$, this new method is preferred by some people for certain problems that we will soon see, including the cube root problem below.

Example 1.5.64

Find $\sqrt[3]{64}$.

Solution. Inside the radical is 64, which we can replace with 4^3, since 4^3 and 64 are equal. That is,

$$\sqrt[3]{64} = \sqrt[3]{4^3}.$$

Note that in the above, we still had to write the third root: we have not processed the root here yet (we have only replaced 64 with something that is equal to 64). Now, we can use the formula $\sqrt[n]{a^n} = a$ to process the radical: $\sqrt[3]{4^3} = 4$. Presenting all the work with equal signs, we have

$$\sqrt[3]{64} = \sqrt[3]{4^3} = 4.$$

In the setting of algebra, we will need to simplify the root of expressions that contain variables.

Example 1.5.65

Simplify $\sqrt{x^{10}}$.

Solution. Since the little number outside the radical is missing, $\sqrt{x^{10}} = \sqrt[2]{x^{10}}$. So, this is asking for the base that when raised to the exponent 2 results in the value of x^{10}. That is, we are being asked to fill in the blank in $(\_\_\_)^2 = x^{10}$. Here are three valid ways that we can think about this:

- We can use the exponent rule $(a^m)^n = a^{m \cdot n}$ in Principle 1.5.3 in reverse. The expression x^{10} can be rewritten as $(x^5)^2$, since $5 \cdot 2 = 10$. The missing base is x^5, since $(x^5)^2 = x^{10}$. Therefore, $\sqrt{x^{10}} = \sqrt[2]{(x^5)^2} = x^5$.

- We can think about what the exponent 2 means (or in other words, we can think about what squaring means). Squaring a thing means to multiply the thing by itself. So, we ask ourselves: what thing can be multiplied by itself to get x^{10}? More visually, how do we fill in the blanks in $(\_\_\_)(\_\_\_) = x^{10}$ with the restriction that both blanks need to be filled in with the same thing? Informed by the exponent rule $a^m a^n = a^{m+n}$ in Principle 1.5.3 which says that when we multiply two things with the same base we add the exponents, both blanks can be filled in with x^5. That is, $x^5 \cdot x^5 = x^{10}$. Therefore, $\sqrt{x^{10}} = \sqrt[2]{(x^5)^2} = x^5$.

- We can also apply the formula $\sqrt[n]{a^n} = a$ in Principle 1.5.62, and emulate the work we did in Example 1.5.63. The content inside the radical, which is x^{10}, can be replaced with $(x^5)^2$, since $(x^5)^2$ and x^{10} are equal. Thus,

$$\sqrt{x^{10}} = \sqrt[2]{x^{10}} = \sqrt[2]{(x^5)^2} = x^5.$$

Example 1.5.66

With proper notation, answer both questions below as similarly as possible, using Principle 1.5.62:

1. Simplify $\sqrt[2]{c^{18}}$.
2. Simplify $\sqrt[2]{49}$.

Solution.

1. $\sqrt[2]{c^{18}} = \sqrt[2]{(c^9)^2} = c^9$
2. $\sqrt[2]{49} = \sqrt[2]{(7)^2} = 7$

Principle 1.5.67

$\sqrt[n]{ab} = \sqrt[n]{a}\,\sqrt[n]{b}$

In words, this formula says that an nth root of a product can be rewritten as the product of the nth roots of each factor. In fact, this may be slightly clearer by writing the same formula, but including the hidden multiplication signs:

$$\sqrt[n]{a \cdot b} = \sqrt[n]{a} \cdot \sqrt[n]{b}.$$

Either version of the formula (with hidden multiplication signs or written multiplication signs) is very useful when we need to simplify a radical that contains a product inside the radical. The key is that all roots have to be the same kind: this formula doesn't allow mixing a square root and a cube root, for example. To be clear, special cases of this formula include the version for square roots:

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$

and the version for cube roots:

$$\sqrt[3]{ab} = \sqrt[3]{a}\,\sqrt[3]{b}$$

but there's also a version for fourth roots and fifth roots and so on. (Each of these special versions can also be written where we write the hidden multiplication signs. I encourage you to write the two previous equations with multiplication signs included, since multiplication needs to be present to apply this formula.)

Note 1.5.68 There are extended versions of these formulas for three factors, namely $\sqrt[n]{abc} = \sqrt[n]{a}\,\sqrt[n]{b}\,\sqrt[n]{c}$ and we show below what this looks like for square roots and cube roots:

- $\sqrt{abc} = \sqrt{a}\sqrt{b}\sqrt{c}$
- $\sqrt[3]{abc} = \sqrt[3]{a}\sqrt[3]{b}\sqrt[3]{c}$

but there's an even more extended version for four factors, namely $\sqrt[n]{abcd} = \sqrt[n]{a}\,\sqrt[n]{b}\,\sqrt[n]{c}\,\sqrt[n]{d}$, and for square roots and cube roots, this means:

- $\sqrt{abcd} = \sqrt{a}\sqrt{b}\sqrt{c}\sqrt{d}$
- $\sqrt[3]{abcd} = \sqrt[3]{a}\sqrt[3]{b}\sqrt[3]{c}\sqrt[3]{d}$

and so on: there are extended versions of this for any number of factors. We just need to ensure that all radicals are of the same kind (that is, we have the same little number on the outside of all radicals), and that we have multiplication everywhere (instead of addition).

Example 1.5.69

Simplify $\sqrt{18}$.

Solution. We examine the content inside the radical (which in this example is 18) and ask how we can write 18 as the product, where one of the factors of the product is a perfect square. We do need a product (since the left side of the formula in Principle 1.5.67 says $\sqrt[n]{ab}$ and not $\sqrt[n]{a+b}$). So, while $18 = 16 + 2$ is true and one of the pieces (namely 16) is a perfect square, this isn't really going to help us. We want to write 18 as a product with one piece being a perfect square, and a useful way to do that is $18 = 9 \cdot 2$. So,

$$\sqrt{18} = \sqrt{9 \cdot 2}.$$

Note that in the work shown above, we still wrote the square root sign, because all we did was replace 18 with the equal value of $9 \cdot 2$. We haven't processed the square root yet. Now, $\sqrt{9 \cdot 2}$ is in the format of the left side of the formula $\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$ So, we can apply this:

$$\sqrt{9 \cdot 2} = \sqrt{9} \cdot \sqrt{2}.$$

Now, in the writing of $\sqrt{9} \cdot \sqrt{2}$, we can simplify $\sqrt{9}$ because 9 is a perfect square, but we have to leave $\sqrt{2}$ alone. That is,

$$\sqrt{9} \cdot \sqrt{2} = 3 \cdot \sqrt{2}.$$

We now present a summary of all the work, writing equal signs where they belong, though we will leave out multiplication symbols where possible to save a little time: $\sqrt{18} = \sqrt{9 \cdot 2} = \sqrt{9}\sqrt{2} = 3\sqrt{2}$.

The example we did suggests the following strategy:

Strategy 1.5.70 Simplifying the square root of a product.

- ***When does this strategy apply?*** This strategy applies in any expression which is a square root, and the contents inside the square root is a product (or can be turned into a product).
- ***How to apply the strategy***
  1. Rewrite the content inside the square root as a product with one factor being a perfect square.
  2. Apply the formula $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$.
  3. In the resulting expression of the form $\sqrt{a} \cdot \sqrt{b}$, the factor which is the square root of a perfect square can be simplified into an expression that does not require writing a square root. (However, the square root of an expression which is not a perfect square will always still have a square root symbol.)

Example 1.5.71

Simplify $\sqrt{75}$.

Solution. We need to write the content inside the radical (which in this example is 75 as the product where one factor is a perfect square. Note $75 = 25 \cdot 3$.

$$\sqrt{75} = \sqrt{25 \cdot 3}.$$

Note that in the work shown above, we still wrote the square root sign, because all we did was replace 75 with the equal value of $25 \cdot 3$. We haven't processed the square root yet. Applying $\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$, we have:

$$\sqrt{25 \cdot 3} = \sqrt{25} \cdot \sqrt{3}.$$

In $\sqrt{25} \cdot \sqrt{3}$, we can simplify $\sqrt{25}$ because 25 is a perfect square, but we have to leave $\sqrt{3}$ alone. That is,

$$\sqrt{25} \cdot \sqrt{3} = 5 \cdot \sqrt{3}.$$

We now present a summary of all the work, writing equal signs where they belong: $\sqrt{75} = \sqrt{25 \cdot 3} = \sqrt{25}\sqrt{3} = 5\sqrt{3}$.

Example 1.5.72

Simplify $\sqrt{1800}$.

Solution 1. Note that $1800 = 100 \cdot 18$. So, we have:

$$\sqrt{1800} = \sqrt{100 \cdot 18} = \sqrt{100}\sqrt{18} = 10\sqrt{18}.$$

While the work that we have shown above is true and correct, we're not quite done yet, because 18 has a factor which is a perfect square, namely 9. So let's take the last expression $10\sqrt{18}$ and zoom in to rewrite the portion that is $\sqrt{18}$:

$$10\sqrt{18} = 10\sqrt{9 \cdot 2} = 10\sqrt{9}\sqrt{2} = 10 \cdot 3\sqrt{2} = 30\sqrt{2}.$$

Presenting all the work from the beginning with equal signs, we have $\sqrt{1800} = \sqrt{100 \cdot 18} = \sqrt{100}\sqrt{18} = 10\sqrt{18} = 10\sqrt{9 \cdot 2} = 10\sqrt{9}\sqrt{2} = 10 \cdot 3\sqrt{2} = 30\sqrt{2}$.

Solution 2. Alternatively, we could have started by writing $1800 = 900 \cdot 2$. (The number 900 is a larger perfect square than 100, and the benefit of the work we did in the previous answer was to see perfect square factors of 100 and 9.) Thus,

$$\sqrt{1800} = \sqrt{900 \cdot 2} = \sqrt{900}\sqrt{2} = 30\sqrt{2}.$$

We gave two answers to the previous question. The second answer gave us a shorter path to the same final answer of $30\sqrt{2}$. However, when the contents inside the radical get so large, I struggle to see the "best" (that is, largest) perfect square that is a factor of the contents inside the radical. In these situations, it's good to just find some perfect square factor, even if it's not the largest one. At some point, the original expression gets rewritten as a whole number multiplied by the root of a smaller number. Then, we keep going as long as this smaller number inside the radical has a perfect square factor.

We can apply Strategy 1.5.70 even when the content inside the radical contains variables. Before we get started, there is often a major source of confusion about perfect squares, so let us first clear that up.

Definition 1.5.73

An expression is a **perfect square** if it is the result of squaring an expression.

Example 1.5.74

- 0 is a perfect square because $0 = (0)^2$.
- 1 is a perfect square because $1 = (1)^2$.
- 4 is a perfect square because $4 = (2)^2$.
- 9 is a perfect square because $9 = (3)^2$.
- 16 is a perfect square because $16 = (4)^2$.

- 25 is a perfect square because $25 = (5)^2$.
- 36 is a perfect square because $36 = (6)^2$.

This list continues, following a pattern.

Example 1.5.75

- x^0 is a perfect square because $x^0 = (x^0)^2$.
- x^2 is a perfect square because $x^2 = (x^1)^2$.
- x^4 is a perfect square because $x^4 = (x^2)^2$.
- x^6 is a perfect square because $x^6 = (x^3)^2$.
- x^8 is a perfect square because $x^8 = (x^4)^2$.
- x^{10} is a perfect square because $x^{10} = (x^5)^2$.
- x^{12} is a perfect square because $x^{12} = (x^6)^2$.

This list continues, following a pattern.

Note 1.5.76 It is easy to mishear the discussion above. We said that x^8 is a perfect square. We never said that 8 is a perfect square. The point is that x^8 is should not be treated the same as 8 itself. The number 8 is not a perfect square (it is missing from the list in Example 1.5.74, where we jump from the perfect square 4 to the next perfect square 9). However, the expression x^8 is a perfect square (it is present in the list in Example 1.5.75). We are saying that when we have x as the base, and this is raised to the power 8, with emphasis that 8 is the exponent, which is just part of the expression x^8.

To further clarify, examine this template:

$$(\underline{\quad\quad})^2 = \underline{\quad\quad}$$

When we fill in the result of the first blank, this will force the value of the second blank on the right side of the equal sign. The content on the right side of the equal sign is the perfect square. For example, from the equation

$$(9)^2 = 81$$

we conclude that 81 is a perfect square, and in a similar way, from the equation

$$(x^{13})^2 = x^{26}$$

we conclude that x^{26} is a perfect square. (Note, we are saying that x^{26} is a perfect square, not that 26 is a perfect square.)

Now that we have identified a square root, we should clarify how we simplify the square root of a perfect square. What we are about to say is just technically a special case of Definition 1.5.51, but it is worth stating explicitly to clear up some confusion:

Strategy 1.5.77 How to simplify the square root of a perfect square.

- ***When does this strategy apply?*** This strategy applies in any expression which is a square root, and the contents inside the square root is a perfect square.
- ***How to apply the strategy***
  1. Given an expression of the form $\sqrt{a}$, identify if a is a perfect square. This means that there is some expression b such that $b^2 = a$.
  2. From $b^2 = a$, we get $\sqrt{a} = b$.

Example 1.5.78

- From $(7)^2 = 49$, we get $\sqrt{49} = 7$.
- From $(8)^2 = 64$, we get $\sqrt{64} = 8$.
- From $(9)^2 = 81$, we get $\sqrt{81} = 9$.

In all three examples above, the result of the square root was the contents inside the parentheses.

Example 1.5.79

- From $(0)^2 = 0$, we get $\sqrt{0} = 0$.
- From $(1)^2 = 1$, we get $\sqrt{1} = 1$.

Note 1.5.80 The two examples we just presented follow the exact same format as the three examples before. The example of $\sqrt{0} = 0$ looks confusing because 0 is written twice. For the reason, the example $\sqrt{1} = 1$ looks weird to people, because 1 is written twice. For all of us (myself included), an example like $\sqrt{49} = 7$ is easier to mentally process. We presented more familiar examples in Example 1.5.78 before presenting the more confusing examples in Example 1.5.79. However, I hope the two examples in Example 1.5.79 are convincing, since we did follow exactly the same format as in Example 1.5.78.

Through the examples of $\sqrt{0} = 0$ and $\sqrt{1} = 1$, we see that a number can be its own square root. (In fact, these are the only two numbers that are their own square roots.) This all stems from the fact that 0 and 1 are the only two numbers that are their own squares.

Example 1.5.81

- From $(x^1)^2 = x^2$, we get $\sqrt{x^2} = x^1$.
- From $(x^2)^2 = x^4$, we get $\sqrt{x^4} = x^2$.
- From $(x^3)^2 = x^6$, we get $\sqrt{x^6} = x^3$.
- From $(x^4)^2 = x^8$, we get $\sqrt{x^8} = x^4$.
- From $(x^5)^2 = x^{10}$, we get $\sqrt{x^{10}} = x^5$.
- From $(x^6)^2 = x^{12}$, we get $\sqrt{x^{12}} = x^6$.

In all six examples above, the result of the square root was the contents inside the parentheses. The work was presented in exactly the same format as in Example 1.5.78 as much as possible. The pattern here continues.

Try it 1.5.82

When an expression is of the format base raised to an exponent, what do you notice about the exponent when the entire expression is a perfect square?

Example 1.5.83

Simplify $\sqrt{20x^9}$.

Solution. To follow Strategy 1.5.70, we examine the contents inside the radical, namely $20x^9$ with the goal of rewriting this as a product, looking for perfect squares:

$$20x^9 = 4 \cdot 5 \cdot x^8 \cdot x.$$

In the work above, $x^8 \cdot x = x^8 \cdot x^1 = x^9$ because of Note 1.5.2 and the exponent rule $a^m a^n = a^{m+n}$ in Principle 1.5.3. So,

$$\sqrt{20x^9} = \sqrt{4 \cdot 5 \cdot x^8 \cdot x}.$$

Since the expression on the right is the root of a product, we can turn this into a product of roots as mentioned in Note 1.5.68:

$$\sqrt{4 \cdot 5 \cdot x^8 \cdot x} = \sqrt{4} \cdot \sqrt{5} \cdot \sqrt{x^8} \cdot \sqrt{x}.$$

The square roots of perfect squares will simplify:

$$\sqrt{4} \cdot \sqrt{5} \cdot \sqrt{x^8} \cdot \sqrt{x} = 2 \cdot \sqrt{5} \cdot x^4 \cdot \sqrt{x}.$$

And while we can leave our final answer like this, we can also write this in a slightly more compact way:

$$2 \cdot \sqrt{5} \cdot x^4 \cdot \sqrt{x} = 2x^4\sqrt{5x}.$$

It is typical to write the content that is not in any radical first, primarily to avoid writing a radical symbol too large, and intentionally placing content inside a radical that should not be placed inside a radical. We now present a summary of all the work, with the work slightly reorganized, writing equal signs where they belong: $\sqrt{20x^9} = \sqrt{4x^8 \cdot 5x} = \sqrt{4}\sqrt{x^8} \cdot \sqrt{5x} = 2x^4 \cdot \sqrt{5x}$.

Example 1.5.84

Simplify $\sqrt{8a^{24}b^{25}c^{26}d^{27}}$.

Solution. We can rewrite 8 as $4 \cdot 2$. Note that a^{24} and c^{26} are already perfect squares since these exponents are even, but b^{25} and d^{27} are not perfect squares since these exponents are odd. To fix this, we can rewrite b^{25} as $b^{24} \cdot b^1$, and we can rewrite d^{27} as $d^{26} \cdot d^1$. (Please note that this example was carefully chosen to have 25 as an exponent. For details on why b^{25} is not a perfect square, see Note 1.5.76.)

So, we have

$$\begin{aligned}
&\sqrt{8a^{24}b^{25}c^{26}d^{27}} \\
&= \sqrt{4 \cdot 2 \cdot a^{24} \cdot b^{24} \cdot b \cdot c^{26} \cdot d^{26} \cdot d} \\
&= \sqrt{4a^{24}b^{24}c^{26}d^{26} \cdot 2bd} \\
&= \sqrt{4} \cdot \sqrt{a^{24}} \cdot \sqrt{b^{24}} \cdot \sqrt{c^{26}} \cdot \sqrt{d^{26}} \cdot \sqrt{2bd} \\
&= 2a^{12}b^{12}c^{13}d^{13}\sqrt{2bd}
\end{aligned}$$

Example 1.5.85

Simplify $\sqrt{10a^4b^5n^5}\sqrt{6a^4b^4n^{20}}$.

Solution. We can start by applying $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$ from Principle 1.5.67 to get

$$\sqrt{10a^4b^5n^5} \cdot \sqrt{6a^4b^4n^{20}} = \sqrt{(10a^4b^5n^5)(6a^4b^4n^{20})} = \sqrt{60a^8b^9n^{25}}.$$

Note that this is the first time we have used this formula to turn something in the form of $\sqrt{a}\sqrt{b}$ into $\sqrt{ab}$: we had previously used this formula to turn the root of a product into the product of roots, but we can also use this formula the way we just did to turn the product of roots into the root of a product.

For each of the three factors 60 and a^8 and b^9 and n^{25}, we look for perfect squares that are factors. (Recall that an expression in the format of base raised to an exponent is a perfect square if the exponent is a multiple of 2, or is in other words even.)

$$\sqrt{60a^8b^9n^{25}} = \sqrt{4 \cdot 15 \cdot a^8 \cdot b^8 \cdot b^1 \cdot n^{24} \cdot n^1} = \sqrt{4}\sqrt{a^8}\sqrt{b^8}\sqrt{n^{24}}\sqrt{15b^1n^1} = 2a^4b^4n^{12}\sqrt{15bn}.$$

Putting all of the work together, we have $\sqrt{10a^4b^5n^5}\sqrt{6a^4b^4n^{20}} = \sqrt{(10a^4b^5n^5)(6a^4b^4n^{20})} = \sqrt{60a^8b^9n^{25}} = 2a^4b^4n^{12}\sqrt{15bn}$. One of the things to note is that there is an n in the problem, but this was not the n in the formula in Principle 1.5.67.

After presenting Principle 1.5.67, we presented the special case of this formula for cube roots, namely

$$\sqrt[3]{ab} = \sqrt[3]{a}\sqrt[3]{b}.$$

Our process for simplifying cube roots is like our process for simplifying square roots, with some minor twists. There are several small twists: we will be looking for cube roots now instead of square roots. To be thorough, let's formally define a perfect cube (analogous to our definition of a perfect square in Definition 1.5.73), give examples, and state the strategy for simplifying.

Definition 1.5.86

An expression is a perfect cube if it is the result of cubing an expression.

Example 1.5.87

- 0 is a perfect cube because $0 = (0)^2$.
- 1 is a perfect cube because $1 = (1)^3$.
- 8 is a perfect square because $8 = (2)^3$.
- 27 is a perfect square because $27 = (3)^3$.
- 81 is a perfect square because $81 = (4)^3$.
- 125 is a perfect square because $125 = (5)^3$.

This list continues, following a pattern.

Similar to Example 1.5.75, we present perfect cubes with variables:

Example 1.5.88

- x^0 is a perfect cube because $x^0 = (x^0)^3$.

- x^3 is a perfect cube because $x^3 = (x^1)^3$.
- x^6 is a perfect cube because $x^6 = (x^2)^3$.
- x^9 is a perfect cube because $x^9 = (x^3)^3$.
- x^{12} is a perfect cube because $x^{12} = (x^4)^3$.
- x^{15} is a perfect cube because $x^{15} = (x^5)^3$.

This list continues, following a pattern.

In Try it 1.5.82, we asked about the pattern for perfect squares. For an expression to be a perfect square, the exponent needed to be even, or you could say that the exponent needed to be a multiple of 2. Notice the exponents in the previous example were $0, 3, 6, 9, 12, 15$, and so on. So for an expression to be a perfect cube, the exponent needs to be a multiple of 3.

Strategy 1.5.89 How to simplify the cube root of a perfect cube.

- ***When does this strategy apply?*** This strategy applies to simplify any expression which is a cube root, and the contents inside the cube root is a perfect cube.
- ***How to apply the strategy***
  1. Given an expression of the form $\sqrt[3]{a}$, identify if a is a perfect cube. This means that there is some expression b such that $b^3 = a$.
  2. From $b^3 = a$, we get $\sqrt[3]{a} = b$.

Example 1.5.90

- From $(0)^3 = 0$, we get $\sqrt[3]{0} = 0$.
- From $(1)^3 = 1$, we get $\sqrt[3]{1} = 1$.
- From $(2)^3 = 8$, we get $\sqrt[3]{8} = 2$.
- From $(3)^3 = 27$, we get $\sqrt[3]{27} = 3$.
- From $(4)^3 = 64$, we get $\sqrt[3]{64} = 4$.
- From $(5)^3 = 125$, we get $\sqrt[3]{125} = 5$.

Example 1.5.91

- From $(x^0)^3 = x^0$, we get $\sqrt[3]{x^0} = x^0$.
- From $(x^1)^3 = x^3$, we get $\sqrt[3]{x^3} = x^1$.
- From $(x^2)^3 = x^6$, we get $\sqrt[3]{x^6} = x^2$.
- From $(x^3)^3 = x^9$, we get $\sqrt[3]{x^9} = x^3$.
- From $(x^4)^3 = x^{12}$, we get $\sqrt[3]{x^{12}} = x^4$.
- From $(x^5)^3 = x^{15}$, we get $\sqrt[3]{x^{15}} = x^5$.

Try it 1.5.92

We alreday saw that for expression in the format of "base raised to an exponent", the exponent needs to be a multiple of 3 for the entire expression to be a perfect cube. Now, examining the examples given earlier, how do you find the result of the cube root of a perfect cube? Is there a pattern? (This is one of those moments in math where the pattern always works!) Is there a uniform process that you can teach me? How would you explain to somebody how to simplify $\sqrt[3]{z^{33}}$?

Strategy 1.5.93 Simplifying the cube root of a product.

- ***When does this strategy apply?*** This strategy applies to simplify any expression which is a cube root, and the contents inside the cube root is a product (or can be turned into a product).
- ***How to apply the strategy***
    1. Rewrite the content inside the cube root as a product with one factor being a perfect cube.
    2. Apply the formula $\sqrt[3]{ab} = \sqrt[3]{a} \cdot \sqrt[3]{b}$.
    3. In the resulting expression of the form $\sqrt[3]{a} \cdot \sqrt[3]{b}$, the factor which is the cube root of a perfect cube can be simplified into an expression that does not require writing a cube root. (However, the cube root of an expression which is not a perfect cube will always still have a cube root symbol.)

Example 1.5.94

Simplify $\sqrt[3]{80a^9b^{10}c^{11}d^{12}e^{13}}$.

Solution. We can rewrite 80 as $8 \cdot 10$. Note that a^9 and d^{12} are already perfect cubes since these exponents are multiples of 3, but b^{10} and c^{11} and e^{13} are not perfect cubes since the exponents are not multiples of 3. To fix this, we need to rewrite b^{10} and c^{11} and e^{13}.

- For b^{10}, based on the exponent 10, we are looking for the largest multiple of 3 that is less than 10, which is 9. We can rewrite b^{10} as $b^9 \cdot b^1$.
- For c^{11}, based on the exponent 11, we are looking for the largest multiple of 3 that is less than 11, which is 9. We can rewrite c^{11} as $c^9 \cdot c^2$.
- For e^{13}, based on the exponent 13, we are looking for the largest multiple of 3 that is less than 13, which is 12. We can rewrite e^{13} as $e^{12} \cdot e^1$.

$$\begin{aligned}
&\sqrt[3]{80a^9b^{10}c^{11}d^{12}e^{13}} \\
&= \sqrt[3]{8 \cdot 10 \cdot a^9 \cdot b^9 \cdot b^1 \cdot c^9 \cdot c^2 \cdot d^{12} \cdot e^{12} \cdot e^1} \\
&= \sqrt[3]{8 \cdot a^9 \cdot b^9 \cdot c^9 \cdot d^{12} \cdot e^{12} \cdot 10bc^2e} \\
&= \sqrt[3]{8} \cdot \sqrt[3]{a^9} \cdot \sqrt[3]{b^9} \cdot \sqrt[3]{c^9} \cdot \sqrt[3]{d^{12}} \cdot \sqrt[3]{e^{12}} \cdot \sqrt[3]{10bc^2e} \\
&= 2a^3b^3c^3d^4e^4\sqrt[3]{10bc^2e}
\end{aligned}$$

Example 1.5.95

Simplify $\sqrt[3]{16x^3y^8z^4}$.

Solution. We can rewrite 16 as $8 \cdot 2$. Note that x^3 is already perfect cubes since the exponent 3 is a multiple of of 3, but y^8 and z^4 are not perfect cubes since the exponents 8 and 4 are not multiples of 3. To fix this, we need to rewrite y^8 and z^4.

- For y^8, based on the exponent 8, we are looking for the largest multiple of 3 that is less than 8, which is 6. We can rewrite y^8 as $y^6 \cdot y^2$.
- For z^4, based on the exponent 4, we are looking for the largest multiple of 3 that is less than 4, which is 3. We can rewrite z^4 as $z^3 \cdot z^1$.

So, we have

$$\begin{aligned}
&\sqrt[3]{16x^3y^8z^4} \\
&= \sqrt[3]{8 \cdot 2 \cdot x^3 \cdot y^6 \cdot y^2 \cdot z^3 \cdot z^1} \\
&= \sqrt[3]{8 \cdot x^3 \cdot y^6 \cdot z^3 \cdot 2y^2z}
\end{aligned}$$

$$= \sqrt[3]{8} \cdot \sqrt[3]{y^6}\,\sqrt[3]{z^3} \cdot \sqrt[3]{2y^2z}$$
$$= 2yy^2z\sqrt[3]{2y^2z}$$

The formula below is a useful tool for converting a radical expression into an exponential expression:

Principle 1.5.96

$\sqrt[n]{a} = a^{1/n}$

This formula is telling us that the radical expression $\sqrt[n]{a}$ can turn into to the exponential expression $a^{1/n}$ and like all our formulas, this also works in reverse: we can turn $a^{1/n}$ into $\sqrt[n]{a}$.

Like with any formula, we should pay attention to which variable appears where. In our formula, a was the content inside the radical and also the base of the exponential expression, and n was the small number outside the radical and also the denominator of the exponent (and note that the exponent was a fraction $\frac{1}{n}$). Then, when we apply this formula to convert $\sqrt[3]{4}$, this expression turns into $4^{1/3}$, and not into $3^{1/4}$.

Example 1.5.97

Rewrite $\sqrt[7]{c}\sqrt[8]{c}$ using an equal expression consisting of one radical.

Solution. Using Principle 1.5.96, we can rewrite the expression as

$$c^{1/7} \cdot c^{1/8}.$$

Since we now have the multiplication of two exponential expressions with the same base we can use $a^m a^n = a^{m+n}$ to add the exponents:

$$c^{1/7} \cdot c^{1/8} = c^{1/7+1/8} = c^{15/56}.$$

Finally, using Principle 1.5.96 in reverse, we rewrite this as a radical:

$$c^{15/56} = \sqrt[56]{c^{15}}.$$

Putting all of the work together, from the first expression to the last expression, writing equal signs where they belong, we have $\sqrt[7]{c}\sqrt[8]{c} = c^{1/7} \cdot c^{1/8} = c^{1/7+1/8} = c^{15/56} = \sqrt[56]{c^{15}}$.

Example 1.5.98

Write $\frac{\sqrt[3]{x}}{\sqrt[4]{x}}$ with only one radical.

Solution. Using Principle 1.5.96, we can rewrite the expression as

$$\frac{x^{1/3}}{x^{1/4}}.$$

Since we now have the fraction of two exponential expressions with the same base we can use $\frac{a^m}{a^n} = a^{m-n}$ to subtract the exponents:

$$\frac{x^{1/3}}{x^{1/4}} = x^{1/3-1/4} = x^{1/12}.$$

Finally, using Principle 1.5.96 in reverse, we rewrite this as a radical:

$$x^{1/12} = \sqrt[12]{x}$$

Putting all of the work together, $\frac{\sqrt[3]{x}}{\sqrt[4]{x}} = \frac{x^{1/3}}{x^{1/4}} = x^{1/3-1/4} = x^{1/12} = \sqrt[12]{x}$.

The formula in Principle 1.5.96 has two natural extensions, which we can discover by examining:

$$\sqrt[n]{a^m} = (a^m)^{1/n} = a^{m\cdot\frac{1}{n}} = a^{\frac{m}{1}\cdot\frac{1}{n}} = a^{\frac{m}{n}}$$

as well as

$$(\sqrt[n]{a})^m = (a^{1/n})^m = a^{\frac{1}{n}\cdot m} = a^{\frac{1}{n}\cdot\frac{m}{1}} = a^{\frac{m}{n}}.$$

In both computations above, when we zoomed in to the exponent, we saw the product of a fraction of a non-fraction, so we converted the non-fraction to a fraction and multiplied straight across, as our formula for multiplying fractions tells us in Principle 1.2.56. In the end we see that $\sqrt[n]{a^m}$ and $(\sqrt[n]{a})^m$ both turn into $a^{m/n}$.

Principle 1.5.99

- $\sqrt[n]{a^m} = a^{m/n}$
- $(\sqrt[n]{a})^m = a^{m/n}$

As earlier, the denominator of the fractional exponent becomes the small number outside the radical. What about the numerator of the fractional exponent? The numerator of the exponent stays as a numerator (since content that is not part of a fraction can be thought of as a fraction over 1). In reality, the expressions $\sqrt[n]{a^m}$ and $(\sqrt[n]{a})^m$ only differ in the order in which the actions are performed: between raising to the mth power and taking an nth root, the placement of the writing (and parentheses) just specifies which action is performed first. But in the end, both expressions are equal to $a^{m/n}$.

Example 1.5.100

Simplify $16^{3/4}$.

Solution 1. We can apply $\sqrt[n]{a^m} = a^{m/n}$ from Principle 1.5.99 backwards:

$$16^{3/4} = \sqrt[4]{16^3} = \sqrt[4]{4096} = 8.$$

Solution 2. Alternatively, we can apply $(\sqrt[n]{a})^m = a^{m/n}$ from Principle 1.5.99 backwards:

$$16^{3/4} = (\sqrt[4]{16})^3 = (2)^3 = 8.$$

Both answers above worked, giving the same final answer of 8. However, pause to stare at the steps in both answers, and see which one you prefer. Can I share my thoughts? In the first answer, we had to simplify 16^3. Once we did that, we had to take the fourth root of 4096: it was really a trial-and-error that I did to take a number, raise it to the fourth power, and see if I got 4096, and I was thankful that I only had to stop at 8. If you want my honest preference, I prefer the second answer. The reason is that our first arithmetic step was to find the fourth root of 16. Once we did that, we had to cube 2. Overall, the arithmetic there was easier.

Stepping back from the details of the arithmetic, I want to point out that both answers were correct. However, the steps in the second answer were easier for me to do. This is a good reminder that in math, there are often multiple ways to get to the same answer. In the specific case of an expression in the form $a^{m/n}$, we can apply either formula in Principle 1.5.99, and they just differ in whether we should process the nth root first or the mth power first (note the nth root comes from the fact that we can say $\frac{1}{n}$ is one factor of the exponent). If it is possible to simplify the nth root, let's try to do this first, because this will lower the overall size of the numbers involved in the arithmetic.

Example 1.5.101

Simplify $32^{-2/5}$.

Solution. This problem is almost in the format of the question we did, but the exponent introduces a minus sign. We can use $a^{-n} = \frac{1}{a^n}$ from Principle 1.5.36 to rewrite $32^{-2/5}$ as $\frac{1}{32^{2/5}}$. Then, we can apply $(\sqrt[n]{a})^m = a^{m/n}$ from Principle 1.5.99 backwards. $32^{-2/5} = \frac{1}{32^{2/5}} = \frac{1}{(\sqrt[5]{32})^2} = \frac{1}{(2)^2} = \frac{1}{4}$.

There are other possible ways to write steps do to this problem, but this problem involves smaller numbers when prioritizing the 5th root before the 2 power. The extra complication with the negative exponent makes this a good time to bring up a common error that occurs. Let's start a section on clarifications and new techniques, and start by bringing up the error, and then explain.

1.5.4 Clarifications and techniques

Though we have now seen all the tools we need, the last exercise we did pointed out a common error. So, let's start with that error:

Warning 1.5.102 $a^{1/n} \neq \frac{1}{a^n}$.

Example 1.5.103

This warning tells us that $x^{1/3} \neq \frac{1}{x^3}$.

Example 1.5.104

This warning tells us that $25^{1/2} \neq \frac{1}{25^2}$.

The kind of error that's occuring when someone incorrectly turns $25^{1/2}$ into $\frac{1}{25^2}$ or vice versa is an attempt at rearranging the things present, by trying to keep the location of certain things in place (for example, the 2 in the denominator of the exponent in $25^{1/2}$ was moved to becoming the numeator exponent of an expression where the exponent appeared in the denominator of a fraction.) That explanation may have been confusing, but there was a general attempt to move things around in places that seemed reasonable, but we instead need to follow our formulas. Let's dig into an explanation that I encourage you to read:

Note 1.5.105 We've presented formulas (just a few at a time, so that it's not as overwhelming) and then practiced, but now it's time to place two formulas that we've seen one right after the other, so that we can compare the formulas and make some comments:

- $a^{1/n} = \sqrt[n]{a}$ from Principle 1.5.96
- $\frac{1}{a^n} = a^{-n}$ from Principle 1.5.36

Both formulas have a fraction in them, but we need to pay close attention to where those fractions are. Speaking a little informally: The first formula tells us that when the exponent is a fraction, then the entire expression can be replaced with a radical. The second formula tells us that when the entire expression is a fraction (and we do mean entire expression being a fraction: the left side of the second formula doesn't have a fraction for the exponent) then the entire expression can be replaced with an expression with a negative exponent.

Let's see examples. The expression $3^{1/4}$ can be rewritten as $\sqrt[4]{3}$. The expression $\frac{1}{3^4}$ is in its entirety of writing a fraction, and this becomes 3^{-4}, which we should note has no fractions in it whatsoever.

These comments apply when reading both formulas from right to left as well:

- $\sqrt[n]{a} = a^{1/n}$, so we can turn an expression that is a radical into an exponential expression, where we end up writing a fraction as the exponent.
- $a^{-n} = \frac{1}{a^n}$, so we can turn an expression that has a negative exponent into a fraction, where we end up writing a fraction as the entire replacement expression.

We stated earlier that the warning tells us $25^{1/2} \neq \frac{1}{25^2}$. The left side legitimately can be rewritten $25^{1/2} = \sqrt[2]{25} = \sqrt{25} = 5$, while the right side can be rewritten as $\frac{1}{25^2} = \frac{1}{625}$. Just as the warning let us know, these numbers 5 and $\frac{1}{625}$ are not equal.

There are some typical techniques that are used to speed up the simplification of certain expressions with exponents. These techniques are not new formulas, but rather they are combinations of formulas that we've already seen. Let's introduce one of these techinques now. If we take $a^{-n} = \frac{1}{a^n}$ and $\frac{1}{a^{-n}} = a^n$ from Principle 1.5.36 and combine these with $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$ from Principle 1.2.56, we have the following, which we describe informally:

Principle 1.5.106

- *A factor in the denominator can be turned into a factor of the numerator by negating its exponent.*
- *A factor in the numerator can be turned into a factor of the denominator by negating its exponent.*

In a similar manner, by combining $\frac{a^m}{a^n} = a^{m-n}$ from Principle 1.5.27 and Principle 1.2.56:

Principle 1.5.107

Given a fraction with x^m as a factor in the numerator and x^n as a factor in the denominator, the two factors can be replaced with a single factor of x^{m-n} in the numerator.

Example 1.5.108

Simplify $(-3x^{-2})(4x^4)$.

Solution 1. All parentheses here can be dropped, since there are no plus or minus signs inside the parentheses. So we have

$$-3 \cdot x^{-2} \cdot 4 \cdot x^4.$$

We can apply Principle 1.5.106 to replace the x^{-2} in the numerator with an x^2 in the denominator, so we have

$$\frac{-3 \cdot 4 \cdot x^4}{x^2}.$$

We can simplify the arithmetic of $-3 \cdot 4$, and our expression is:

$$\frac{-12x^4}{x^2}.$$

One option is to now apply $\frac{a^m}{a^n} = a^{m-n}$ from Principle 1.5.16 to subtract the exponents of x, giving us

$$-12x^2.$$

Putting this work altogether, we have: $(-3x^{-2})(4x^4) = -3 \cdot x^{-2} \cdot 4 \cdot x^4 = \frac{-3 \cdot 4 \cdot x^4}{x^2} = \frac{-12x^4}{x^2} = -12x^2$.

Solution 2. Alternatively, we take the expression

$$-3 \cdot x^{-2} \cdot 4 \cdot x^4$$

from the first step of our previosu answer and apply $a^m a^n = a^{m+n}$ from Principle 1.5.3 to add the exponents of x, giving us

$$-3 \cdot 4 \cdot x^{-2+4} = -3 \cdot 4 \cdot x^2.$$

We can simplify the arithmetic of $-3 \cdot 4$, and our expression is:

$$-12x^2.$$

Putting this work altogether, we have: $(-3x^{-2})(4x^4) = -3 \cdot 4 \cdot x^{-2+4} = -3 \cdot 4 \cdot x^2 = -12x^2$.

Example 1.5.109

Rewrite $\frac{a^3b^5c^7d^{10}}{a^8c^7d^6}$ as an expression with no negative exponents.

Solution 1. We can apply Principle 1.5.107 to replace the factors of a, c, and d in the numerator and denominator with a single factor in the numerator:

$$\frac{a^3b^5c^7d^{10}}{a^8c^7d^6} = a^{8-3}b^5d^{10-6} = a^{-5}b^5d^4.$$

Now, to heed to the request that there are no negative exponents, we can replace a^{-5} in the numerator with a^5 in the denominator using Principle 1.5.106:

$$a^{-5}b^5d^4 = \frac{b^5d^4}{a^5}.$$

Putting this work altogether, we have: $\frac{a^3b^5c^7d^{10}}{a^8c^7d^6} = a^{8-3}b^5d^{10-6} = a^{-5}b^5d^4 = \frac{b^5d^4}{a^5}$.

Solution 2. In case you're a little weary of Principle 1.5.107, let's try this question a little slower:

$$\frac{a^3b^5c^7d^{10}}{a^8c^7d^6} = \frac{a^3}{a^8} \cdot \frac{b^5}{1} \cdot \frac{c^7}{c^7} \cdot \frac{d^{10}}{d^6}$$

because we can take the expression after the equal sign (a product of fractions) and multiply staright across. In the new expression, we apply Principle 1.5.27 to get

$$a^{-5}b^5d^4$$

which matches the expression we saw in the previous answer. (Note that we could say that $\frac{c^7}{c^7} = 1$ since any nonzero number divided by itself is 1.) Then, this turns into $\frac{b^5d^4}{a^5}$ using Principle 1.5.106.

Example 1.5.110

Rewrite $\frac{2x^{-3}y^4}{5x^2y^{-6}}$ as an expression with no negative exponents.

Solution 1. We can apply Principle 1.5.106 to replace the factor of x^{-3} in the numerator with a factor of x^3 in the denominator, and also replace the factor of y^{-6} in the denominator with a factor of y^6 in the numerator: $\frac{2x^{-3}y^4}{5x^2y^{-6}} = \frac{2y^4y^6}{5x^2x^3} = \frac{2y^{4+6}}{5x^{2+3}} = \frac{2y^{10}}{5x^5}$.

Solution 2. We can instead apply Principle 1.5.107 to replace the factor of x^{-3} in the numerator and the factor of x^2 in the denominator with a single factor of $x^{-3-2} = x^{-5}$ in the numerator, and also replace the factor of y^4 in the numerator and the factor of y^{-6} in the denominator with a single factor of $y^{4-(-6)} = y^{10}$ in the numerator: $\frac{2x^{-3}y^4}{5x^2y^{-6}} = \frac{2x^{-5}y^{10}}{5} = \frac{2y^{10}}{5x^5}$.

One key source of confusion is how we got y^{10}. Note that we need to take the exponent of 4 in the numerator and subtract the exponent of -6 in the denominator. This is why we wrote $y^{4-(-6)}$. It is common here to make the error of writing y^{4-6}. Note that $\frac{a^m}{a^n} = a^{m-n}$ from Principle 1.5.27 tells us we need to subtract exponents, and the need to subtract exponents remains true even when the exponent in the denominator is negative. (See a similar example earlier in Example 1.5.35, with a more thorough discussion there.)

For this exercise, we presented two different answers (one starting with Principle 1.5.106 and one starting with Principle 1.5.107) and both answers require basically the same amount of writing. In fact, I think in any room full of people, folks will be split 50-50 as to which way they'd like to think about this type of question!

Note 1.5.111 We should be careful to state that both Principle 1.5.106 and Principle 1.5.107 require that we are seeing factors.

- For example, Principle 1.5.106 doesn't apply to $\frac{3}{4+b^{-5}}$ because b^{-5} is not a factor of the denominator.
- Similarly, Principle 1.5.107 doesn't apply to $\frac{x^7}{x^2+5}$, because x^2 is not a factor of the denominator (though we say x^7 is a factor of the numerator, since we can rewrite the numerator as $x^7 \cdot 1$.)

Let's discuss a common error for radicals which is very much like the error for exponents that we discussed in Warning 1.5.50:

Warning 1.5.112 Radicals do not "distribute" over addition or subtraction.

- $\sqrt[n]{x+y} \neq \sqrt[n]{x} + \sqrt[n]{y}$
- $\sqrt[n]{x-y} \neq \sqrt[n]{x} - \sqrt[n]{y}$

As examples of what this warning is trying to say,

- $\sqrt{g+h}$ does not become $\sqrt{g} + \sqrt{h}$.
- $\sqrt{g-h}$ does not become $\sqrt{g} - \sqrt{h}$.
- $\sqrt[3]{g+h}$ does not become $\sqrt[3]{g} + \sqrt[3]{h}$.
- $\sqrt{1+x^2}$ does not become $1+x$. It seems like someone started with $\sqrt{1+x^2}$ then erroneously distributed the radical to get $\sqrt{1} + \sqrt{x^2}$, and from this expression got $1+x$. While it is true that $\sqrt{1} + \sqrt{x^2}$ becomes $1+x$, the problem is that $\sqrt{1+x^2}$ is not equal to $\sqrt{1} + \sqrt{x^2}$.
- $\sqrt{x^2+y^2}$ does not become $x+y$.
- $\sqrt{4+x}$ does not become $2+\sqrt{x}$.
- $\sqrt{(x-3)^2+(y-4)^2}$ does not become $(x-3)+(y-4)$.

The warning is adressing the common error of trying to distribute a radical over addition or subtraction. However, we can turn $\sqrt{4 \cdot x}$ into $2\sqrt{x}$ by applying Principle 1.5.62 then simplifying: $\sqrt{4 \cdot x} = \sqrt{4} \cdot \sqrt{x} = 2\sqrt{x}$.

Warning 1.5.113 Do not treat a^n as if it is $a \cdot n$.

It is not usually the case that someone will turn 3^2 into 6, but when someone does this, I believe they were reading too quickly, and turned 3^2 into $3 \cdot 2$ by just going too fast.

A more typical version of this error happens when the exponent is a fraction, especially a fraction whose numerator is 1. For example, it is not correct to turn $16^{1/2}$ into 8. I believe the way that someone gets there is by thinking the following: $16^{1/2}$ was accidentally turned into $16 \cdot \frac{1}{2}$ which does equal 8, but the problem was going from $16^{1/2}$ to $16 \cdot \frac{1}{2}$. It was probably this line of reasoning that got someone to produce a final value of 8, even if they didn't actually write $16 \cdot \frac{1}{2}$ on paper. That is, even if someone didn't write $16 \cdot \frac{1}{2}$ on paper, they were seeing this in their mind, because 16 multiplied by $\frac{1}{2}$ is equal to 8. The problem is that we can't just treat the exponent like it's a factor, which is exactly what is warning is trying to address. Now that we've discussed what the error was, in the case of this expression $16^{1/2}$, let's talk about what we could do instead. Our first step will apply Principle 1.5.96, and we get $16^{1/2} = \sqrt{16} = 4$, which is a different final answer than 8.

We now end the section by doing a mixed variety of practice problems. Some of these can be done multiple ways, so we present several answers (but know that many of these have even more answer than we show you).

Example 1.5.114

Simplify $(-6x^{7/5})(2x^{8/5})$.

Solution. In this expression, there is no operation of addition or subtraction inside parentheses, so the Distributive Law does not apply. In fact, we can drop all parentheses, it is helpful (for me, anyway) to write in all multiplication symbols:

$$-6 \cdot x^{7/5} \cdot 2 \cdot x^{8/5}.$$

We can juggle the order of the four factors because we have the Commutative Property of Multiplication:

$$-6 \cdot 2 \cdot x^{7/5} \cdot x^{8/5}.$$

Applying arithmetic in the first two factors gives us -12, and we can apply $a^m a^n = a^{m+n}$ for the third and fourth factors:

$$-12x^{15/5}$$

and finally, the exponont is a reducible fraction:

$$-12x^3.$$

In summary, $(-6x^{7/5})(2x^{8/5}) = -6 \cdot x^{7/5} \cdot 2 \cdot x^{8/5} = -6 \cdot 2 \cdot x^{7/5}x^{8/5} = -12x^{15/5} = -12x^3$.

Example 1.5.115

Simplify $\frac{4a^2b}{a^3b^3} \cdot \frac{40a^2b}{2b^4}$ as an expression using only positive exponents.

Solution 1. We can start by multiplying across to get

$$\frac{160a^4b^2}{2a^3b^7}.$$

To briefly explain this work, we'll describe the numerator in detail: $(4a^2b) \cdot (40a^2b) = 4 \cdot a^2 \cdot b \cdot 40 \cdot a^2 \cdot b = (4 \cdot 40) \cdot (a^2 \cdot a^2) \cdot (b \cdot b) = 160a^{2+2}b^{1+1} = 160a^4b^2$. In $(4a^2b) \cdot (40a^2b)$, everything is multiplication (no additions or subtractions) so we can drop parentheses, rearrange the factors through the Commutative Property of Multiplication, and apply $a^m a^n = a^{m+n}$. The denominator is similar, so I invite you to think about it. In

$$\frac{160a^4b^2}{2a^3b^7},$$

we can rewrite this fraction as:

$$\frac{160}{2} \cdot \frac{a^4}{a^3} \cdot \frac{b^2}{b^7}.$$

We often take the product of fractions and multiply across to simplify, and we just did that process in reverse here. The benefit is that we can simplify each of the three fractions separately. While we have some new tools, we can choose to rewrite $\frac{a^4}{a^3} = \frac{a \cdot a \cdot a \cdot a}{a \cdot a \cdot a}$ and then cancel three factors of a, leaving us with just a in the numerator. Similarly, when we rewrite $\frac{b^2}{b^7}$, we can cancel two factors of b, leaving us with b^5 in the denominator. In the end, we're left with

$$80 \cdot a \cdot \frac{1}{b^5} = \frac{80a}{b^5}.$$

Putting this work altogether, we have: $\frac{4a^2b}{a^3b^3} \cdot \frac{40a^2b}{2b^4} = \frac{160a^4b^2}{2a^3b^7} = \frac{160}{2} \cdot \frac{a^4}{a^3} \cdot \frac{b^2}{b^7} = 80 \cdot a \cdot \frac{1}{b^5} = \frac{80a}{b^5}$.

Solution 2. Picking up our previous answer from

$$\frac{160a^4b^2}{2a^3b^7},$$

we can simplify the arithmetic of $\frac{160}{2}$ to get

$$\frac{80a^4b^2}{a^3b^7}.$$

Then, we can apply Principle 1.5.107 to get

$$80a^1b^{-5},$$

then rewrite the b^{-5} in the numerator as b^5 in the denominator using Principle 1.5.106, giving us

$$80ab^{-5} = \frac{80a}{b^5}.$$

Putting all work together (including the work borrowed from the earlier answer), $\frac{4a^2b}{a^3b^3} \cdot \frac{40a^2b}{2b^4} = \frac{160a^4b^2}{2a^3b^7} = 80a^{4-3}b^{2-7} = 80a^1b^{-5} = \frac{80a}{b^5}$.

Solution 3. Our second answer was built off our first answer. Here's a way to go back to the beginning and do something totally different with the original expression

$$\frac{4a^2b}{a^3b^3} \cdot \frac{40a^2b}{2b^4}.$$

We can look at each fraction separately. For example, by applying Principle 1.5.107 we can turn $\frac{4a^2b}{a^3b^3}$ into $4a^{-1}b^{-2}$. In the same way, we can turn $\frac{40a^2b}{2b^4}$ into $20a^2b^{-3}$. Then, we can multiply these two expressions together to get

$$(4a^{-1}b^{-2})(20a^2b^{-3}) = 80a^{(-1)+2}b^{(-2)+(-3)} = 80a^1b^{-5} = \frac{80a}{b^5}.$$

Putting this work altogether, we have: $\frac{4a^2b}{a^3b^3} \cdot \frac{40a^2b}{2b^4} = (4a^{-1}b^{-2})(20a^2b^{-3}) = 80a^{(-1)+2}b^{(-2)+(-3)} = 80a^1b^{-5} = \frac{80a}{b^5}$.

Example 1.5.116

Simplify $\left(\frac{y^{3/2}}{y^{-1/3}}\right)^{-3}$.

Solution 1. We can start by applying $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$ with to get $a = y^{3/2}$ and $b = y^{-1/3}$ and $n = -3$ to get

$$\left(\frac{y^{3/2}}{y^{-1/3}}\right)^{-3} = \frac{(y^{3/2})^{-3}}{(y^{-1/3})^{-3}}.$$

Then, we can apply $(a^m)^n = a^{mn}$ to both the numerator and denominator:

$$\frac{(y^{3/2})^{-3}}{(y^{-1/3})^{-3}} = \frac{y^{(3/2)(-3)}}{y^{(-1/3)(-3)}} = \frac{y^{-9/2}}{y^1}.$$

Finally, we can apply Principle 1.5.107 to get

$$y^{-9/2-1} = y^{-11/2} = -\frac{1}{y^{11/2}}.$$

Putting this work altogether, we have: $\left(\frac{-y^{3/2}}{y^{-1/3}}\right)^{-3} = \frac{(-y^{3/2})^{-3}}{(y^{-1/3})^{-3}} = \frac{-1 \cdot y^{(3/2)(-3)}}{y^{(-1/3)(-3)}} = \frac{-1 \cdot y^{-9/2}}{y^1} = -1 \cdot y^{-9/2-1} = -1 \cdot y^{-11/2} = -y^{-11/2} = -\frac{1}{y^{11/2}}$.

Solution 2. Instead of replacing the entire expression right away, we can "zoom in" slightly (ignoring the outer exponent of -3) and focus on replacing $\frac{y^{3/2}}{y^{-1/3}}$ with something equal. For this fraction, we can apply Principle 1.5.107 to get $y^{\frac{3}{2}-(-\frac{1}{3})} = y^{\frac{3}{2}+\frac{1}{3}} = y^{\frac{9}{6}+\frac{2}{6}} = y^{11/6}$. So, the original expression is equal to $(y^{11/6})^{-3}$. (The point is that we zoomed in to focus on a portion of the original expression, but we still need to account for the -3 exponent on the outside which didn't change.) We can apply $(a^m)^n = a^{mn}$ to $(y^{11/6})^{-3}$ to get

$$y^{\frac{11}{6}\cdot\frac{-3}{1}} = y^{-\frac{33}{6}} = y^{-\frac{11}{2}} = -\frac{1}{y^{11/2}}.$$

Putting this work together, we have: $\left(\frac{y^{3/2}}{y^{-1/3}}\right)^{-3} = (y^{\frac{3}{2}-(-\frac{1}{3})})^{-3} = (y^{\frac{11}{6}})^{-3} = y^{-\frac{33}{6}} = y^{-\frac{11}{2}} = -\frac{1}{y^{11/2}}$.

Example 1.5.117

Simplify $\sqrt{3a^2b^3}\sqrt{6a^5b}$.

Solution. We can start by applying $\sqrt{a}\cdot\sqrt{b} = \sqrt{ab}$ from Principle 1.5.67 to get

$$\sqrt{3a^2b^3}\cdot\sqrt{6a^5b} = \sqrt{(3a^2b^3)(6a^5b)} = \sqrt{18a^7b^4}.$$

For each of the three factors 18 and a^7 and b^4, we look for perfect squares that are factors. (Recall that an expression in the format of base raised to an exponent is a perfect square if the exponent is a multiple of 2, or is in other words even.)

$$\sqrt{18a^7b^4} = \sqrt{9\cdot 2\cdot a^6\cdot a^1\cdot b^4} = \sqrt{9}\sqrt{a^6}\sqrt{b^4}\sqrt{2a} = 3a^3b^2\sqrt{2a}.$$

Example 1.5.118

Simplify $\sqrt[3]{320}$.

Solution. Because 320 is such a large number, I am finding it difficult to know off the top of my head the largest factor of 320 that is a perfect cube. I will instead just look for any factor that is a perfect cube. I'm drawn to 32 because $32\cdot 10 = 320$ and because I know that 32 has 8 as a factor, and 8 is a perfect cube. From $320 \div 8 = 40$, I might choose do this as a first step:

$$\sqrt[3]{320} = \sqrt[3]{8\cdot 40} = \sqrt[3]{8}\cdot\sqrt[3]{40} = 2\sqrt[3]{40}.$$

Now, I can look at 40 and see if it has any perfect cube factors. It doesn't, so I can stop here.

1.5.5 Applications Revisited

Let's revisit the applications we introduced at the beginning of this section. Please keep in mind that some of the applications below will bring in equations from various sectors that may not be familiar to you. That's okay! We can still see embedded inside the work the use of the exponential and radical concepts that we discussed! The point here isn't to burden you with introducing a formula in the middle of an answer as if you already have to know it and what each part means. No, that's not what we're trying to do here. Instead, we're saying that there are real applications that use the concepts from exponents and radicals, and to be authentic, we just had to temporarily introduce one other thing to fully talk through an example. I would encourage you to read through these examples for where the exponent and radical skills get used, but I think it would be unreasonable for you or anyone else to expect you to have come up with these answers which often required bringing up equations from finance and science and so on.

Example 1.5.119

You deposit 100 dollars into an account earning 5% interest compounded continuously. How much money is in the account at the end of 3 years, and what is the factor by which the amount in the account grows on an annual basis?

Solution. We will need the Continuous Compound Interest formula $A = Pe^{rt}$, where P is the initial balance, r is the annual interest rate (as a decimal), t is the time in years, and A is the balance at the time given by t. Thus, we have $P = 100$, $r = 0.05$, and $t = 3$, meaning the generic equation $A = Pe^{rt}$ becomes

$$A = 100e^{0.05\cdot 3}.$$

To answer the first qestion (what is the account balance in 3 years), we can evaluate $100e^{0.05\cdot 3} \approx 116.18$, though in a calculator, we need to insert parentheses as in `1000 * e^(0.05 * 3)` since exponentiation is simplified before multiplication in the Order of Operations. So the account's balance at the end of 3 years is approximately 116.18 dollars.

To answer the second question (what is the factor by which the amount grows on an annual basis), we need to see what factor is multiplied each year, no longer needing the constant 3 for t. That is, in

$$A = 100e^{0.05t}$$

how can we see what the base is when t is the exponent? We can apply $(a^m)^n = a^{mn}$ with $a = e$, $m = 0.05$, and $n = t$ to get

$$A = 100 \cdot (e^{0.05})^t,$$

thus the base is $e^{0.05}$ which is approximately 1.05127. Thus, on a year-to-year basis, the account balance will grow to 1.051227 times the previous year's balance. In other words, if we know the balance of the account at a certain point in time, then to get the balance exactly one year later, multiply the known balance by 1.051227.

Example 1.5.120

As an immunologist, you start with a culture of 1000 bacteria. The population doubles every 3 hours. How many bacteria are there after 12 hours, and by what factor does the population grow in a single hour?

Solution. We'll need the formula $P = I \cdot 2^{t/3}$, where I is the initial population, t is time in hours, and P is the population at time t. Thus, we have $I = 1000$ and $t = 12$, meaning the generic equation $P = I \cdot 2^{t/3}$ becomes

$$P = 1000 \cdot 2^{12/3} = 1000 \cdot 2^4 = 1000 \cdot 16 = 16000.$$

So the population after 12 hours is 16000 bacteria.

To find the factor by which the population grows in a single hour, we need to see what factor is multiplied each hour, no longer needing the constant 12 for t. That is, in

$$P = 1000 \cdot 2^{t/3}$$

how can we see what the base is when t is the exponent? We can apply $(a^m)^n = a^{mn}$ with $a = 2$, $m = \frac{1}{3}$, and $n = t$ to get

$$P = 1000 \cdot (2^{1/3})^t,$$

thus the base is $2^{1/3}$, which can also be written $\sqrt[3]{2}$ due to Principle 1.5.96. It will actually be more convenent to put $2^{1/3}$ in a calculator. This needs to be typed in as `2^(1/3)` with the parentheses due to Order of Operations, because typing in `2^1/3` will be interpreted as $\frac{2^1}{3}$. Using a calculator, $2^{1/3} \approx 1.25992$, so take the number of bacteria at a given moment, and to get the number of bacteria an hour later, multiply by 1.25992.

Example 1.5.121

You are filling small and large boxes to move apartments. Each small box is a cube with side length s. If you double the side length, by what factor does the volume of the cube increase? This will tell us how much more can fit into the larger box over the smaller box.

Solution. Volume of a rectangular prism is given by $V = \ell \cdot w \cdot h$. We can substitute s for ℓ and w and h to get the volume of the small box: $V = s \cdot s \cdot s = s^3$.

Now, for the larger box, we can substitute $2s$ for ℓ and w and h to get the volume of the large box: $V = (2s)\cdot(2s)\cdot(2s) = (2s)^3$. The final expression $(2s)^3$ can be rewritten using $(ab)^n = a^n b^n$ from Principle 1.5.16 to get $(2s)^3 = 2^3 \cdot s^3 = 8s^3$.

Comparing s^3 for the volume of the small box and $8 \cdot s^3$ for the volume of the big box, we see that the big box can hold 8 times as much as the small box.

Example 1.5.122

Now that you're in your new apartment, you want to build in a bookshelf as large as you can. You will bring in a slab of wood "diagonally" through a door that has a width of 3 feet and a height of 6 feet. What is the diagonal length of the door, and thus what is the longest piece of wood you can bring in through the door?

Solution. The diagonal of the door is the hypotenuse of a right triangle with legs of 3 feet and 6 feet. If we denote the diagonal of the door by d then the Pythagorean Theorem gives

$$3^2 + 6^2 = d^2$$

so $d^2 = 45$ which means that $d = \sqrt{45} = \sqrt{9\cdot 5} = \sqrt{9}\cdot\sqrt{5} = 3\sqrt{5}$ feet. The longest piece of lumber that will fit diagonlly is $3\sqrt{5}$ feet, which is approximately 6.71 feet.

Example 1.5.123

Continuing with the bookshelf, you have identical slabs of wood coming from a previous project. These slabs have their width determined by taking the hypotenuse of a right triangle with side lengths 1 foot and 2 feet. (The measurements of 1 foot and 2 feet came from the height and width of a storage space in your previous apartment.) How many of these slabs should be glued together side-by-side to create the back of the bookshelf in the previous question?

Solution. Each slab's measurement is the hypotenuse of a right triangle with legs of 1 foot and 2 feet. If we denote the length of each slab by x, then the Pythagorean Theorem gives

$$1^2 + 2^2 = x^2$$

which gives $x^2 = 5$ and so $x = \sqrt{5}$ feet.

Since each slab has a measurement of $\sqrt{5}$ feet, and the back of the bookshelf needs to be $3\sqrt{5}$ feet, we can see that 3 slabs are needed.

1.5.6 SUMMARY

1. Various exponential laws such as $a^m a^n = a^{m+n}$ and $(a^m)^n = a^{mn}$ and $(ab)^n = a^n b^n$ and $\frac{a^m}{a^n} = a^{m-n}$ and $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$ can all be explained by keeping in mind that exponentiation can be interpreted as repeated multiplication. This is helpful when we know we need one of these formulas but can't recall both sides: we can substitute small values for the exponents and recover how the formulas should look just by counting the number of times a base appears.

2. In the situation of $(ab)^n$ and $\left(\frac{a}{b}\right)^n$ respectively, the exponent can apply to each base, so we get $a^n b^n$ and $\frac{a^n}{b^n}$ respectively. However, as soon as there is an addition symbol or subtraction symbol as in $(a+b)^n$ and $(a-b)^n$, we cannot "distribute" the exponent here.

3. While some expressions with radicals are fine to simplify using radical formulas, we always have the option of converting radical expressions to exponential expressions using $\sqrt[n]{a} = a^{1/n}$ and applying exponent formulas instead.

4. Exponent formulas such as $(ab)^n = a^n b^n$ have analogous versions for radicals, i.e., $\sqrt[c]{ab} = \sqrt[c]{a}\sqrt[c]{b}$.

5. When an expression is written in the form of base raised to exponent, the exponent must be even for the expression to be a perfect square. This is as true for x^{12} as it is for 25 once we rewrite this as 5^2.

6. When an expression is written in the form of base raised to an exponent, the exponent must be a multilpe of three to be a perfect cube.

1.5.7 EXERCISES

1. Simplify the following expressions containing exponents: when a negative exponent appears, rewrite the expression using an exponent law so that the final expression has no negative exponents.

   (a) Simplify $x^3 \cdot x^4$.
   (b) Simplify $a^5 \div a^2$.
   (c) Simplify $(y^3)^2$.
   (d) Simplify $(2x)^3$.
   (e) Simplify $x^0 + 3x^2$.
   (f) Simplify $x^{-3} \cdot x^5$.
   (g) Simplify $\frac{m^7}{m^{10}}$.
   (h) Simplify $(3a^2b^3)^2$.
   (i) Simplify $\frac{4x^3y^2}{2x^5y}$.
   (j) Simplify $(x^2y^{-3})^3$.
   (k) Simplify $\frac{(a^3b^2)^2}{a^4b^3}$.

(l) Simplify $\frac{(2x^{-1}y^3)^2}{4x^{-2}y}$.

(m) Simplify $(x^3y^2)^{-2}$.

(n) Simplify $\frac{(p^4q^{-2})^3}{p^2q^{-4}}$.

(o) Simplify $(x^{1/2}y^3)^4$.

(p) Simplify $\frac{x^{3/2}}{x^{1/4}}$.

(q) Simplify $\left(\frac{x^2y^{-1}}{z^3}\right)^2$.

(r) Simplify $\frac{(a^{-2}b^3)^{-1}}{b^{-2}a}$.

(s) Simplify $(x^{-1}y^2z^{-3})^3$.

(t) Simplify $\frac{(2x^{-2}y^3z^{-1})^2}{4x^{-1}y^{-2}z^3}$.

2. Simplify the following expressions containing radicals.

(a) Simplify $\sqrt{49}$.

(b) Simplify $\sqrt{12}$.

(c) Simplify $\sqrt{18x^2}$.

(d) Simplify $3\sqrt{8}$.

(e) Simplify $\sqrt{50a^4}$.

(f) Simplify $\sqrt{27} \cdot \sqrt{3}$.

(g) Simplify $\sqrt{5} \cdot \sqrt{20}$.

(h) Simplify $\sqrt{2x} \cdot \sqrt{8x^3}$.

(i) Simplify $\sqrt{75a^2b^4}$.

(j) Simplify $\sqrt{3x^2y^4} \cdot \sqrt{12x^3}$.

(k) Simplify $\sqrt[3]{8}$.

(l) Simplify $\sqrt[3]{54x^3}$.

(m) Simplify $\sqrt[3]{16a^5b^2}$.

(n) Simplify $\sqrt[3]{8x^3} \cdot \sqrt[3]{27x^6}$.

(o) Simplify $\sqrt[4]{16x^8}$.

(p) Simplify $\sqrt{25x^2y^6}$.

(q) Simplify $2x^{1/2} \cdot 3x^{3/2}$.

(r) Simplify $(x^{1/2}y^{1/3})^6$.

(s) Simplify $\sqrt[3]{x^4y^2} \cdot \sqrt[3]{x^2y^5}$.

(t) Simplify $\sqrt{x^3y^5} \cdot \sqrt[3]{x^6y^3}$.

1.6 Factoring and Expansion

Objectives

In this section, we learn how to:

1. Expand products of binomials.

2. Factor expressions into products of binomials

1.6.1 Applications

Intro:

1. Text

2. Text

3. Text

1.6.2 Expansion

In this section, we will look at products of binomials, which are expressions of the format $(a+b)(c+d)$. As before, a product means the result of multiplication, and that multiplication happens between the value of $(a+b)$ and the vaule of $(c+d)$. By **binomial**, we mean $(a+b)$ is a binomial, and $(c+d)$ is a binomial. We start with expansion, where we turn this product of binomials $(a+b)(c+d)$ into an expression of the format Ax^2+Bx+C, known as a **trinomial**. Because the highest power is a 2, we note that Ax^2+Bx+C is a **quadratic**. In Ax^2+Bx+C, we will call A the **quadratic coefficient**, call B the **linear coefficient**, and call C the **constant term**. For example, in $5x^2+6x+7$, the quadratic coefficient is 5, the linear coefficient is 6, and the constant term is 7.

We will start with $(a+b)(c+d)$ where both a and c are the variable x. This will lead to the quadratic coefficient A in Ax^2+Bx+C always being 1. In these all such examples, we will start with expressions of the form $(a+b)(c+d)$, but then look at the expansion Ax^2+Bx+C, comparing the expansions from several examples to see if there are any patterns. Spoiler alert: some of the patterns can be come about from examining which coefficients in Ax^2+Bx+C are positive and which coefficients are negative.

Let's take our initial Distributive Law $a(b+c)=ab+ac$, for the sake of clarity, temporarily replace the letters a and b, and c with x and y and z instead. It is technically the same formula to have

$$x(y+z)=xy+xz$$

but in this version of the Distributive Law, we will subsitute $x=a+b$ and $y=c$ and $z=d$ to get

$$(a+b)(c+d)=(a+b)c+(a+b)d.$$

Note that in the equation above, the left side is $(a+b)(c+d)$ instead of $a+b(c+d)$ due to the Order of Operations: we really needed the value of x, which is $a+b$, to be multiplied by $(c+d)$. Due to the fact that the Order of Operations says to do multiplication before addition, we need to use parentheses to protect the value of $a+b$, since it is the value of $a+b$ that needs to be multiplied by $c+d$.

On the right side of the equation above, we have $(a+b)c+(a+b)d$, and we can apply the Distributive Law to rewrite $(a+b)c$ as $ac+bc$, and rewrite $(a+b)d$ as $ad+bd$. Putting all the pieces together, we have

$$(a+b)(c+d) = ac+bc+ad+bd.$$

Principle 1.6.1 Expanding the Product of Two Binomials.

$$(a+b)(c+d) = ac+ad+bc+bd$$

Example 1.6.2

Expand and simplify $(x+3)(x+5)$.

Solution. We can use Principle 1.6.1 with $a=x$ and $b=3$ and $c=x$ and $d=5$ to get

$$(x+3)(x+5) = x\cdot x + x\cdot 5 + 3\cdot x + 3\cdot 5 = x^2+5x+3x+15 = x^2+8x+15.$$

Note 1.6.3 Because it will be very useful later, we examine the final expression $x^2+8x+15$ identifying that it is in the form Ax^2+Bx+C by noting the quadratic coefficient is $A=1$, the linear coefficient $B=8$, and the constant term is $C=15$. It is very important to note that the linear coefficient 8 is the sum of the numbers 3 and 5 in the original expression, while the constant term 15 is the product of the numbers 3 and 5 in the original expression.

Example 1.6.4

Expand and simplify $(x+4)(x+10)$.

Solution. We can use Principle 1.6.1 with $a=x$ and $b=4$ and $c=x$ and $d=10$ to get

$$(x+4)(x+10) = x\cdot x + x\cdot 10 + 4\cdot x + 4\cdot 10 = x^2+10x+4x+40 = x^2+14x+40.$$

Note 1.6.5 Because it will be very useful later, we examine the final expression $x^2+14x+40$ identifying that it is in the form Ax^2+Bx+C by noting the quadratic coefficient is $A=1$, the linear coefficient $B=14$, and the constant term is $C=40$. It is very important to note that the linear coefficient 14 is the sum of the numbers 4 and 10 in the original expression, while the constant term 40 is the product of the numbers 4 and 10 in the original expression.

Example 1.6.6

Expand and simplify $(x+5)(x+5)$.

Solution. $(x+5)(x+5) = x^2+5x+5x+25 = x^2+10x+25$.

Warning 1.6.7 Given the expression $(x+8)^2$ this is not x^2+64. For more information, see Warning 1.5.50.

Example 1.6.8

Expand and simplify $(x+8)^2$.

Solution. $(x+8)^2 = (x+8)(x+8) = x^2+8x+8x+64 = x^2+16x+64$.

Example 1.6.9

INSERT preview to factor by grouping

Solution.

Principle 1.6.10

$$(a-b)(c-d) = ac-ad-bc+bd$$

Whether you choose to use this version $(a-b)(c-d) = ac-ad-bc+bd$ directly or think via Principle 1.6.1 is a matter of personal choice. Let's demonstrate the two ways to think about this through an example.

Example 1.6.11

Expand and simplify $(x-3)(x-8)$

Solution 1. The given expression $(x-3)(x-8)$ fits the format of the left side $(a-b)(c-d)$ in Principle 1.6.10 with $a=x$ and $b=3$ and $c=x$ and $d=8$. In doing these identifications, the confusion may be why the stating of b and d do not have minus signs, but this is because the format of $(a-b)(c-d)$ already includes minus signs. Then, by Principle 1.6.10,

$$(x-3)(x-8) = x\cdot x - x\cdot 8 - 3\cdot x + 3\cdot 8 = x^2-8x-3x+24 = x^2-11x+24.$$

Solution 2. We can rewrite the expression $(x-3)(x-8) = (x+(-3))(x+(-8))$

because $(x+(-3))(x+(-8))$ fits the format of the left side $(a+b)(c+d)$ in Principle 1.6.1. In this format, $a=x$, and b should consist of content right after the first plus sign, so $b=-3$. Similarly, $c=x$, and d should be all the content after the second plus sign, so $d=-8$. Then by Principle 1.6.1,

$$(x+(-3))(x+(-8)) = x{\cdot}x+x{\cdot}(-8)+(-3){\cdot}x+(-3)(-8) = x^2-8x-3x+24 = x^2-11x+24.$$

Note 1.6.12 Because it will be very useful later, we examine the final expression $x^2-11x+24$ identifying that it is in the form Ax^2+Bx+C will need us to first slightly rewrite the final expression as $x^2+(-11)x+24$. by noting the quadratic coefficient is $A=1$, the linear coefficient $B=-11$, and the constant term is $C=24$. It is very important to note that the linear coefficient -11 is the sum of the numbers -3 and -8 in the original expression, while the constant term 24 is the product of the numbers -3 and -8 in the original expression. (While we noted $(-3)+(-8)=-11$ we can also note $3+8=11$.)

Before moving on to another example, we note it is a matter of personal taste whether there really was a need to introduce a separate formula as Principle 1.6.10. You may prefer to apply principles such as a negative times a negative results in a positive, without needing to introduce a variant of Principle 1.6.1.

Example 1.6.13

Expand and simplify $(x-4)(x-6)$

Solution 1. The given expression $(x-4)(x-6)$ fits the format of the left side $(a-b)(c-d)$ in Principle 1.6.10 with $a=x$ and $b=4$ and $c=x$ and $d=6$. In doing these identifications, the confusion may be why the stating of b and d do not have minus signs, but this is because the format of $(a-b)(c-d)$ already includes minus signs. Then, by Principle 1.6.10,

$$(x-4)(x-6) = x\cdot x - x\cdot 6 - 4\cdot x + 4\cdot 6 = x^2-6x-4x+24 = x^2-10x+24.$$

Solution 2. We can rewrite the expression $(x-4)(x-6) = (x+(-4))(x+(-6))$ because $(x+(-4))(x+(-6))$ fits the format of the left side $(a+b)(c+d)$ in Principle 1.6.1. In this format, $a=x$, and b should consist of content right after the first plus sign, so $b=-4$. Similarly, $c=x$, and d should be all the content after the second plus sign, so $d=-6$. Then by Principle 1.6.1,

$$(x+(-4))(x+(-6)) = x{\cdot}x+x{\cdot}(-6)+(-4){\cdot}x+(-4)(-6) = x^2-6x-4x+24 = x^2-10x+24.$$

Note 1.6.14 In $x^2 - 10x + 24$, we can rewrite the final expression as $x^2 + (-10)x + 24$, which has linear coefficient -10 and constant term 24. Note that the constant 24 is the product of the numbers (-4) and (-6) in the original expression (or we can point out that the product of the numbers after the minus signs, namely 4 and 6 multiply to 24). Thus the linear coefficient -10 is the sum of the numbers -4 and -6, (or we can point out that the numbers after the minus signs 4 and 6 add to 10, the number after the minus sign in $x^2 - 10x + 24$).

Example 1.6.15

Expand and simplify $(x-5)(x-5)$.

$(x-5)(x-5) = x^2 - 5x - 5x + 25 = x^2 - 10x + 25.$

Example 1.6.16

Expand and simplify $(x-8)^2$.

Solution. $(x-8)^2 = (x-8)(x-8) = x^2 - 8x - 8x + 64 = x^2 - 16x + 64.$

We present two formulas together because they are similar: in each, the left side is a product of two binomials, with one binomial having a plus sign in between, and one binomial having a minus sign in between.

Principle 1.6.17

- $(a+b)(c-d) = ac - ad + bc - bd$
- $(a-b)(c+d) = ac + ad - bc - bd$

Example 1.6.18

Expand and simplify $(x+7)(x-2)$.

Solution. We can use the first formula in Principle 1.6.17 with $a = x$ and $b = 7$ and $c = x$ and $d = 2$ to get

$$(x+7)(x-2) = x \cdot x - x \cdot 2 + 7 \cdot x - 7 \cdot 2 = x^2 - 2x + 7x - 14 = x^2 + 5x - 14.$$

Note 1.6.19 For the first time in this section, we end with a quadratic expression $x^2 + 5x - 14$ where the constant term is negative. This happened because we multiplied $+7$ with -2. Recall the product of a real numbers with opposite signs results in a negative number. In the original expression, let's examine the 7 and the 2 which appear after the plus sign and the minus sign. The difference between these numbers is 5, and note the linear coefficient was 5.

Example 1.6.20

Expand and simplify $(x-7)(x+2)$.

Solution. We can use the second formula in Principle 1.6.17 with $a = x$ and $b = 7$ and $c = x$ and $d = 2$ to get

$$(x-7)(x+2) = x \cdot x + x \cdot 2 - 7 \cdot x - 7 \cdot 2 = x^2 + 2x - 7x - 14 = x^2 - 5x - 14.$$

Note 1.6.21 As in our previous example, the final expression $x^2 - 5x - 14$ has a negative constant term, for the same reason as earlier: this was the result of multiplying a negative number (namely -7) together with a positive number (namely $+2$). The opposite of the linear coefficient -5 is 5, and that is the difference between the numbers 7 and 2 in the original expression. When we subtract 2 and 7 in a certain order, we get -5.

Example 1.6.22

Expand and simplify $(x-3)(x+10)$.

Solution. $(x-3)(x+10) = x^2 + 10x - 3x - 30 = x^2 + 7x - 30.$

Example 1.6.23

Expand and simplify $(x+9)(x-9)$.

Solution. $(x+9)(x-9) = x^2 - 9x + 9x - 81 = x^2 + 0x - 81 = x^2 - 81.$

Note 1.6.24 We have been looking at expanding and simplifying the product of binomials. Typically in these questions, the expansion creates four terms and then collecting of like terms leads to a total of three terms. In this case, we only have two terms in $x^2 - 81$ because the middle terms $-9x + 9x$ simplified to zero.

1.6.3 Factoring

Our goal here is to take a trinomial in the format of $Ax^2 + Bx + C$ such as $x^2 + 7x - 30$ and $x^2 + 14x + 40$ and factor this, producing a product of binomials. We start with the situation where $A = 1$. It is helpful to see what happens based on whether the constant term is positive or negative.

Try it 1.6.25

Compare the examples in the previous portion on expansion. What patterns do you notice about the constant term C in the final expression $x^2 + Bx + C$?

Example 1.6.26

Will the two binomials in the factored form of $x^2 + 8x + 15$ have the same sign or opposite signs?

Solution. Because the constant term is positive (namely 15), the two binomials will have the same sign.

Example 1.6.27

Will the two binomials in the factored form of $x^2 + 8x - 20$ have the same sign or opposite signs?

Solution. Because the constant term is negative (namely -20), the two binomials will have opposite signs.

Strategy 1.6.28 Factoring a quadratic trinomial into a product of binomials.

Suppose we are given Ax^2+Bx+C with $A = 1$, meaning we have x^2+Bx+C. To factor $x^2 + Bx + C$ as the product of binomials:

- If the costant term C is positive, then the signs contained in both binomials match.
- If the costant term C is negative, then the signs contained in both binomials differ.

Now, let's dig in further to see what happens when the constant term is postive. First, let's note that when we have the consstant term positive, then the two binomials have to have the same sign, meaning that both signs are positive or both signs are negative. But now, we can see two situations: the linear term can be positive or negative. For example, we can have $x^2 + 9x + 20$ or we can have $x^2 - 9x + 20$.

It seems like a reasonable guess (and we'll check if it works) that if we have a $+9x$ term, that the matching sign in both binomials is a plus, while if we have a $-9x$ term, then the matching sign in both binomials is a minus.

Let's dig in to see if we can find the missing values in

$$(x + \_\_\_)(x + \_\_\_) = x^2 + 9x + 20.$$

To match the notation earlier, let's fill in these blanks with b and d. That is, let's find the b and d in

$$(x+b)(x+d) = x^2 + 9x + 20$$

by expanding the left side:

$$x^2 + dx + bx + bd = x^2 + 9x + 20.$$

Now, the middle two terms dx and bx are like terms, and we can "collect like terms" here, though what we'll really do is apply Note 1.1.84 and factor out x for the middle two terms:

$$x^2 + (d+b)x + bd = x^2 + 9x + 20.$$

The last term on the left bd has no x, so this must be what matches up with 20 on the right. The linear coefficient (the number in front of x) on the left is $d+b$ and on the right is 9. This combination of things tells us what we need. The two numbers we need to find (called b and d here) need to multiply to 20 and need to add to 9. Those two numbers are 4 and 5 so we can factor $x^2 + 9x + 20$ as either

$$(x+4)(x+5)$$

or

$$(x+5)(x+4).$$

These are essentially the same (just switching the order of the factors), and we can check our factoring by expanding.

For $x^2 - 9x + 20$, we'll need minus signs inside both binomials. That is, our set up looks like:

$$(x - \underline{\quad})(x - \underline{\quad}) = x^2 - 9x + 20.$$

Now, the minus signs are already drawn in, so we just need to fill in numbers after the minus signs. Based on the 20 and the number after the minus sign but before the x being 9, we are again looking for two numbers whose product is 20 and whose sum is 9, and these are again 4 and 5, which fill in the two blanks:

$$(x-4)(x-5)$$

or alternately,

$$(x-5)(x-4).$$

It is always a little more confusing to describe, which is why we are using the following language: in

$$(x - \underline{\quad})(x - \underline{\quad}) = x^2 - 9x + 20$$

we have already written in the minus signs, so we just need to include write after the minus signs the two positive numbers that add to 9 and multiply to 20.

Strategy 1.6.29 Factoring a quadratic trinomial with a positive constant term.

To factor $x^2 + Bx + C$ with constant term $C > 0$,

- If the linear term B is positive, then the signs contained in both binomials are positive signs.

  In this template

  $$(x + \_\_\_)(x + \_\_\_) = x^2 + Bx + C$$

  we need to fill the blanks with positive numbers that multiply to C and add to B.

- If the constant term C is negative, then the signs contained in both binomials differ.

  In this template

  $$(x - \_\_\_)(x - \_\_\_) = x^2 + Bx + C$$

  the minus signs are already drawn in the factoring, and we need to fill the blanks with positive numbers that multiply to C and add to $|B|$.

Example 1.6.30

Factor $x^2 + 11x + 24$.

Solution. Because we have a quadratic trinomial with positive constant term 24 and we are being asked to factor, Strategy 1.6.29 applies.

FIX: strategy box missing for print version. Numbering is off in print

Add warning that we shouldn't set an expression equal to zero and create an equation to solve: gently preview what the student is confusing this with Factor by grouping

Principle 1.6.31 text.

text

Example 1.6.32

Statement text

Solution. Solution text

Try it 1.6.33

Text

Note 1.6.34 Text

Title.

Text

Warning 1.6.35 text

Investigate!

While walking through a fictional forest, you encounter three trolls guarding a bridge. Each is either a *knight*, who always tells the truth, or a *knave*, who always lies. The trolls will not let you pass until you correctly identify each as either a knight or a knave. Each troll makes a single statement:

Troll 1: If I am a knave, then there are exactly two knights here.

Troll 2: Troll 1 is lying.

Troll 3: Either we are all knaves, or at least one of us is a knight.

Which troll is which?

Try it 1.6.36

Spend a few minutes thinking about the Investigate problem above. What could you conclude if you knew Troll 1 really was a knave (i.e., their statement was false)? Share your initial thoughts on this.

Definition 1.6.37 Argument.

An **argument** is a sequence of statements, the last of which is called the **conclusion** and the rest of which are called **premises**.

An argument is said to be **valid** provided the conclusion must be true whenever the premises are all true. An argument is **invalid** if it is not valid; that is, all the premises can be true, and the conclusion could still be false.

An argument is **sound** provided it is valid and all the premises are true. A **proof** of a statement is a sound argument whose conclusion is the statement.

By the way... Our definitions of **argument**, **valid argument**, and **sound argument** are the same ones used in philosophy, the other primary academic discipline concerned with logic and reasoning.

Example 1.6.38

Consider the following two arguments:

$$\begin{array}{ll} & \text{If Edith eats her vegetables, then she can have a cookie.} \\ & \text{Edith eats her vegetables.} \\ \hline \therefore & \text{Edith gets a cookie.} \end{array}$$

$$\begin{array}{ll} & \text{Florence must eat her vegetables to get a cookie.} \\ & \text{Florence eats her vegetables.} \\ \hline \therefore & \text{Florence gets a cookie.} \end{array}$$

(The symbol "$\therefore$" means "therefore")

Are these arguments valid?

Solution. Do you agree that the first argument is valid but the second argument is not? We will soon develop a better understanding of the logic involved in this analysis, but if your intuition agrees with this assessment, then you are in good shape.

Notice the two arguments look almost identical. Edith and Florence both eat their vegetables. In both cases, there is a connection between the eating of vegetables and cookies. Yet we claim that it is valid to conclude that Edith gets a cookie, but not that Florence does. The difference must be in the connection between eating vegetables and getting cookies. We need to be skilled at reading and comprehending these sentences. Do the two sentences mean the same thing?

Unfortunately, in everyday language we are often sloppy, and you might be tempted to say they are equivalent. But notice that just because Florence *must* eat her vegetables, we have not claimed that doing so would be *enough* (she might also need to clean her room, for example). In everyday (non-mathematical) practice, you might be tempted to say this "other direction" is implied. In mathematics, we never get that luxury.

Remark 1.6.39 The arguments in the example above illustrate another important point: Even if you don't care about the advancement of human knowledge in the field of mathematics, becoming skilled at analyzing arguments is useful. And even if you don't want to give your grandmother a cookie. If you are *using* mathematics to solve problems in some other discipline, it is still necessary to demonstrate that your solution is correct. You better have a good argument that it is!

1.6.4 Applications Revisited

Let's revisit the applications we introduced at the beginning of this section.

Example 1.6.40

Text

Solution. Answer

1.6.5 Summary

1. Summary point.
2. Summary point.
3. Summary point.

1.6.6 Exercises

1. Text.
   (a) Subpart
   (b) Subpart
   (c) Subpart
2. Text.
   (a) Subpart
   (b) Subpart
   (c) Subpart
3. Text.
   (a) Subpart
   (b) Subpart
   (c) Subpart

1.7 Rational Expressions

Text of section.

1.8 Solving Equations Revisited

Text of section. More applications, more challenging (up to linear) equations, and fractional equations (checking denominator). Applications where the variable appears more than once: transfer bonuses, mixtures, distance, percentages Literal equations (multiple variables) Systems of equations Justifying cross-multiplication

A flight requires 12,500 miles, but your credit card offers a 10% bonus when transferring. How many miles should you transfer to meet the requirement while taking advantage of the transfer bonus?